

# The Question Book

General Physics II · every question, solved and explained · 13 August 2026

All **253 questions** from the lecture slides, the reference sheets and both real tests. Grouped by topic, and within each topic ordered **easy first**, so the definitions come before the calculations that assume them.

Every question carries the source it came from. Where the same question was set twice it appears once and says where else it was asked. The answer is ticked, and every other option carries a line saying why it fails.

---

## Contents

1. **Charge, Coulomb and the Electric Field** 48 questions
2. **Electric Potential and Energy** 14 questions
3. **Electric Flux and Gauss** 17 questions
4. **Capacitance, Dielectrics and Energy** 32 questions
5. **Current, Resistance and Networks** 43 questions
6. **EMF and Internal Resistance** 14 questions
7. **Magnetic Force on Charges and Wires** 30 questions
8. **Sources of a Magnetic Field** 19 questions
9. **Electromagnetic Induction** 19 questions
10. **Transformers** 6 questions
11. **Maxwell and Electromagnetic Waves** 11 questions

## 1 Charge, Coulomb and the Electric Field

**Speed tip.** Convert to SI before anything else. mg to kg,  $\mu\text{C}$  to C, cm to m. Then ask: one charge or two? One gives a field, two gives a force.

1.1

EASY

Reference sheet R.3

**Step 1 of the seven-step method is to read the question twice and underline what is asked for. Why does it come first?**

✓ Because many marks are lost answering a question that was not set

Correct, and it is the reason the step exists.

✗ Because it saves time later

Wrong. It costs time and buys accuracy.

✗ Because the formula cannot be chosen until the units are known

Wrong. That describes steps 3 and 4.

✗ Because examiners award marks for underlining

Wrong. The underlining is for you, not the marker.

✗ Because it is the only step that does not need a calculator

Wrong. That is not a reason for its order.

Answering the wrong question scores zero however perfect the arithmetic. Test 1 question 13 is the standing example: it gives two charges and asks for the field produced by the NEGATIVE one, and using the other charge produces an answer that is on the option list, so the mistake is invisible unless you read carefully.

1.2

EASY

Reference sheet R.3

**Step 2 is to list what you know with symbols and units, and to write the unknown with a  $\{\{1\}\}$  mark.**

Answer:

A question mark. Writing the unknown down as its own line, with a question mark, is what makes step 4 possible: you cannot choose the formula containing what you know and what you want until both are on the page in front of you.

1.3

EASY

Reference sheet R.4

**What distinguishes a conductor from an insulator?**

✓ In a conductor charge moves freely; in an insulator charge is bound and cannot move freely

Correct. Mobility, not quantity.

✗ A conductor has charge and an insulator has none

Wrong. Both are full of charge; an insulator's simply cannot travel.

✗ A conductor is always metal and an insulator is always plastic

Wrong. Those are examples, not the definition.

✗ A conductor stores charge and an insulator releases it

Wrong. That describes a capacitor, and inaccurately.

✗ An insulator has no electrons

Wrong. Every atom has electrons, bound ones included.

Whether the charge can MOVE. Both have plenty of charge in them; the difference is mobility. It is why a conductor has no interior field, the free charges rearranging until it cancels, and why an insulator can instead polarise, its bound charges shifting slightly to form dipoles without ever leaving home.

1.4

EASY

Reference sheet R.1

**What is the value of the Coulomb constant  $k$ ?**

✓  $8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

Correct.  $k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ , taken as  $9 \times 10^9$  in practice.

✗  $8.85 \times 10^{-12} \text{ N m}^2 \text{ C}^{-2}$

Wrong. That is  $\epsilon_0$ , the permittivity of free space, in F/m.

✗  $9.11 \times 10^{-31} \text{ N m}^2 \text{ C}^{-2}$

Wrong. That is the electron mass in kilograms.

✗  $1.602 \times 10^{-19} \text{ N m}^2 \text{ C}^{-2}$

Wrong. That is the elementary charge in coulombs.

✗  $4 \pi \times 10^{-7} \text{ N m}^2 \text{ C}^{-2}$

Wrong. That is  $\mu_0$ , the permeability of free space, in H/m.

$k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ , usually rounded to  $9 \times 10^9$  in a calculation. It is the constant in Coulomb's law and in  $E = kQ/r^2$ , and it is related to the permittivity of free space by  $k = 1/(4 \pi \epsilon_0)$ .

1.5

EASY

Reference sheet R.1

**The Coulomb constant  $k$  has the value  $8.99 \times 10$  to the power  $\{\{1\}\}$   $\text{N m}^2$  per  $\text{C}^2$ .**

Answer:

The exponent is positive 9. A common slip is to borrow the negative exponent of  $\epsilon_0$ ;  $k$  is a large number precisely because the electric force is strong.

1.6

EASY

Reference sheet R.1

What is the value of the permittivity of free space,  $\epsilon_0$ ?

✓  $8.85 \times 10^{-12} \text{ F m}^{-1}$

Correct.  $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ .

✗  $8.99 \times 10^9 \text{ F m}^{-1}$

Wrong. That is the Coulomb constant  $k$ .

✗  $4 \pi \times 10^{-7} \text{ F m}^{-1}$

Wrong. That is  $\mu_0$ , the magnetic counterpart.

✗  $1.602 \times 10^{-19} \text{ F m}^{-1}$

Wrong. That is the elementary charge.

✗  $1.67 \times 10^{-27} \text{ F m}^{-1}$

Wrong. That is the proton mass.

$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ . It appears in the parallel plate formula  $C = \epsilon r \epsilon_0 A/d$  and in Gauss's law, and it is tied to  $k$  by  $k = 1/(4 \pi \epsilon_0)$ .

1.7

EASY

Reference sheet R.1

What is the value of the elementary charge  $e$ ?

✓  $1.602 \times 10^{-19} \text{ C}$

Correct.  $e = 1.602 \times 10^{-19} \text{ C}$ .

✗  $1.602 \times 10^{19} \text{ C}$

Wrong sign on the exponent. A charge that size is impossible for one electron.

✗  $9.11 \times 10^{-31} \text{ C}$

Wrong. That is the electron MASS in kilograms, not its charge.

✗  $1.67 \times 10^{-27} \text{ C}$

Wrong. That is the proton mass in kilograms.

✗  $8.85 \times 10^{-12} \text{ C}$

Wrong. That is  $\epsilon_0$ .

$e = 1.602 \times 10^{-19} \text{ C}$ . Every charge is a whole multiple of it, which is what  $q = ne$  means. Its reciprocal is about  $6.24 \times 10^{18}$ , which is why a system of roughly  $6.2 \times 10^{18}$  electrons carries close to one coulomb.

1.8

EASY

Reference sheet R.1

**Which statement is the starting point of all electrostatics?**

✓ Like charges repel and unlike charges attract

Correct. Opposites attract; the same signs push apart.

✗ Like charges attract and unlike charges repel

Wrong, and exactly reversed.

✗ All charges attract, regardless of sign

Wrong. Gravity always attracts; the electric force does not.

✗ All charges repel, regardless of sign

Wrong, for the same reason.

✗ Charges neither attract nor repel unless they are moving

Wrong. Electrostatics is the study of charges AT REST exerting forces on each other.

Like repels, unlike attracts. Everything electrostatic starts here, and the sign of the Coulomb force follows from it: a positive product of charges means repulsion, a negative product means attraction.

1.9

EASY

Reference sheet R.2

**Two electrons placed near each other will attract.**

Answer: **false**

They repel. Both carry negative charge, so they are like charges, and like charges repel.

A charge of  $+1.5 \times 10^{-9} \text{ C}$  is placed in a uniform electric field of magnitude  $2500 \text{ N/C}$ . What is the magnitude of the force acting on the charge?

✗  $4.0 \times 10^{-6} \text{ N}$

Wrong. This would need a charge of  $1.6 \times 10^{-9} \text{ C}$ .

✗  $1.0 \times 10^{-6} \text{ N}$

Wrong. This drops the 1.5 and uses the charge as  $4 \times 10^{-10} \text{ C}$ .

✓  $3.75 \times 10^{-6} \text{ N}$

Correct.  $F = qE = 1.5 \times 10^{-9} \times 2500 = 3.75 \times 10^{-6} \text{ N}$ .

✗  $2.0 \times 10^{-6} \text{ N}$

Wrong. This is the answer for a charge of  $8 \times 10^{-10} \text{ C}$ .

✗  $3.0 \times 10^{-6} \text{ N}$

Wrong. This rounds 3.75 down to 3, which the options do not intend.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

The electric field is defined as the force felt per unit charge, so it is force divided by charge. Turn that definition around and it becomes a way of finding the force: put a charge in a known field and the force is simply the field times the charge. The field belongs to the region and exists whether or not anything is there to feel it; the force only appears when a charge is placed in it.

#### GIVEN

- $q = +1.5 \times 10^{-9} \text{ C}$
- $E = 2500 \text{ N C}^{-1}$  (uniform, so the same everywhere)

**Find:** The magnitude of the force  $F$  on the charge, in newtons

#### FORMULA

$$F = qE$$

**Why this one.** ONE charge is named, not two, and a field is given rather than a second charge and a distance. That count decides it: two charges and a separation would need Coulomb's law, but one charge sitting in a ready-made field needs only this. Uniform means  $E$  does not change from place to place, so no distance is needed at all.

#### WHAT EACH SYMBOL MEANS

- $F$  = the electric force, in newtons
- $q$  = the charge placed in the field, in coulombs
- $E$  = the field strength, in  $\text{N C}^{-1}$ , which is the same unit as  $\text{V m}^{-1}$

#### THE WORKING, LINE BY LINE

1.  $F = q \times E$
2.  $F = 1.5 \times 10^{-9} \times 2500$
3. Write 2500 in standard form:  $2.5 \times 10^3$
4. Front numbers:  $1.5 \times 2.5 = 3.75$
5. Powers of ten:  $10^{-9} \times 10^3 = 10^{-6}$
6.  $F = 3.75 \times 10^{-6} \text{ N}$

#### CHECK IT

No distance appears anywhere in the question, which is the giveaway that no inverse square is involved. If you found yourself squaring something, you reached for Coulomb's law by reflex. The answer is a few micronewtons, which is right for a nanocoulomb sitting in an ordinary laboratory field.

$F = qE$ , the definition of the electric field as force per unit charge rearranged.  $F = 1.5 \times 10^{-9} \times 2500 = 3.75 \times 10^{-6}$  N. Nothing else is needed: the field is given, so Coulomb's law does not come into it.

1.11

EASY

Reference sheet R.1

**Charge is quantised as  $q = ne$ , where  $n$  must be a **number**.**

Answer:

A whole number. Charge comes in coins, not in liquid: you can have 3 electrons or 4, never three and a half. That is why the charge of any object is always an exact multiple of  $1.602 \times 10^{-19}$  C.

1.12

EASY

Reference sheet R.3

**Step 1 of the method is to read the question twice and underline what is asked for.**

Answer: **true**

True, and the reason is blunt: many marks are lost answering a question that was not set. Test 1 question 13 asks for the field produced by the **NEGATIVE** charge, and using the other one gives an answer that appears on the option list, so the error is invisible without careful reading.

1.13

EASY

Reference sheet R.4

**In a conductor charge moves freely, while in an insulator charge is **not** and cannot move freely.**

Answer:

Bound. Both materials are full of charge; the difference is whether it can travel. That single word explains why a conductor has no interior field, its free charges rearranging until everything cancels, and why an insulator instead polarises, its bound charges shifting slightly into dipoles without leaving home.

1.14

EASY

Lecture slides, module 1, slide 60

**The S.I. unit of electric field is:**

✓  $\text{NC}^{-1}$

Correct. Force per unit charge.

✗  $\text{Vm}^{-2}$

Wrong. The other valid form is  $\text{Vm}^{-1}$ , volts per metre, not per metre squared.

✗ Cm

Wrong. Coulomb metre is the unit of **DIPOLE MOMENT**,  $p = qd$ .

✗ Vm

Wrong. Volt metre inverts the correct volts per metre.

✗ NC

Wrong. Newton coulomb multiplies where the definition divides.

Read it off the definition.  $E = F/q$ , a force in newtons divided by a charge in coulombs, giving newtons per coulomb,  $\text{NC}^{-1}$ . The slide also derives it the other way, from  $E = \text{minus } dV/dx$ , a voltage divided by a distance, giving volts per metre,  $\text{Vm}^{-1}$ . The two are the same unit, which is why a field may be quoted in either N/C or V/m.

1.15

EASY

Lecture slides, module 1, slide 68

The magnitude of the electric intensity is represented by  $E = 3i + 4j$ , where  $i$  and  $j$  are unit vectors along the X and Y axes respectively. The Z component of the electric field is:

✓  $0 \text{ Vm}^{-1}$

Correct. No k term means no z component.

✗  $5 \text{ Vm}^{-1}$

Wrong. 5 is the MAGNITUDE of this vector, from the square root of  $9 + 16$ , not its z component.

✗  $7 \text{ Vm}^{-1}$

Wrong. 7 is  $3 + 4$ , added as though the components were plain numbers.

✗  $1 \text{ Vm}^{-1}$

Wrong. Nothing here gives 1.

✗  $3 \text{ Vm}^{-1}$

Wrong. 3 is the x component.

There is no k term in the expression, and k is the unit vector along z. A component that is not written is zero. The whole field lies in the x-y plane. The question looks like it wants a calculation and does not: it is checking that you can read a vector.

1.16

MEDIUM

Reference sheet R.3

Step 3 says convert everything to SI immediately, before anything else is done.

Answer: **true**

True, and it is one of the two steps that save the most marks. cm to metres, mg to kilograms,  $\mu\text{F}$  to farads, all before a single number enters a formula. In this course 5 cm left unconverted costs a factor of 100 once it is squared, and both the wrong answers it produces appear on the option lists.

1.17

MEDIUM

Reference sheet R.3

Step 4 is to choose the formula that contains what you know and what you want. What does the method say to do if no single formula does?

✓ You need an intermediate step

Correct, and it is the standard shape of the harder questions on this paper.

✗ Guess the closest option

Wrong. The method exists to avoid guessing.

✗ The question cannot be answered

Wrong. It usually just needs two steps rather than one.

✗ Use any formula with the right units

Wrong. Matching units is a check, not a way to choose a formula.

✗ Go back and convert again

Wrong. Converting is step 3 and is already done.

An intermediate step: find some quantity that links the two. Test 2 question 4 is exactly this. No formula takes you from a terminal p.d. straight to the emf, so you compute the current first,  $I = V/R$ , and that current is the bridge into  $E = I(R + r)$ .

1.18

MEDIUM

Reference sheet R.3

**Step 5 is to rearrange in  $\{\{1\}\}$  first, and substitute numbers only at the end.**

Answer:

In symbols. Rearranging a formula that already has numbers in it is where sign and reciprocal errors breed. Getting  $E = mg/q$  on the page before any digit appears means the only thing left to do is arithmetic.

1.19

MEDIUM

Reference sheet R.3

**Step 6 is to compute, keeping track of the powers of ten  $\{\{1\}\}$  from the front numbers.**

Answer:

Separately, in two columns. Multiply the front numbers, then add the exponents. It is what turns a frightening expression like  $9.1 \times 10^{-31} \times 4 \times 10^6$  into two easy operations, and it is why the exponent stops slipping.

1.20

MEDIUM

Reference sheet R.3

**Step 7 is to check. Select the three checks the method names.**

✓ Is the size believable?

Correct, and it kills the positive-exponent capacitance in Test 1 question 6 immediately.

✓ Does it have the right unit?

Correct, and it catches a force answered where a field was asked.

✓ Does the sanity check apply, such as parallel being less than the smallest?

Correct, and the sanity checks are free.

✗ Did you use a calculator?

Wrong. Not one of the checks, and a calculator will happily give a wrong answer.

✗ Is the answer one of the offered options?

Wrong, and dangerous: the examiner deliberately puts the common wrong answers on the list.

Size, unit, sanity check. Together they catch most of what goes wrong: a capacitance of 4.77 farads is not believable, an answer in newtons when the question wanted volts is not the right unit, and a parallel resistance larger than its smallest member is impossible. Steps 3 and 7 are the two that save the most marks and the two most often skipped.

1.21

MEDIUM

Reference sheet R.2

**A field points away from a positive charge and  $\{\{1\}\}$  a negative one.**

Answer:

Towards. The memory hook is the fan and the vacuum: a positive charge is a fan blowing outward, a negative charge is a vacuum sucking inward, and the field is which way the air moves. Stand between them and both send you the same way, so their fields ADD. Two fans fight, so they cancel.

1.22

MEDIUM

Reference sheet R.1

**The permittivity of free space is written  $\epsilon_0$  and is measured in farads per  $\{\{1\}\}$ .**

Answer:

Farads per metre. Reading the unit is itself a check: a permittivity multiplied by an area over a distance gives a capacitance in farads, which is exactly the parallel plate formula.

1.23

MEDIUM

Reference sheet R.1

The elementary charge is  $1.602 \times 10^{-19}$  coulombs.

Answer:

Minus 19. Keep it apart from the electron mass at  $10^{-31}$  kg and the proton mass at  $10^{-27}$  kg: one is a charge, the others are masses.

1.24

MEDIUM

Reference sheet R.1

The hook for  $q = ne$  is that charge comes in coins, not in liquid. What does that mean in practice?

✓ A charge is always a whole multiple of  $e$ , never a fraction of an electron

Correct.  $n$  is a whole number, so a fraction of an electron is not available.

✗ Charge can take any value at all

Wrong, and this is what quantisation denies.

✗ Charge can be created and destroyed freely

Wrong. That contradicts conservation of charge, a separate property.

✗ Charge depends on the speed of the object

Wrong. Charge is invariant: it does not depend on speed or reference frame.

✗ Charge is a vector quantity

Wrong. Charge is a scalar; the  $+$  and  $-$  signs indicate type, not direction.

Quantisation: charge exists only in whole multiples of the elementary charge, so  $q = ne$  with  $n$  a whole number. You can have  $3e$  or  $4e$  but never  $3.5e$ , exactly as you can have three coins or four but not three and a half.

1.25

MEDIUM

Reference sheet R.2

In  $q = ne$ , the letter  $n$  must be a whole number.

Answer:

A whole number. That is the whole content of quantisation, and it is why the answer to a charge-from-electrons question is the electron count multiplied by  $e$ .

1.26

MEDIUM

Reference sheet R.1

**Coulomb's law is an inverse square law. If the distance between two charges is TRIPLED, what happens to the force?**

✓ It falls to one ninth

Correct. Three squared is nine, so the force drops to a ninth.

✗ It falls to one third

Wrong, and it is the trap: falling to a third is what the POTENTIAL does, because  $V$  goes as  $1/r$  rather than  $1/r^2$ .

✗ It falls to one sixth

Wrong. No inverse power gives a sixth for a tripling.

✗ It becomes three times larger

Wrong direction. Moving charges apart weakens the force.

✗ It is unchanged

Wrong. Distance is in the formula, so it certainly changes.

$F = k q_1 q_2 / r^2$ , so the force depends on the SQUARE of the distance. Tripling  $r$  multiplies  $r^2$  by 9, so the force falls to one ninth. The same reasoning applies to the electric field  $E = kQ/r^2$ , but NOT to the potential  $V = kQ/r$ , which would fall only to one third.

1.27

MEDIUM

Reference sheet R.2

**Tripling the distance in Coulomb's law drops the force to one  $\frac{1}{9}$ .**

Answer:

One ninth, because the distance is squared. Compare the potential, which would fall only to one third.

1.28

MEDIUM

Reference sheet R.1

**What does the principle of superposition tell you to do, and what does it apply to?**

✓ Compute each contribution one at a time and then add them, which works for forces, fields and potentials

Correct, and the fact that it covers all three is what makes it so useful.

✗ Compute all contributions at once, which works only for forces

Wrong on both counts: the method is one at a time, and it is not restricted to forces.

✗ Add the charges first and then apply one formula, which works only for fields

Wrong. Adding the charges first only works in special symmetric cases and is not the principle.

✗ Take only the largest contribution and ignore the rest

Wrong. Every contribution counts.

✗ Multiply the individual contributions together

Wrong. Contributions add, they do not multiply.

One at a time, then add. Superposition is what lets a many-charge problem be broken into a series of two-charge problems, and it holds for forces, for fields and for potentials alike. Remember that forces and fields are VECTORS, so they add as vectors, while potentials are scalars and add as plain numbers.

1.29

MEDIUM

Reference sheet R.2

**Superposition applies to electric forces and fields but not to electric potentials.**

Answer: **false**

It applies to all three. Potentials superpose too, and being scalars they are the easiest of the three to add, since no directions are involved.

Calculate the electric field in which a charged oil drops of mass 8 mg and charged +9  $\mu\text{C}$  will be held stationary.

✗ 5.3  $\text{Vm}^{-1}$

Wrong. No route to 5.3 from these numbers.

✗ 4.35  $\text{Vm}^{-1}$

Wrong. This is roughly half the correct value, which is what you get from using 4 mg or double the charge.

✓ 8.89  $\text{Vm}^{-1}$

Correct.  $E = mg/q = (8 \times 10^{-6} \times 10) / (9 \times 10^{-6}) = 8.89 \text{ Vm}^{-1}$ .

✗ 7.26  $\text{Vm}^{-1}$

Wrong. No route to 7.26 from these numbers.

✗ 6.62  $\text{Vm}^{-1}$

Wrong. No route to 6.62 from these numbers.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Anything that hangs still is not free of forces; it has forces that cancel. A charged oil drop in an electric field has two: gravity pulling it down, and the electric force pushing it up. When they are equal in size and opposite in direction, the drop hangs. This is the whole of Millikan's famous experiment, which is how the charge on the electron was first measured.

#### GIVEN

- $m = 8 \text{ mg} = 8 \times 10^{-6} \text{ kg}$  (milligram, so  $\div 1000$  twice)
- $q = +9 \mu\text{C} = 9 \times 10^{-6} \text{ C}$
- $g = 10 \text{ m s}^{-2}$  (this examiner uses 10, not 9.8)

**Find:** The electric field  $E$  that holds it stationary, in  $\text{V m}^{-1}$

#### FORMULA

$$qE = mg, \text{ so } E = mg / q$$

**Why this one.** The words HELD STATIONARY are the whole instruction. Stationary means the net force is zero, so the upward electric force  $qE$  must exactly equal the downward weight  $mg$ . Setting them equal and rearranging for  $E$  is the entire question. Suspended, in equilibrium and does not fall all mean the same thing.

#### WHAT EACH SYMBOL MEANS

- $q$  = the charge on the drop, in coulombs
- $E$  = the electric field strength, in  $\text{V m}^{-1}$  (the same unit as  $\text{N C}^{-1}$ )
- $m$  = the mass of the drop, in kilograms
- $g$  = the acceleration due to gravity, in  $\text{m s}^{-2}$

#### THE WORKING, LINE BY LINE

1.  $m = 8 \text{ mg} = 8 \times 10^{-6} \text{ kg}$
2.  $q = 9 \mu\text{C} = 9 \times 10^{-6} \text{ C}$
3.  $mg = 8 \times 10^{-6} \times 10 = 8 \times 10^{-5} \text{ N}$
4.  $E = mg / q = (8 \times 10^{-5}) / (9 \times 10^{-6})$
5. The powers of ten:  $10^{-5} \div 10^{-6} = 10^1$
6.  $E = (8 / 9) \times 10 = 0.889 \times 10 = 8.89 \text{ V m}^{-1}$

#### CHECK IT

Both quantities were  $\times 10^{-6}$  before  $g$  was applied, so the prefixes cancel and only the 10 from  $g$  survives. That is why the answer is a clean single-digit number rather than something tiny.

Stationary means the electric force balances the weight, so  $qE = mg$  and  $E = mg/q$ . Given  $m = 8 \text{ mg} = 8 \times 10^{-6} \text{ kg}$  and  $q = 9 \times 10^{-6} \text{ C}$ ,  $E = (8 \times 10^{-6} \times 10) / (9 \times 10^{-6}) = 80 \times 10^{-6} / 9 \times 10^{-6} = 8.89 \text{ Vm}^{-1}$ . Note the examiner is using  $g = 10 \text{ ms}^{-2}$ ; with  $g = 9.8$  the answer would be 8.71, which is not offered.

**What is the charge of a system that has  $6.2 \times 10^{18}$  electrons?**

**X** 1.99 C

Wrong, and about double: this comes from using  $3.2 \times 10^{-19}$  for the electronic charge.

**X** 0.89 C

Wrong. A near miss that does not follow from  $6.2 \times 10^{18}$  electrons.

**✓** 0.99 C

Correct.  $q = ne = 6.2 \times 10^{18} \times 1.602 \times 10^{-19} = 0.99$  C.

**X** 0.79 C

Wrong. A near miss that does not follow from  $6.2 \times 10^{18}$  electrons.

**X** 1.07 C

Wrong. This would need about  $6.7 \times 10^{18}$  electrons.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Charge is not continuous; it comes in lumps. Every lump is the charge on one electron,  $e = 1.602 \times 10^{-19}$  C, and no smaller amount has ever been found free. That is called QUANTISATION of charge. So the total charge of anything is just how many lumps it has, times the size of one lump.

#### GIVEN

- $n = 6.2 \times 10^{18}$  electrons
- $e = 1.602 \times 10^{-19}$  C (the elementary charge)

**Find:** The total charge  $Q$ , in coulombs

#### FORMULA

$$Q = n e$$

**Why this one.** The question gives a COUNT of electrons and asks for a charge, so the two are linked by the size of one electron's charge. Nothing else in the question is needed, and no distances or fields appear.

#### WHAT EACH SYMBOL MEANS

- $Q$  = the total charge, in coulombs
- $n$  = the number of elementary charges, a pure count with no unit
- $e$  = the elementary charge,  $1.602 \times 10^{-19}$  C

#### THE WORKING, LINE BY LINE

1.  $Q = n \times e$
2.  $Q = 6.2 \times 10^{18} \times 1.602 \times 10^{-19}$
3. Front numbers:  $6.2 \times 1.602 = 9.93$
4. Powers of ten:  $10^{18} \times 10^{-19} = 10^{-1}$
5.  $Q = 9.93 \times 10^{-1} = 0.993 \approx 0.99$  C

#### CHECK IT

$1 \div e$  is about  $6.24 \times 10^{18}$ , so  $6.2 \times 10^{18}$  electrons is very nearly exactly one coulomb. An answer just under 1 C is what you should expect, and anything far from 1 means a slipped power of ten.

Charge is quantised:  $q = ne$ , where  $e = 1.602 \times 10^{-19}$  C. So  $q = 6.2 \times 10^{18} \times 1.602 \times 10^{-19} = 0.993$  C, which rounds to 0.99 C. The number of electrons is close to  $1/e$ , which is why the answer sits just under 1 coulomb.

A charge of  $8 \times 10^{-8} \text{ C}$  is 0.40 m from a charge of  $-2.0 \times 10^{-8} \text{ C}$ . Determine the electric field produced by the negative charge at the location of the positive charge.

✗ 1114.07 N/C

Wrong. A near miss designed to catch a rounding slip in the division.

✓ 1125.75 N/C

Correct.  $E = kQ/r^2$  using only the negative charge:  $9 \times 10^9 \times 2 \times 10^{-8} / 0.16 = 1125 \text{ N/C}$ .

✗ 1523.23 N/C

Wrong. No route to this value from these numbers.

✗ 3124.76 N/C

Wrong. This is close to four times the answer, which is what using the  $8 \times 10^{-8} \text{ C}$  charge instead would give.

✗ 2314.56 N/C

Wrong. No route to this value from these numbers.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A field is created by a charge and exists in the space around it whether or not anything is there to feel it. So when a question asks for the field PRODUCED BY one charge AT the position of another, the second charge is only a place marker. It tells you where to stand and nothing more. Its own size never enters, because a charge does not produce a field on itself.

#### GIVEN

- The field-producing charge:  $Q = 2.0 \times 10^{-8} \text{ C}$ , negative
- $r = 0.40 \text{ m}$
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- The  $8 \times 10^{-8} \text{ C}$  charge is only the LOCATION, and is not used

**Find:** The electric field  $E$  at the position of the positive charge, in  $\text{N C}^{-1}$

#### FORMULA

$$E = \frac{kQ}{r^2}$$

**Why this one.** Only the charge that PRODUCES the field goes into the formula. Reading the question decides which one that is: it says the field produced by the negative charge, so  $Q$  is  $2.0 \times 10^{-8} \text{ C}$ . Had it asked for the FORCE between them, both charges would have been needed and Coulomb's law would apply instead. Field means one charge, force means two.

#### WHAT EACH SYMBOL MEANS

- $E$  = the field strength at that point, in  $\text{N C}^{-1}$
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- $Q$  = the charge producing the field, in coulombs
- $r$  = the distance from the producing charge to the point, and it is SQUARED

#### THE WORKING, LINE BY LINE

1. Square the distance first:  $r^2 = 0.40^2 = 0.16 \text{ m}^2$
2. Top:  $kQ = 9 \times 10^9 \times 2.0 \times 10^{-8}$
3. Front numbers:  $9 \times 2 = 18$
4. Powers of ten:  $10^9 \times 10^{-8} = 10^1$
5. Top =  $18 \times 10^1 = 180$
6.  $E = 180 \div 0.16$
7.  $E \approx 1125 \text{ N C}^{-1}$ , and the direction is TOWARDS the negative charge

#### CHECK IT

Two habits protect this mark. First, if you find yourself using both charges, re-read the question, because it says produced BY the negative charge. Second,  $r^2 = 0.16$  and not  $0.4$ , which is the slip that would give  $450$  and appears nowhere in the options. The direction is towards the negative charge, since field lines always end on negative charge.

Read the question carefully: it asks for the field produced by the NEGATIVE charge, so only that charge and the separation are used. The  $8 \times 10^{-8} \text{ C}$  charge is where you are measuring, not what is producing.  $E = kQ/r^2 = 9 \times 10^9 \times 2 \times 10^{-8} / (0.40)^2 = 180 / 0.16 = 1125 \text{ N/C}$ , which the examiner gives as  $1125.75$ .

1.33

MEDIUM

Reference sheet R.1

**Coulomb's law is an inverse square law, so tripling the distance between two charges drops the force to a ninth.**

Answer: **true**

True.  $F = \frac{k q_1 q_2}{r^2}$ , and three squared is nine. It is the standard sanity check on any force answer: doubling the distance

quarters the force, halving it quadruples the force. The same square appears in the FIELD but not in the potential.

1.34

MEDIUM

Reference sheet R.1

**The electric field  $E = F/q$  may be quoted in newtons per coulomb or, equivalently, in volts per  $\{\{1\}\}$ .**

Answer:

Volts per metre. The two units are the same thing seen from two definitions:  $E = F/q$  gives N/C, and  $E = V/d$  gives V/m. Test 1 uses  $\text{Vm}^{-1}$  while Test 2 uses N/C, so both must be recognised on sight.

1.35

MEDIUM

Reference sheet R.3

**Step 5 says to substitute the numbers first and then rearrange the formula.**

Answer: **false**

The other way round. Rearrange in SYMBOLS first, and substitute numbers only at the end. Rearranging an expression that already contains numbers is where reciprocal and sign errors breed: get  $E = mg/q$  on the page before a single digit appears, and all that remains is arithmetic.

1.36

MEDIUM

Reference sheet R.3

**Step 7 asks three things: is the size believable, is the  $\{\{1\}\}$  right, and does the sanity check apply?**

Answer:

The unit. Together with size and the sanity checks it catches most of what goes wrong: a capacitance of 4.77 farads is not believable, an answer in newtons where volts were wanted has the wrong unit, and a parallel resistance above its smallest member is impossible. Steps 3 and 7 save the most marks and are skipped the most often.

Using the fan and vacuum picture, a positive charge is a fan blowing outward and a negative charge is a vacuum sucking inward. What happens midway between one of each?

✓ Both send you the same way, so the two fields ADD

Correct, and it is why the answer doubles rather than vanishing.

✗ They push you in opposite directions, so the fields cancel

Wrong. That is what two charges of the SAME sign would do.

✗ Neither has any effect at the midpoint

Wrong. Both act at the midpoint; they simply act the same way.

✗ The fan wins because it is positive

Wrong. Equal magnitudes at equal distances give equal contributions.

✗ The field points at right angles to the line joining them

Wrong. Both contributions lie along the line joining the charges.

The fan blows you away from itself, to the right, and the vacuum pulls you towards itself, also to the right. Same direction, so they add, which is why Test 1 question 8 comes to  $2.16 \times 10^7$  rather than zero. Two FANS would fight and cancel. Opposite signs add, same signs cancel, and most people guess that backwards.

**What force would a  $1.4 \times 10^{-5}$  C negative charge experience at 1 m distance from a 1 C positive charge, and would the force be directed toward or away from the 1 C charge?**

✓  $1.26 \times 10^5$  N, directed toward the 1 C charge

Correct.  $1.26 \times 10^5$  N, and attraction because the signs are opposite.

✗  $1.26 \times 10^5$  N, directed away from the 1 C charge

Wrong on direction only. Opposite charges attract, so the force points toward, not away.

✗  $1.4 \times 10^{-5}$  N, directed toward the 1 C charge

Wrong. This is the charge itself copied out, with no force calculated.

✗  $9 \times 10^9$  N, directed away from the 1 C charge

Wrong. That is Coulomb's constant, not a force, and the direction is wrong too.

✗  $1.26 \times 10^{-5}$  N, directed toward the 1 C charge

Wrong by ten orders of magnitude: the exponent should be positive, since  $9 \times 10^9 \times 1.4 \times 10^{-5}$  is  $1.26 \times 10^5$ .

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Coulomb's law works in exactly the same shape as gravity, but with one extra freedom: charges can push as well as pull. Like signs repel, unlike signs attract, and the size of the force does not care which is which. So the routine is always two separate jobs: get the size from the formula, then get the direction from the signs.

**GIVEN**

- $Q_1 = 1 \text{ C}$ , positive
- $Q_2 = 1.4 \times 10^{-5} \text{ C}$ , negative
- $r = 1 \text{ m}$
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

**Find:** The size of the force on the negative charge, and its direction

**FORMULA**

$$F = \frac{kQ_1Q_2}{r^2}$$

**Why this one.** Two charges and a separation are named, which is the signature of Coulomb's law rather than  $F = qE$ . The signs are deliberately left out of the arithmetic: putting a minus into the formula gives a negative force, which is meaningless as a magnitude. Handle the sign in words instead.

**WHAT EACH SYMBOL MEANS**

- $F$  = the electrostatic force, in newtons
- $k$  = the Coulomb constant,  $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- $Q_1, Q_2$  = the sizes of the two charges, in coulombs
- $r$  = their separation, in metres, and it is SQUARED

**THE WORKING, LINE BY LINE**

1.  $F = (9 \times 10^9 \times 1 \times 1.4 \times 10^{-5}) \div 1^2$
2.  $r^2 = 1^2 = 1$ , so the bottom does nothing here
3. Front numbers:  $9 \times 1.4 = 12.6$
4. Powers of ten:  $10^9 \times 10^{-5} = 10^4$
5.  $F = 12.6 \times 10^4 = 1.26 \times 10^5 \text{ N}$
6. Direction: one charge is positive and the other negative, so they ATTRACT
7. The force on the negative charge points TOWARDS the 1 C charge

**CHECK IT**

Four of the five options carry a plausible-looking number, so the direction is doing half the work of this question. Read the signs before you read the numbers: unlike signs means towards, like signs means away. Note also how large  $1.26 \times 10^5 \text{ N}$  is, about the weight of a lorry, which is why a whole coulomb of charge is never found sitting still in one place.

Coulomb's law needs BOTH charges, since a force acts between two objects.  $F = k q_1 q_2 \text{ over } r^2 = (9 \times 10^9 \times 1 \times 1.4 \times 10^{-5}) / 1^2 = 1.26 \times 10^5 \text{ N}$ . Then the direction, decided by the signs and not by the arithmetic: the charges are opposite, so they attract, and the force on the negative charge points TOWARD the 1 C positive charge. Note how large the answer is. A whole coulomb is an enormous charge, which is why real problems use  $\mu\text{C}$ .

If  $F$  be the electrostatic force between two charges in air, then the force between the charges when air is replaced by a medium of dielectric constant  $K$  is:

✓  $F/K$

Correct. The dielectric screens the charges from each other, so the force falls, and it falls by exactly  $K$  because  $\epsilon$  sits on the bottom of Coulomb's law.

✗  $F/\sqrt{K}$

Wrong. The direction is right, the force does fall, but nothing here takes a square root.  $K$  divides the force once, not by  $\sqrt{K}$ .

✗  $\sqrt{K} F$

Wrong twice: it makes the force BIGGER, and it uses a square root that appears nowhere in Coulomb's law.

✗  $KF$

Wrong, and it is the trap.  $K$  multiplying the force is what happens to CAPACITANCE, where  $\epsilon$  sits on top. Here it sits underneath, so  $K$  divides instead.

✗  $F$ , unchanged

Wrong. Something physical changed in the gap: the molecules polarise and screen the charges, so the force cannot be unaffected.

Picture it first, then the formula.

Two charges sit in air, a certain distance apart, pulling on each other with force  $F$ . Now slide a slab of glass or plastic into the gap between them. Nothing else changes: same charges, same distance.

The slab is made of molecules, and each molecule has a positive end and a negative end. Put them near a charge and they all TURN, lining up with their opposite ends facing it. That is called POLARISATION, and the effect is a thin layer of opposite charge sitting between the two originals, partly hiding each one from the other. Each charge now sees less of its partner than it did through air.

So the force goes DOWN. That alone rules out three of the five options.

By how much? By exactly  $K$ , the dielectric constant of the slab.  $F$  becomes  $F/K$ . Glass has  $K$  around 5, so the force drops to about a fifth.

The formula agrees, and here is the one line that makes it obvious. In  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$ , the permittivity  $\epsilon$  sits UNDERNEATH.

Adding the slab multiplies  $\epsilon$  by  $K$ . Multiplying the bottom of a fraction by  $K$  divides the whole thing by  $K$ . So  $F$  becomes  $F/K$ .

The contrast worth carrying: in capacitance,  $C = \epsilon A/d$ , that same  $\epsilon$  sits ON TOP, so the same slab MULTIPLIES the capacitance by  $K$  instead. Same material, same  $K$ , opposite effect, purely because of which side of the fraction  $\epsilon$  is on.

The magnitude of the electric intensity represented by  $E = 3i + 4j - 2k$ , where  $i$ ,  $j$  and  $k$  are unit vectors along the  $x$ ,  $y$  and  $z$  axes respectively, is:

✓  $5.4 \text{ Vm}^{-1}$

Correct. The square root of 29.

✗  $5 \text{ Vm}^{-1}$

Wrong, and it is the trap: 5 is the answer if you drop the  $k$  component and use only 3 and 4, the familiar 3-4-5 triangle.

✗  $9 \text{ Vm}^{-1}$

Wrong. 9 is  $3^2$ , one term of the sum rather than the result.

✗  $81 \text{ Vm}^{-1}$

Wrong. 81 is  $9^2$ , squaring something that was already a square.

✗  $29 \text{ Vm}^{-1}$

Wrong. 29 is the sum of the squares, before taking the square root.

The magnitude of a vector is the square root of the sum of the squares of its components.  $E = \text{square root of } (3^2 + 4^2 + \text{minus } 2^2) = \text{square root of } (9 + 16 + 4) = \text{square root of } 29 = 5.39$ , which rounds to  $5.4 \text{ Vm}^{-1}$ . Note that  $-2^2$  is PLUS 4: squaring destroys the sign, which is why a magnitude can never come out negative.

The magnitude bars in  $E = k|Q|/r^2$  mean you drop the sign of the charge, because the formula returns the **SIZE** of the field while the sign fixes its direction instead.

Answer: **true**

True. Feed a negative  $Q$  into the arithmetic and you get a negative field strength, which means nothing: a field has a direction, not a negative size. So use the magnitude to get the number, then use the sign separately to decide the direction, away from a positive charge and towards a negative one.

Two charges  $Q_1 = +3.0 \times 10^{-6} \text{ C}$  and  $Q_2 = -3.0 \times 10^{-6} \text{ C}$ , are placed  $0.10 \text{ m}$  apart in air. Find the net electric field at the midpoint between the charges.

✗  $1.99 \times 10^7 \text{ N/C}$

Wrong. No route to 1.99 from these numbers.

✗  $1.08 \times 10^7 \text{ N/C}$

Wrong. This is the field from ONE charge only, and it is the trap: the two contributions add rather than cancel because the charges are opposite in sign.

✓  $2.16 \times 10^7 \text{ N/C}$

Correct. Two equal contributions of  $1.08 \times 10^7 \text{ N/C}$  pointing the same way, so  $2.16 \times 10^7 \text{ N/C}$ .

✗  $3.0 \times 10^7 \text{ N/C}$

Wrong. This would need a separation smaller than the one given.

✗  $1.07 \times 10^7 \text{ N/C}$

Wrong, and it is a near miss on the single-charge value, which is itself only half the answer.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Two equal and opposite charges a short distance apart make a dipole, the arrangement at the heart of every polar molecule. The field lines leave the positive charge and land on the negative one, so between the two charges they all point the same way, from plus to minus. That is why the two fields at the midpoint **REINFORCE** instead of cancelling, which is the exact opposite of what most people guess on first reading.

**GIVEN**

- $Q_1 = +3.0 \times 10^{-6} \text{ C}$
- $Q_2 = -3.0 \times 10^{-6} \text{ C}$
- separation = 0.10 m, so the midpoint is at  $r = 0.05 \text{ m}$  from EACH charge
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

**Find:** The net electric field at the midpoint, in  $\text{N C}^{-1}$

**FORMULA**

$$E = \frac{kQ}{r^2}, \text{ then add the two as vectors}$$

**Why this one.** This is a FIELD question, so  $r$  is squared, and there are two sources, so the two fields must be combined by direction rather than simply added as numbers. Here the directions happen to coincide, so the combination reduces to doubling. Getting the direction right before the arithmetic is what makes this a hard question rather than an easy one.

**WHAT EACH SYMBOL MEANS**

- $E$  = the field due to one charge, in  $\text{N C}^{-1}$
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- $Q$  = the size of that charge, sign handled by the direction argument
- $r$  = the distance from that charge to the point, and it is SQUARED
- $r = 0.05 \text{ m}$  here, HALF the separation, not the whole 0.10 m

**THE WORKING, LINE BY LINE**

1. Distance from each charge to the midpoint:  $r = 0.10 \div 2 = 0.05 \text{ m}$
2. Square it once, for both:  $r^2 = 0.05^2 = 0.0025 \text{ m}^2$
3. Field from the positive charge, pointing away from it, so towards  $Q_2$ :
4.  $E_1 = (9 \times 10^9 \times 3.0 \times 10^{-6}) \div 0.0025$
5. Top:  $9 \times 3 = 27$ , and  $10^9 \times 10^{-6} = 10^3$ , so top =  $27 \times 10^3 = 2.7 \times 10^4$
6.  $E_1 = (2.7 \times 10^4) \div 0.0025 = 1.08 \times 10^7 \text{ N C}^{-1}$
7. Field from the negative charge, pointing TOWARDS it, which is the same direction:
8.  $E_2 = 1.08 \times 10^7 \text{ N C}^{-1}$ , identical in size because the charges are equal
9. Same direction, so ADD:  $E = 1.08 \times 10^7 + 1.08 \times 10^7$
10.  $E = 2.16 \times 10^7 \text{ N C}^{-1}$ , directed from the positive charge towards the negative

**CHECK IT**

The two traps are both in the option list.  $1.08 \times 10^7$  is what you get by finding one field and forgetting to add the second, and using the full 0.10 m instead of the half would divide the answer by four. Remember the rule: between **UNLIKE** charges the fields add, between **LIKE** charges they cancel and the midpoint field is zero.

The midpoint is 0.05 m from each charge. Each contributes  $E = kQ/r^2 = 9 \times 10^9 \times 3 \times 10^{-6} / (0.05)^2 = 27000 / 0.0025 = 1.08 \times 10^7 \text{ N/C}$ . The field points **AWAY** from the positive charge and **TOWARDS** the negative one, so at the midpoint both contributions point the same way and they add:  $2 \times 1.08 \times 10^7 = 2.16 \times 10^7 \text{ N/C}$ .

**Calculate the magnitude of the electric field intensity represented by****X**  $10.2 \text{ Vm}^{-1}$ 

Cannot be judged. The figure the stem depends on is missing from the captured paper.

**X**  $12 \text{ Vm}^{-1}$ 

Cannot be judged. The figure the stem depends on is missing from the captured paper.

**✓**  $10 \text{ Vm}^{-1}$ This is the option selected in the sitting, and the examiner marked it **WRONG**. It is recorded here only to keep the paper complete.**X**  $9 \text{ Vm}^{-1}$ 

Cannot be judged. The figure the stem depends on is missing from the captured paper.

**X**  $8 \text{ Vm}^{-1}$ 

Cannot be judged. The figure the stem depends on is missing from the captured paper.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

A field has a direction as well as a size, so it is a **VECTOR**. Writing it as  $3\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$  is just saying how far it reaches along three perpendicular directions at once:  $\mathbf{i}$  is along  $x$ ,  $\mathbf{j}$  along  $y$ ,  $\mathbf{k}$  along  $z$ . Asking for the magnitude is asking how long that arrow is overall, which is Pythagoras extended into three dimensions.

**GIVEN**

- $\mathbf{E} = 3\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ , where  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  are the unit vectors along  $x$ ,  $y$  and  $z$

**Find:** The magnitude  $|\mathbf{E}|$ , a single number with no direction**FORMULA**

$$|\mathbf{E}| = \sqrt{x^2 + y^2 + z^2}$$

**Why this one.** The components are at right angles to each other, so they combine by Pythagoras rather than by simple addition. In two dimensions 3 and 4 give 5; the third component just adds one more square under the root.

**WHAT EACH SYMBOL MEANS**

- $|\mathbf{E}|$  = the magnitude, always positive
- $x, y, z$  = the components along  $\mathbf{i}, \mathbf{j}$  and  $\mathbf{k}$

**THE WORKING, LINE BY LINE**

1. Square each component:  $3^2 = 9$ ,  $4^2 = 16$ ,  $(-2)^2 = 4$
2. Add them:  $9 + 16 + 4 = 29$
3. Take the square root:  $|\mathbf{E}| = \sqrt{29}$
4.  $|\mathbf{E}| = 5.39$

**CHECK IT**

A magnitude is always **LARGER** than the biggest single component and **SMALLER** than their sum. Here  $4 < 5.39 < 9$ . If your answer fails that, you added instead of squaring. Note the minus vanished when squared, which is why a magnitude can never be negative.

**QUOTED AS PRINTED, AND INCOMPLETE.** The stem ends at "represented by" and the figure it refers to is not in the captured paper, so the question cannot be worked from what survives. This is also the one question on Test 1 marked 0 out of 1: the option selected in the sitting was  $10 \text{ Vm}^{-1}$ , and the examiner marked it wrong, so the true key is not recoverable from the capture either. The answer is set to the selected option only so the paper stays answerable end to end; treat this question as evidence that a figure was lost, not as a fact to learn. Everything the question tests,  $E = V/d$  and reading a field from a diagram, is drilled properly elsewhere in this topic.

Why does the field formula take the **MAGNITUDE** of the charge, written  $E = k|Q|/r^2$ ?

✓ Because the formula returns the size of the field, while the sign is used separately to fix its direction

Correct. Size from the magnitude, direction from the sign, in two separate steps.

✗ Because charges are always positive

Wrong. Charges are routinely negative, and Test 1 question 13 uses one.

✗ Because the field of a negative charge is zero

Wrong. A negative charge has a perfectly ordinary field; it simply points inward.

✗ Because the bars mean the same as brackets

Wrong. Bars mean absolute value, brackets merely group.

✗ Because  $r^2$  is always positive

Wrong. That is true but has nothing to do with why the bars are on the  $Q$ .

A field has a direction, not a negative size. Feeding a negative  $Q$  into the arithmetic would give a negative field strength, which means nothing. So the magnitude gives the number, and the sign is spent afterwards on the direction: away from a positive charge, towards a negative one.

Point charge A has a charge of  $-1.0 \text{ nC}$  and point charge B has a charge of  $4.0 \text{ nC}$ . They are separated by  $1.0 \text{ cm}$ . What are the magnitude and direction of the electric forces on charges A and B?

✓  $3.6 \times 10^{-4} \text{ N}$  on each, attractive, equal and opposite

Correct, and the equal-and-opposite part is the physics being tested.

✗  $3.6 \times 10^{-4} \text{ N}$  on A but  $1.4 \times 10^{-3} \text{ N}$  on B, because B has more charge

Wrong, and it is the trap. A bigger charge does not feel a bigger force:  $q_1$  and  $q_2$  appear as a product, so both charges see the same number.

✗  $3.6 \times 10^{-6} \text{ N}$  on each, attractive

Wrong by a factor of 100, which is what leaving the separation as  $1.0 \text{ cm}$  rather than  $0.010 \text{ m}$  does.

✗  $3.6 \times 10^{-4} \text{ N}$  on each, repulsive

Wrong on direction. One charge is negative and the other positive, so they attract.

✗  $3.6 \times 10^{-2} \text{ N}$  on each, attractive

Wrong by a factor of 100 the other way.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Newton's third law does not stop being true for electric forces. Whatever A does to B, B does back to A with equal size and opposite direction, no matter how unequal the two charges are. This surprises people, because a 4 nC charge feels like it ought to push harder than a 1 nC one. It does not, and the formula shows why: both charges appear in the SAME product, so there is only one force size to be had.

### GIVEN

- $Q_A = 1.0 \text{ nC} = 1.0 \times 10^{-9} \text{ C}$ , negative
- $Q_B = 4.0 \text{ nC} = 4.0 \times 10^{-9} \text{ C}$ , positive
- $r = 1.0 \text{ cm} = 0.01 \text{ m}$
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

**Find:** The size and direction of the force on A, and the force on B

### FORMULA

$$F = \frac{kQ_1Q_2}{r^2}$$

**Why this one.** Two charges and a separation again, so Coulomb's law. The important reading is that the formula produces ONE number, not one per charge. That single number is the force each feels, which is Newton's third law falling out of the algebra rather than being quoted separately.

### WHAT EACH SYMBOL MEANS

- $F$  = the force each charge feels, in newtons
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- $Q_1, Q_2$  = the two charges, here 1.0 nC and 4.0 nC, in coulombs
- $r$  = their separation, in metres, SQUARED
- $1 \text{ nC} = 10^{-9} \text{ C}$ , and  $1 \text{ cm} = 0.01 \text{ m}$

### THE WORKING, LINE BY LINE

1. Convert first:  $1.0 \text{ nC} = 1.0 \times 10^{-9} \text{ C}$ ,  $4.0 \text{ nC} = 4.0 \times 10^{-9} \text{ C}$ ,  $1.0 \text{ cm} = 0.01 \text{ m}$
2. Square the distance:  $r^2 = 0.01^2 = 1 \times 10^{-4} \text{ m}^2$
3. Top:  $9 \times 10^9 \times 1.0 \times 10^{-9} \times 4.0 \times 10^{-9}$
4. Front numbers:  $9 \times 1 \times 4 = 36$
5. Powers of ten:  $10^9 \times 10^{-9} \times 10^{-9} = 10^{-9}$
6. Top =  $36 \times 10^{-9} = 3.6 \times 10^{-8}$
7.  $F = (3.6 \times 10^{-8}) \div (1 \times 10^{-4}) = 3.6 \times 10^{-4} \text{ N}$
8. Direction: one negative, one positive, so ATTRACTIVE, each pulled towards the other

### CHECK IT

The wrong option that says B feels more because it has more charge is the exact misconception this question exists to catch. Squash it with the formula:  $Q_A$  and  $Q_B$  are multiplied together, so swapping which charge you call which changes nothing. Watch the centimetre too, since 1 cm squared is  $10^{-4}$  and not  $10^{-2}$ , and that slip moves the answer by two powers of ten straight into another option.

Convert first:  $1.0 \text{ cm} = 0.010 \text{ m}$ . Then  $F = k q_1 q_2 \text{ over } r^2 = (9 \times 10^9 \times 1.0 \times 10^{-9} \times 4.0 \times 10^{-9}) / (0.010)^2 = 3.6 \times 10^{-8} / 1.0 \times 10^{-4} = 3.6 \times 10^{-4} \text{ N}$ . The signs are opposite so the force is attractive, and by Newton's third law the two charges feel forces of EQUAL size in opposite directions, however unequal the charges themselves are.

The force on an electron of charge  $q$  placed at a point in an electric field where the electric intensity is  $E$ , is given by  $F =$

✓  $qE$  opposite to the direction of the field

Correct. The size is  $qE$  and the electron's negative charge reverses the direction.

✗  $qE$  in the direction of the field

Wrong on direction, and it is the trap: this is the right answer for a POSITIVE charge, not an electron.

✗  $E/q$  in the direction of the field

Wrong on both.  $E/q$  inverts the formula, and the direction is wrong for an electron.

✗  $E/q$  opposite to the direction of the field

Wrong size. The direction is right but  $E/q$  is upside down.

✗  $qE$  at right angles to the field

Wrong. A right-angle force is the MAGNETIC case,  $F = qvB$ , not the electric one.

Two separate things to get right. The SIZE comes from  $F = qE$ , the definition of the field as force per unit charge rearranged. The DIRECTION comes from the sign of the charge, and the question says ELECTRON, which is negative. The field points the way a POSITIVE charge would be pushed, so a negative charge is pushed the opposite way. Hence  $qE$ , opposite to the field.

The electric field in which a charged oil drop of mass 5 mg and charge +10  $\mu\text{C}$  be held stationary is:

✓  $5 \text{ Vm}^{-1}$  upwards

Correct.  $5 \text{ Vm}^{-1}$ , and upwards because a positive charge must be pushed up against its weight.

✗  $5 \text{ Vm}^{-1}$  downwards

Wrong direction. A downward field would push a positive charge down, adding to the weight rather than balancing it.

✗  $5000 \text{ Vm}^{-1}$  upwards

Wrong by a factor of 1000, which is what forgetting to convert mg to kilograms does.

✗  $5000 \text{ Vm}^{-1}$  downwards

Wrong on both the size and the direction.

✗  $50 \text{ Vm}^{-1}$  upwards

Wrong by a factor of ten in the exponent arithmetic.

Stationary means the upward electric force balances the downward weight:  $qE = mg$ , so  $E = mg/q$ . Convert first, because both units are prefixed:  $5 \text{ mg} = 5 \times 10^{-6} \text{ kg}$  and  $10 \mu\text{C} = 10 \times 10^{-6} \text{ C}$ . Then  $E = (5 \times 10^{-6} \times 10) / (10 \times 10^{-6}) = 5 \times 10^{-5} / 1 \times 10^{-5} = 5 \text{ Vm}^{-1}$ . The direction must be UPWARDS, because the charge is positive and the force on a positive charge follows the field, and that force has to point up to hold the weight. This is the same question as Test 1 question 3 with different numbers.

A charge S of  $+8 \mu\text{C}$  feels a force of 320 N upwards from a charge P, and 120 N to the left from a charge R, the two being at right angles. What is the net force on S?

✓ 341.76 N at  $69.44^\circ$  to the 120 N force

Correct. Pythagoras for the size, then the arctangent for the angle.

✗ 440 N at  $45^\circ$

Wrong. 440 N is  $320 + 120$ , adding the sizes as if they pointed the same way. They are at right angles.

✗ 200 N at  $69.44^\circ$

Wrong. 200 N is  $320 - 120$ , subtracting as if they opposed.

✗ 341.76 N at  $20.56^\circ$  to the 120 N force

Wrong angle.  $20.56^\circ$  is the angle to the OTHER force, since the two angles add to 90.

✗ 116800 N at  $69.44^\circ$

Wrong. 116800 is the sum of the squares, before taking the square root.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Forces add as arrows, not as numbers. If two forces pull in different directions you cannot simply total them, because part of each is spent pulling across the other. When the two happen to be at RIGHT ANGLES the arithmetic is at its kindest: neither has any component along the other, so the resultant is just the diagonal of the rectangle they make, which is Pythagoras.

### GIVEN

- $F_1 = 320$  N, directed upwards (from charge P)
- $F_2 = 120$  N, directed to the left (from charge R)
- The angle between them is  $90^\circ$
- The  $+8 \mu\text{C}$  on S is scene-setting and is NOT used

**Find:** The resultant force on S: its size, and the angle it makes

### FORMULA

$$F = \sqrt{(F_1^2 + F_2^2)}, \tan \theta = F_1 / F_2$$

**Why this one.** Two perpendicular forces are being combined, and perpendicular means Pythagoras. The angle comes from the tangent, which is opposite over adjacent, so which force you put on top decides which angle you get. The question asks for the angle to the 120 N force, so the 320 N is the opposite side and goes on top.

### WHAT EACH SYMBOL MEANS

- $F$  = the resultant, the single force that would do the same job as both
- $F_1 = 320$  N, the upward force
- $F_2 = 120$  N, the leftward force
- $\theta$  = the angle measured from the 120 N force round to the resultant

### THE WORKING, LINE BY LINE

1. Square each force:  $320^2 = 102400$ ,  $120^2 = 14400$
2. Add:  $102400 + 14400 = 116800$
3. Square root:  $F = \sqrt{116800} = 341.76$  N
4. Now the angle:  $\tan \theta = 320 \div 120 = 2.667$
5.  $\theta = \tan^{-1}(2.667) = 69.44^\circ$ , measured from the 120 N force

### CHECK IT

116800 is sitting in the option list as an answer, which is what you get by stopping before the square root, so finish the job. And sanity-check the angle by eye: the 320 N force is much the bigger of the two, so the resultant must lean towards it, meaning an angle well over  $45^\circ$  from the 120 N.  $69.44^\circ$  leans the right way; the option offering  $20.56^\circ$  is the same angle measured from the wrong force.

Forces add as VECTORS, not as numbers. At right angles that means Pythagoras for the size:  $F_{\text{net}} = \text{square root of } (320^2 + 120^2) = \text{square root of } (102400 + 14400) = \text{square root of } 116800 = 341.76$  N. Then the angle by trigonometry:  $\tan \theta = 320 \text{ over } 120 = 2.67$ , so  $\theta = 69.44^\circ$ , measured from the 120 N force. That is superposition: work out each force on its own, then combine.

## 2 Electric Potential and Energy

**Speed tip.** Potential has a plain  $r$ , field has  $r$  squared. Four times further out makes the potential a quarter and the field a sixteenth.

2.1

EASY

Reference sheet R.4

**An equipotential is a surface on which the potential is the same everywhere. Which of these follows?**

✓ No work is done moving a charge along it, and field lines cross it at right angles

Correct on both counts.

✗ Work is done moving a charge along it, proportional to the distance

Wrong. No potential difference means no work, however far you go.

✗ Field lines run parallel to it

Wrong, and exactly backwards: they are perpendicular.

✗ Two equipotential surfaces may intersect

Wrong. Intersecting would give one point two potentials.

✗ The field is zero everywhere on it

Wrong. The surface of a charged sphere is an equipotential and the field there is certainly not zero.

$W = qV$ , and along an equipotential there is no CHANGE in  $V$ , so no work is done. Field lines are always perpendicular to equipotentials, because any component of field along the surface would mean a potential difference along it, which would contradict it being equipotential. And they never intersect, since a point cannot hold two different potentials at once.

2.2

EASY

Reference sheet R.1

**The work done moving a charge through a potential difference is  $W = qV$ , charge times voltage.**

Answer: **true**

True, and it is simply the definition of potential difference turned around: a volt IS a joule per coulomb, so multiplying coulombs by volts returns joules. It is the reason no work is done along an equipotential, where  $V$  does not change at all.

2.3

EASY

Reference sheet R.4

**An equipotential is a surface on which the potential is the same everywhere, so moving a charge along it does **no** work.**

Answer:

No work at all.  $W = qV$  depends on the CHANGE in potential, and along an equipotential there is none. Field lines always cross such a surface at right angles, because any component along it would create a potential difference and contradict the definition.

2.4

MEDIUM

Reference sheet R.1

**Electric potential is  $V = \frac{kQ}{r}$ . Unlike the field, the  $r$  here is to the power **1**.**

Answer:

One.  $V = kQ/r$  has a plain  $r$ , while  $E = kQ/r^2$ . Move three times further away and the FIELD falls to a ninth but the POTENTIAL only to a third. Test 1 question 4 depends on this: 5 cm to 20 cm is four times the distance, and the potential falls from 720 kV to 180 kV, exactly a quarter.

**The electric potential energy of two charges is:**

✓  $U = kQq/r$ , which also equals  $qV$

Correct, and the  $qV$  form is the one that makes  $W = qV$  obvious.

✗  $U = kQq/r^2$

Wrong.  $r^2$  belongs to the FORCE and the FIELD, not to energy or potential.

✗  $U = kQ/r$

Wrong. That is the potential  $V$ , which is energy per unit charge, not the energy itself.

✗  $U = qE$

Wrong.  $qE$  is a FORCE in newtons, not an energy in joules.

✗  $U = kQq r$

Wrong.  $r$  divides here, it does not multiply.

$U = kQq/r$ , in joules, and since  $V = kQ/r$  it can be written more simply as  $U = qV$ : the energy is the charge times the potential it sits in. Note the plain  $r$ , matching the potential rather than the field.

**A charge of 3 C moves through a potential difference of 12 V. Calculate the work done.**

✓ 36 J

Correct. 3 coulombs each carried through 12 volts.

✗ 4 J

Wrong, and upside down: 12 divided by 3.

✗ 0.25 J

Wrong. 3 divided by 12.

✗ 15 J

Wrong. This ADDS the two numbers.

✗ 9 J

Wrong. No route to 9 from these numbers.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Potential difference is not really about volts as a substance; it is a statement about ENERGY PER COULOMB. Saying a battery is 12 V is saying that each coulomb passing through it gains 12 joules. So the work done on any amount of charge is just how many coulombs there are, times how many joules each one gets.

#### GIVEN

- $Q = 3 \text{ C}$
- $V = 12 \text{ V}$

**Find:** The work done  $W$ , in joules

#### FORMULA

$$W = QV$$

**Why this one.** This comes straight from the definition of the volt: one volt is one joule per coulomb. Multiply the volts by the coulombs and the coulombs cancel, leaving joules. Nothing else is needed, and in particular no distance or field, because a potential difference already has the distance built into it.

#### WHAT EACH SYMBOL MEANS

- $W$  = the work done, or the energy transferred, in joules
- $Q$  = the charge moved, in coulombs
- $V$  = the potential difference it moves through, in volts

#### THE WORKING, LINE BY LINE

1.  $W = Q \times V$
2.  $W = 3 \times 12$
3.  $W = 36 \text{ J}$

#### CHECK IT

Check by units, which is faster than re-doing the arithmetic. Coulombs  $\times$  (joules per coulomb) = joules, so the units work out and the multiplication was the right operation. Had you divided you would have got 0.25, which is sitting in the option list waiting for exactly that mistake.

$W = qV = 3 \times 12 = 36 \text{ J}$ . That is the definition of potential difference rearranged: a volt is a joule per coulomb, so multiplying coulombs by volts gives joules. This is a class question from the Module 1 slides on potential.

2.7

MEDIUM

Reference sheet R.1

Between parallel plates the field is uniform, and  $E = V$  over  $\{\{1\}\}$ , where that symbol is the separation of the plates.

Answer:

$E = V/d$ . More generally  $E = \text{minus } dV/dr$ , the field being the negative gradient of the potential, but for the uniform field between plates it simplifies to voltage divided by separation. That is also why volts per metre is a valid unit for a field.

2.8

MEDIUM

Reference sheet R.2

The electric field is a vector, but the electric potential is a  $\{\{1\}\}$ .

Answer:

A scalar. It has magnitude and sign but no direction, which is why potentials from several charges are added as ordinary signed numbers.

Find the work done on a charge of  $+6.0 \mu\text{C}$  placed in vacuum as it moves through a distance of 50 cm.

✗  $2 \times 10^{-6} \text{ J}$

Wrong. This is  $q \times d$  with the distance left as  $1/3 \text{ m}$  rather than  $0.50 \text{ m}$ .

✗  $1.08 \times 10^{-6} \text{ J}$

Wrong. No route to this value from a  $6.0 \mu\text{C}$  charge over  $0.50 \text{ m}$ .

✗  $0.5 \times 10^{-6} \text{ J}$

Wrong, and it is the trap: this is the DISTANCE in metres multiplied by  $10^{-6}$ , not the work.

✗  $0.65 \times 10^{-6} \text{ J}$

Wrong. No route to this value from these numbers.

✓  $3 \times 10^{-6} \text{ J}$

Correct.  $W = qEd = 6.0 \times 10^{-6} \times 1 \times 0.50 = 3 \times 10^{-6} \text{ J}$ .

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Coulomb's law is the electric version of a rule you already know. Two masses pull on each other with a force that dies off as the square of the distance; two charges do exactly the same, except that charges can also PUSH. The reason for the square is geometric: the influence of a point charge spreads over the surface of a sphere, and a sphere's area is  $4\pi r^2$ , so anything spreading out evenly gets diluted by  $r^2$  as it goes.

#### GIVEN

- $Q_1$  = the first charge, in coulombs (C)
- $Q_2$  = the second charge, in coulombs (C)
- $r$  = the separation between them, in metres (m)

**Find:** The force  $F$  between them, in newtons (N)

#### FORMULA

$$F = kQ_1Q_2 / r^2$$

**Why this one.** Two charges are named and a distance is given, so this is a FORCE question, not a field question. One charge and a distance would give a field instead. That single count, one charge or two, chooses the formula every time.

#### WHAT EACH SYMBOL MEANS

- $F$  = the electrostatic force, in newtons
- $k$  = the Coulomb constant,  $8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ , often rounded to  $9 \times 10^9$
- $Q_1, Q_2$  = the two charges, in coulombs
- $r$  = their separation in metres, and it is SQUARED

#### THE WORKING, LINE BY LINE

1. Convert both charges to coulombs.  $\mu\text{C}$  means  $\times 10^{-6}$ ,  $\text{nC}$  means  $\times 10^{-9}$ .
2. Convert the distance to metres.  $\text{cm}$  means  $\div 100$ ,  $\text{mm}$  means  $\div 1000$ .
3. Square  $r$  first, on its own, before dividing.
4. Multiply  $k \times Q_1 \times Q_2$  on top.
5. Divide top by bottom.
6. Give the direction: like charges repel, unlike attract.

#### CHECK IT

Double the distance and the force must fall to a QUARTER, not a half. If your answer halves, you forgot to square  $r$ .

Work done moving a charge along a field is  $W = qEd$ . Converting first:  $q = 6.0 \times 10^{-6} \text{ C}$  and  $d = 50 \text{ cm} = 0.50 \text{ m}$ . With the unit field the examiner intends,  $E = 1 \text{ Vm}^{-1}$ ,  $W = 6.0 \times 10^{-6} \times 1 \times 0.50 = 3 \times 10^{-6} \text{ J}$ . The stem does not print a field strength, so the marked answer only follows if  $E$  is taken as  $1 \text{ Vm}^{-1}$ ; the arithmetic that earns the mark is  $q \times d$ .

What is the absolute potential at each of the following distances from a charge of  $+4\ \mu\text{C}$  in air:  $r = 5\ \text{cm}$  and  $r = 20\ \text{cm}$ ?

✗ 750 kV & 180 kV

Wrong on the first value. The second is right, so this catches anyone who computes only one and matches on it.

✗ 150 kV & 75 kV

Wrong. This pair falls by half, not by a quarter, so it cannot come from a  $1/r$  law over a fourfold change in distance.

✓ 720 kV & 180 kV

Correct.  $V = kQ/r$  gives 720 kV at 5 cm and 180 kV at 20 cm.

✗ 250 kV & 50 kV

Wrong. This pair falls to a fifth and neither value follows from  $kQ = 36000$ .

✗ 500 kV & 150 kV

Wrong. Neither value follows from  $kQ/r$  with these numbers.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Potential answers a different question from field. Field asks how hard a charge placed here would be pushed; potential asks how much energy it took to bring it here from far away. Because energy is a scalar there is no direction to worry about, only a sign. The key structural difference to hold on to: potential has a plain  $r$  underneath, field has  $r$  SQUARED. That one difference decides most questions in this topic.

**GIVEN**

- $Q = +4 \mu\text{C} = 4 \times 10^{-6} \text{ C}$
- $r_1 = 5 \text{ cm} = 0.05 \text{ m}$
- $r_2 = 20 \text{ cm} = 0.20 \text{ m}$
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

**Find:** The absolute potential  $V$  at each of the two distances, in volts

**FORMULA**

$$V = \frac{kQ}{r}$$

**Why this one.** The question asks for POTENTIAL, so  $r$  appears to the first power. Absolute potential means measured against infinity, where the potential is taken as zero, which is exactly what this formula gives, so no difference between two points is being asked for. Only one charge is named, so nothing has to be added vectorially.

**WHAT EACH SYMBOL MEANS**

- $V$  = the absolute potential at that point, in volts
- $k$  = the Coulomb constant,  $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- $Q$  = the charge creating the potential, in coulombs
- $r$  = the distance from the charge, in metres, NOT squared

**THE WORKING, LINE BY LINE**

1.  $kQ$  is the same for both parts, so compute it once:
2.  $kQ = 9 \times 10^9 \times 4 \times 10^{-6} = 36 \times 10^3 = 3.6 \times 10^4$
3. At  $r_1$ :  $V_1 = (3.6 \times 10^4) \div 0.05$
4.  $V_1 = 720\,000 \text{ V} = 720 \text{ kV}$
5. At  $r_2$ :  $V_2 = (3.6 \times 10^4) \div 0.20$
6.  $V_2 = 180\,000 \text{ V} = 180 \text{ kV}$

**CHECK IT**

The second distance is FOUR times the first, and potential goes as  $1/r$ , so the second potential must be exactly a QUARTER of the first.  $720 \div 4 = 180$ , which it is. Had the question asked for the field instead, the same four times further would have made it SIXTEEN times weaker, because of the square. Computing  $kQ$  once and dividing twice also saves half a minute under exam pressure.

Absolute potential is  $V = kQ/r$ , which falls as  $1/r$ , not as  $1/r^2$ . With  $kQ = 9 \times 10^9 \times 4 \times 10^{-6} = 36000$ : at  $r = 0.05 \text{ m}$ ,  $V = 36000/0.05 = 720000 \text{ V} = 720 \text{ kV}$ ; at  $r = 0.20 \text{ m}$ ,  $V = 36000/0.20 = 180000 \text{ V} = 180 \text{ kV}$ . The distance quadruples, so the potential falls to a quarter, which is the check to run on the pair.

**You move from 5 cm to 20 cm away from a point charge, four times further out. What happens to the POTENTIAL?**

✓ It falls to a quarter, because  $V = kQ/r$  has a plain  $r$

Correct, and the printed answers on the paper confirm the quarter.

✗ It falls to a sixteenth, because of the inverse square

Wrong. A sixteenth is what the FIELD would do, not the potential.

✗ It is unchanged, because potential is a scalar

Wrong. Being a scalar means it has no direction, not that it ignores distance.

✗ It rises four times

Wrong direction: moving away lowers the potential.

✗ It falls to a half

Wrong. A half would need  $r$  to appear as a square root.

A quarter. The potential carries a plain  $r$ , so four times the distance means a quarter of the value. Only the FIELD is squared. Test 1 question 4 confirms it: 720 kV at 5 cm becomes 180 kV at 20 cm, exactly a quarter. Had you squared it you would have looked for a sixteenth and found no such option.

**The potential energy of a charge  $q$  sitting at a potential  $V$  is  $U = qV$ , measured in  $\{1\}$ .**

Answer:

Joules. Coulombs times volts gives joules, because a volt is defined as a joule per coulomb. That is also why  $W = qV$  computes a work in joules from a charge and a voltage.

A uniform field exists between two parallel plates 2 mm apart with 60 V across them. What is the field strength?

✓ 30000 V/m

Correct. 60 divided by 0.002.

✗ 30 V/m

Wrong by a factor of 1000, from leaving the gap in mm.

✗ 120 V/m

Wrong. This multiplies rather than divides.

✗ 0.033 V/m

Wrong, and upside down: 0.002 divided by 60.

✗ 3000 V/m

Wrong by a factor of ten in the conversion.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Between two parallel plates the field is UNIFORM, which means it has the same strength everywhere between them, unlike the field of a point charge which weakens with distance. That uniformity is what makes the plate formula so simple: the voltage is shared out evenly across the gap, so the field is just volts per metre. The unit  $\text{V m}^{-1}$  is telling you the formula.

### GIVEN

- $V = 60 \text{ V}$  across the plates
- $d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

**Find:** The field strength  $E$  between the plates, in  $\text{V m}^{-1}$

### FORMULA

$$E = \frac{V}{d}$$

**Why this one.** Parallel plates with a stated voltage and separation is the signature of this formula. There is no inverse square here at all, because the field does not spread out: the plates are flat and the lines run straight across, so nothing gets diluted with distance.

### WHAT EACH SYMBOL MEANS

- $E$  = the uniform field strength, in  $\text{V m}^{-1}$ , the same as  $\text{N C}^{-1}$
- $V$  = the potential difference across the plates, in volts
- $d$  = the plate separation, in METRES, which is where the mark is lost

### THE WORKING, LINE BY LINE

1. Convert the gap:  $2 \text{ mm} = 2 \times 10^{-3} \text{ m}$
2.  $E = V \div d$
3.  $E = 60 \div (2 \times 10^{-3})$
4. Front numbers:  $60 \div 2 = 30$
5. Dividing by  $10^{-3}$  is multiplying by  $10^3$ :  $E = 30 \times 10^3$
6.  $E = 30000 \text{ V m}^{-1}$

### CHECK IT

The option  $30 \text{ V m}^{-1}$  is what you get by leaving the gap in millimetres, and it is the commonest answer given by people who did the physics correctly. Dividing by a SMALL number must give a BIG answer, so if your result looks similar in size to the 60 you started with, the conversion did not happen.

Convert first:  $2 \text{ mm} = 0.002 \text{ m}$ . Then  $E = V/d = 60 / 0.002 = 30000 \text{ V/m}$ . Note the size: a modest 60 V across a very small gap gives an enormous field, which is exactly why capacitors are built with plates close together and why they eventually break down.

**What is the key difference between  $E = kQ/r^2$  and  $V = kQ/r$ ?**

✓ The field goes as one over  $r^2$  and is a vector, while the potential goes as one over  $r$  and is a scalar

Correct on both the power of  $r$  and the vector or scalar nature.

✗ The field goes as one over  $r$  and is a scalar, while the potential goes as one over  $r^2$  and is a vector

Wrong, and it swaps the two entirely.

✗ Both go as one over  $r^2$ , but only the potential is a vector

Wrong. The potential goes as  $1/r$ , and it is a scalar not a vector.

✗ Both go as one over  $r$ , and both are vectors

Wrong. The field goes as  $1/r^2$ .

✗ They are the same quantity written two ways

Wrong. They are different quantities with different units, N/C against volts.

Field is squared, potential is not. The field is a vector, so contributions add with direction; the potential is a scalar, so contributions add as plain signed numbers. That difference is why a potential question is usually the easier of the two.

### 3 Electric Flux and Gauss

**Speed tip.** For flux, the angle is measured from the NORMAL, not the surface. Face on means the angle is 0 and the cosine is 1, so the flux is just EA.

3.1

EASY

Reference sheet R.4

**A Gaussian surface is an  $\{\{1\}\}$  closed surface, chosen to make Gauss's law easy.**

Answer:

Imaginary. It is a mathematical construction, not a physical object, and you choose its shape to match the symmetry of the problem: a sphere around a point charge, a cylinder around a line charge. The Amperian loop in magnetism is the same idea.

**A uniform electric field  $E = 200 \text{ N/C}$  passes through a surface of area  $2 \text{ m}^2$  at  $0^\circ$ . Find Flux.**

**X** 450 400  $\text{N/m}^2/\text{C}$

Wrong, and the printed option runs two numbers together. Neither follows from  $200 \times 2$ .

**X** 300  $\text{N/m}^2/\text{C}$

Wrong. This would need an area of  $1.5 \text{ m}^2$ .

**✓** 400  $\text{N/m}^2/\text{C}$

Correct.  $\Phi = EA \cos 0 = 200 \times 2 \times 1 = 400$ .

**X** 120  $\text{N/m}^2/\text{C}$

Wrong. This is  $200 \times 2 \times 0.3$ , which is no cosine of any angle offered.

**X** 100  $\text{N/m}^2/\text{C}$

Wrong. This halves the field instead of doubling it by the area.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Flux is a counting idea, not a force idea. Picture the field as a set of parallel arrows and the surface as a hoop held in their path. Flux asks: how many arrows go THROUGH the hoop? Hold the hoop face on and you catch the most. Tilt it and you catch fewer. Turn it edge on and the arrows slide past without crossing it at all, so the flux is zero.

#### GIVEN

- $E = 200 \text{ N C}^{-1}$
- $A = 2 \text{ m}^2$
- $\theta = 0^\circ$  (the angle from the NORMAL to the surface)

**Find:** The electric flux  $\Phi$ , in  $\text{N m}^2 \text{ C}^{-1}$

#### FORMULA

$$\Phi = EA \cos \theta$$

**Why this one.** A uniform field and a flat area is the simplest flux case, so the full Gauss machinery is not needed. The cosine is there to take only the part of the field that is perpendicular to the surface, since only that part actually passes through.

#### WHAT EACH SYMBOL MEANS

- $\Phi$  = the electric flux, in  $\text{N m}^2 \text{ C}^{-1}$
- $E$  = the field strength, in  $\text{N C}^{-1}$
- $A$  = the area of the surface, in  $\text{m}^2$
- $\theta$  = the angle between the field and the NORMAL to the surface, not the surface itself

#### THE WORKING, LINE BY LINE

1.  $\theta = 0^\circ$ , so  $\cos \theta = \cos 0 = 1$
2.  $\Phi = E \times A \times \cos \theta$
3.  $\Phi = 200 \times 2 \times 1$
4.  $\Phi = 400 \text{ N m}^2 \text{ C}^{-1}$

#### CHECK IT

At  $0^\circ$  the surface faces the field squarely, which is the maximum possible flux, so the answer must simply be  $E \times A$ . If the question had said  $60^\circ$ ,  $\cos 60 = 0.5$  and the flux would halve to 200.

Flux is the field times the area times the cosine of the angle between the field and the surface NORMAL:  $\Phi = EA \cos(\theta)$ . At  $0^\circ$  the field is straight through,  $\cos 0 = 1$ , so  $\Phi = 200 \times 2 \times 1 = 400$ . The angle being zero is what makes this a one-line question; at  $90^\circ$  the flux would be zero.

3.3

EASY

Reference sheet R.4

**A Gaussian surface is a real physical shell that must be built around the charge before Gauss's law can be applied.**

Answer: **false**

It is IMAGINARY, a mathematical surface you draw on the diagram and choose to suit the symmetry: a sphere around a point charge, a cylinder around a line. Nothing is constructed. The Amperian loop in magnetism is the same idea, and both are chosen purely to make the algebra come out.

3.4

MEDIUM

Reference sheet R.1

**Electric flux is  $\Phi = EA \cos \theta$ , where  $\theta$  is measured from the  $\{\{1\}\}$  to the surface.**

Answer:

From the normal, the line sticking straight out of the surface. Facing the field squarely gives  $\theta = 0$  and  $\cos 0 = 1$ , the maximum. Edge on gives  $\theta = 90$  and  $\cos 90 = 0$ , nothing through. Measuring from the surface instead inverts every answer.

**A closed spherical surface encloses a charge of  $3\text{ }\mu\text{C}$ . A separate  $10\text{ }\mu\text{C}$  charge sits just outside it. What is the total electric flux through the surface?**

✓  $3.39 \times 10^5\text{ N m}^2/\text{C}$

Correct. Only the enclosed  $3\text{ }\mu\text{C}$  counts.

✗  $1.47 \times 10^6\text{ N m}^2/\text{C}$

Wrong. This adds the outside charge, which contributes nothing.

✗  $1.13 \times 10^6\text{ N m}^2/\text{C}$

Wrong. This uses the  $10\text{ }\mu\text{C}$  instead of the 3.

✗ Zero, because the charges have opposite effects

Wrong. There is enclosed charge, so there is net flux.

✗  $2.66 \times 10^{-17}\text{ N m}^2/\text{C}$

Wrong. This multiplies by  $\epsilon_0$  instead of dividing.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Gauss's law makes a claim that sounds too good to be true: the total flux out of a closed surface depends only on the charge INSIDE it, and not at all on charges outside, nor on where inside the charge sits, nor on the shape of the surface. The reason is that a field line from an outside charge that enters the surface must also leave it, so it contributes once as negative flux and once as positive and cancels itself out.

**GIVEN**

- Charge enclosed:  $Q = 3 \mu\text{C} = 3 \times 10^{-6} \text{ C}$
- A separate  $10 \mu\text{C}$  charge sits OUTSIDE the surface
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$

**Find:** The total electric flux  $\Phi$  through the closed surface, in  $\text{N m}^2 \text{ C}^{-1}$

**FORMULA**

$$\Phi = \frac{Q}{\epsilon_0}$$

**Why this one.** The surface is CLOSED, which is the word that summons Gauss's law rather than  $\Phi = EA \cos \theta$ . The outside charge is a deliberate distractor: it certainly changes the field at every point on the surface, but it adds nothing to the TOTAL flux, because whatever enters also leaves.

**WHAT EACH SYMBOL MEANS**

- $\Phi$  = the total flux through the closed surface, in  $\text{N m}^2 \text{ C}^{-1}$
- $Q$  = the ENCLOSED charge only, in coulombs
- $\epsilon_0$  = the permittivity of free space,  $8.85 \times 10^{-12} \text{ F m}^{-1}$

**THE WORKING, LINE BY LINE**

1. Enclosed charge only:  $Q = 3 \mu\text{C} = 3 \times 10^{-6} \text{ C}$
2. The  $10 \mu\text{C}$  is outside, so it contributes zero net flux and is discarded
3.  $\Phi = (3 \times 10^{-6}) \div (8.85 \times 10^{-12})$
4. Front numbers:  $3 \div 8.85 = 0.339$
5. Powers of ten:  $10^{-6} \div 10^{-12} = 10^6$
6.  $\Phi = 0.339 \times 10^6 = 3.39 \times 10^5 \text{ N m}^2 \text{ C}^{-1}$

**CHECK IT**

Two traps are in the options. Adding both charges gives  $1.47 \times 10^6$ , and the zero option tempts anyone who thinks the outside charge cancels the inside one, which it does not: it cancels only itself. Ask one question of any Gauss problem before anything else, which is what is INSIDE, and the rest is one division.

Gauss's law counts ENCLOSED charge only:  $\Phi = Q \text{ enclosed over } \epsilon_0 = 3 \times 10^{-6} / 8.85 \times 10^{-12} = 3.39 \times 10^5 \text{ N m}^2 \text{ per C}$ . The  $10 \mu\text{C}$  outside contributes exactly nothing, because every field line it sends into the surface comes back out again.

**3.6**

MEDIUM

Reference sheet R.2

**Asked twice.** Also set as: Reference sheet R.2.

**The electric field inside a hollow conductor is zero, provided no charge is enclosed within that region.**

**Answer: true**

True, and it is the Faraday cage. Charge on a conductor sits on the outer surface, so a Gaussian surface drawn inside encloses no charge, and by Gauss's law the flux, and therefore the field, is zero. It is why a car protects you from lightning and why sensitive electronics are wrapped in metal.

3.7

MEDIUM

Reference sheet R.2

In Gauss's law the flux out equals the charge in divided by  $\epsilon_0$ .

Answer:

$\epsilon_0$ , the permittivity of free space.  $\mu_0$  is its magnetic counterpart,  $\sigma$  is a surface charge density and  $\lambda$  a line charge density.

3.8

MEDIUM

Reference sheet R.1

What is the electric field inside a conductor in electrostatic equilibrium, and what is the everyday name for the consequence?

✓ Zero, and the consequence is the Faraday cage

Correct on both.

✗ Zero, and the consequence is the Gaussian surface

Wrong on the name. A Gaussian surface is the imaginary surface you draw to apply the law, not a consequence of it.

✗ Equal to  $kQ/r^2$ , and the consequence is the Faraday cage

Wrong. The field inside is zero, not an inverse square.

✗ Equal to  $\frac{\sigma}{\epsilon_0}$ , and the consequence is polarisation

Wrong.  $\frac{\sigma}{\epsilon_0}$  is a sheet field, and polarisation happens in insulators.

✗ Undefined, because a conductor has no interior

Wrong. The interior is exactly where the field is zero.

$E = 0$  inside a conductor. Any charge on it moves to the surface, so a Gaussian surface drawn inside encloses no charge, and no enclosed charge means no field. The practical consequence is the Faraday cage: the inside of a conducting shell is shielded from outside fields.

3.9

MEDIUM

Reference sheet R.2

Asked twice. Also set as: Reference sheet R.2.

The electric field inside a hollow conductor is zero provided no charge is enclosed within that region.

Answer: **true**

True, and the qualification matters: it is the absence of ENCLOSED charge that forces the field to zero, by Gauss's law.

**A conducting sphere of radius 0.25 m carries a total charge of  $6 \times 10^{-6}$  C. Calculate the electric field at a point 0.50 m from the center of the sphere.**

**X**  $2.92 \times 10^5 \text{ NC}^{-1}$

Wrong. No route to this value from these numbers.

**✓**  $2.16 \times 10^5 \text{ NC}^{-1}$

Correct.  $E = kQ/r^2 = 9 \times 10^9 \times 6 \times 10^{-6} / 0.25 = 2.16 \times 10^5 \text{ NC}^{-1}$ .

**X**  $3.53 \times 10^5 \text{ NC}^{-1}$

Wrong. No route to this value from these numbers.

**X**  $5 \times 10^5 \text{ NC}^{-1}$

Wrong. This drops the  $0.25 \text{ m}^2$  denominator to a bare division and loses the factor entirely.

**X**  $1.47 \times 10^5 \text{ NC}^{-1}$

Wrong. This is close to what you get measuring 0.50 m from the SURFACE, that is  $r = 0.75 \text{ m}$ , which the question does not ask for.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Gauss's law delivers one enormously useful result for spheres: from OUTSIDE, a uniformly charged sphere behaves exactly as though all its charge were squeezed into a point at its centre. So its radius is irrelevant to the field outside it. Inside a conducting sphere the field is zero instead, because the charges rearrange until they cancel any field within the metal, which is why a car protects you in a thunderstorm.

**GIVEN**

- Sphere radius  $R = 0.25 \text{ m}$
- $Q = 6 \times 10^{-6} \text{ C}$
- The point of interest is  $r = 0.50 \text{ m}$  from the CENTRE
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

**Find:** The electric field  $E$  at that point, in  $\text{N C}^{-1}$

**FORMULA**

$$E = \frac{kQ}{r^2}$$

**Why this one.** The first job is to check where the point is.  $r = 0.50 \text{ m}$  is greater than the radius  $0.25 \text{ m}$ , so the point lies OUTSIDE the sphere, and outside a sphere the point-charge formula applies without modification. Had the point been inside the conductor, the answer would have been zero instead, whatever the charge.

**WHAT EACH SYMBOL MEANS**

- $E$  = the field at the point, in  $\text{N C}^{-1}$
- $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- $Q$  = the total charge on the sphere, in coulombs
- $r$  = the distance from the CENTRE of the sphere, not from its surface

**THE WORKING, LINE BY LINE**

1. Check first: is  $r > R$ ?  $0.50 > 0.25$ , yes, so the point is outside
2. Square the distance:  $r^2 = 0.50^2 = 0.25 \text{ m}^2$
3. Top:  $kQ = 9 \times 10^9 \times 6 \times 10^{-6}$
4. Front numbers:  $9 \times 6 = 54$
5. Powers of ten:  $10^9 \times 10^{-6} = 10^3$
6. Top =  $54 \times 10^3 = 5.4 \times 10^4$
7.  $E = (5.4 \times 10^4) \div 0.25$
8.  $E = 2.16 \times 10^5 \text{ N C}^{-1}$

**CHECK IT**

The trap is measuring from the surface instead of the centre. Using  $0.50 - 0.25 = 0.25 \text{ m}$  would give four times the answer,  $8.64 \times 10^5$ , which is not offered. Always start a sphere question by asking whether the point is inside or outside, and always measure  $r$  from the centre. Note also that  $0.25$  appears twice in this question meaning two different things, the radius and  $r^2$ , so label your lines.

Outside a charged conducting sphere the field is the same as if all the charge sat at the centre, so  $E = kQ/r^2$  with  $r$  measured from the CENTRE, not from the surface.  $E = 9 \times 10^9 \times 6 \times 10^{-6} / (0.50)^2 = 54000 / 0.25 = 2.16 \times 10^5 \text{ NC}^{-1}$ . The  $0.25 \text{ m}$  radius is there to tell you the point is outside the sphere, and for no other reason.

**A field of 200 N/C crosses an area of 2 square metres at 60° to the normal. What is the flux?**

✓ 200 N m<sup>2</sup>/C

Correct. Half of the 400 you would get face on.

✗ 400 N m<sup>2</sup>/C

Wrong. 400 is the answer at 0°, with the surface square to the field.

✗ 346 N m<sup>2</sup>/C

Wrong. 346 uses SIN 60, not cos, which is the angle-from-the-surface error.

✗ 0 N m<sup>2</sup>/C

Wrong. Zero needs 90°, with the field skimming along the surface.

✗ 100 N m<sup>2</sup>/C

Wrong. This halves it twice over.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Flux counts how much of a field actually passes through a surface. Hold a hoop square-on to the field and you catch everything; tilt it and you catch less; turn it edge-on and nothing goes through at all. The cosine does that job, but only if the angle is measured the way physics defines it, which is from the **NORMAL**, the line sticking straight out of the surface, not from the surface itself.

#### GIVEN

- $E = 200 \text{ N C}^{-1}$
- $A = 2 \text{ m}^2$
- $\theta = 60^\circ$  to the normal

**Find:** The electric flux  $\Phi$ , in  $\text{N m}^2 \text{ C}^{-1}$

#### FORMULA

$$\Phi = EA \cos \theta$$

**Why this one.** The surface is flat and open rather than closed, and the field is uniform, so this is the direct flux formula and not Gauss's law. Cosine rather than sine because at 0°, square-on, the flux must be at its maximum, and  $\cos 0 = 1$  while  $\sin 0 = 0$ .

#### WHAT EACH SYMBOL MEANS

- $\Phi$  = the electric flux, in  $\text{N m}^2 \text{ C}^{-1}$
- $E$  = the field strength, in  $\text{N C}^{-1}$
- $A$  = the area, in  $\text{m}^2$
- $\theta$  = the angle from the **NORMAL** to the surface

#### THE WORKING, LINE BY LINE

1.  $\cos 60^\circ = 0.5$  (worth knowing without a calculator)
2.  $\Phi = E \times A \times \cos \theta$
3.  $\Phi = 200 \times 2 \times 0.5$
4.  $200 \times 2 = 400$
5.  $\Phi = 400 \times 0.5 = 200 \text{ N m}^2 \text{ C}^{-1}$

#### CHECK IT

The option 400 is the flux you would get square-on, so it must be the maximum and cannot be right for a tilted surface: a tilt can only reduce the flux, never increase it. The option 346 is what  $\sin 60^\circ$  would have given, which is the standard punishment for using the wrong trigonometric function.

$\Phi = EA \cos \theta = 200 \times 2 \times \cos 60 = 200 \times 2 \times 0.5 = 200 \text{ N m}^2 \text{ per C}$ . Test 1 question 5 is this same setup at  $0^\circ$ , where  $\cos 0 = 1$  and the answer is the full 400. Tilting to  $60^\circ$  halves it, because  $\cos 60$  is exactly 0.5.

3.12

MEDIUM

Reference sheet R.1

**A charge sitting outside a closed surface contributes nothing to the net flux through it.**

Answer: **true**

True. Every field line an external charge sends into the surface comes out again somewhere else, so the ins and outs cancel exactly. Gauss's law counts only ENCLOSED charge, which is what makes it useful: you can ignore everything outside the surface you drew.

3.13

MEDIUM

Reference sheet R.1

**The field of an infinite line charge is  $E = \lambda \text{ over } 2 \pi \epsilon_0 r$ , which falls as 1 over  $r$  to the power  $\{1\}$ .**

Answer:

One. A line spreads its field over a cylinder whose area grows in proportion to  $r$ , so the field falls as  $1/r$ , more slowly than a point charge's  $1/r^2$ . The magnetic field of a straight wire falls the same way and for the same geometric reason.

3.14

MEDIUM

Reference sheet R.1

**Why is the electric field zero inside a hollow conductor?**

✓ Because no charge is enclosed, so by Gauss's law there is no flux and no field

Correct, and it is a consequence of Gauss's law rather than a separate rule.

✗ Because the metal absorbs the field

Wrong. Nothing is absorbed; the charges rearrange until the interior field cancels.

✗ Because conductors carry no charge at all

Wrong. A charged conductor carries plenty; it just keeps it on the surface.

✗ Because the field is cancelled by the earth

Wrong. It works whether or not the conductor is earthed.

✗ Because air inside is an insulator

Wrong. The effect is the same with a vacuum inside.

Charge on a conductor sits on its OUTER surface, so any Gaussian surface drawn inside encloses nothing, and Gauss's law then forces the field to zero. That is the Faraday cage: it is why a car protects you from lightning and why sensitive electronics are wrapped in metal.

3.15

HARD

Reference sheet R.1

Match how the field falls with distance for a point charge, an infinite line charge, and an infinite plane sheet.

✓ Point  $1/r^2$ , line  $1/r$ , sheet independent of distance

Correct, and the reason is the geometry of how the field spreads.

✗ Point  $1/r$ , line  $1/r^2$ , sheet independent of distance

Wrong, and it swaps the point and line cases.

✗ All three fall as  $1/r^2$

Wrong. Only the point charge falls as an inverse square.

✗ Point  $1/r^2$ , line independent of distance, sheet  $1/r$

Wrong on the line and the sheet.

✗ All three are independent of distance

Wrong. Only the infinite sheet has that property.

The pattern follows the shape. A point spreads its field over a sphere whose area grows as  $r^2$ , so  $E = kQ/r^2$ . A line spreads over a cylinder whose area grows as  $r$ , so  $E = \lambda$  over  $2\pi\epsilon_0 r$ , falling as  $1/r$ . A plane does not spread at all, so  $E = \sigma$  over  $2\epsilon_0$  is the SAME however far away you stand, which is startling but correct for an infinite sheet.

3.16

HARD

Reference sheet R.1

A charge sits OUTSIDE a closed surface. What is its contribution to the net flux through that surface?

✓ Nothing, because only enclosed charge contributes

Correct. Every line an outside charge sends in comes back out, so the net is zero.

✗ The same as if it were inside

Wrong. If that were so, the law could not be written in terms of enclosed charge alone.

✗ Half as much as if it were inside

Wrong. The contribution is zero, not a fraction.

✗ It makes the net flux negative

Wrong. It contributes nothing at all, of either sign.

✗ It doubles the net flux

Wrong, for the same reason.

Flux out equals charge in over  $\epsilon_0$ . An external charge sends just as many field lines into the surface as out of it, so its net contribution is exactly zero. That is why Gauss's law says  $Q_{\text{ENCLOSED}}$  and nothing else.

3.17

HARD

Reference sheet R.1

The field of an infinite charged plane sheet,  $E = \frac{\sigma}{2\epsilon_0}$ , is the same however far from the sheet you stand.

Answer: **true**

True, and it is genuinely surprising. There is no  $r$  in the formula at all. The reason is geometric: a point spreads its field over a growing sphere and a line over a growing cylinder, but an infinite plane does not spread it at all, so moving away gains you nothing. It is what makes the uniform field between capacitor plates possible.

## 4 Capacitance, Dielectrics and Energy

**Speed tip.** Capacitors are the opposite of resistors: parallel adds, series is the reciprocal. Energy has V squared in it.

4.1

EASY

Reference sheet R.1

For a parallel plate capacitor,  $C = \epsilon_r \frac{\epsilon_0 A}{d}$ , where A is the plate area and d is the distance between the plates.

Answer:

The separation, that is the gap. Closer plates mean a bigger capacitance, since d is in the denominator, which is why real capacitors use very thin insulating layers. Note that d is the gap, not the thickness of the metal.

4.2

EASY

Reference sheet R.4

Capacitance is the charge stored per volt applied, and its unit is the farad, which is one coulomb per volt.

Answer:

One coulomb per volt, straight from  $C = Q/V$ . The farad is a huge unit in practice, which is why real capacitors are quoted in  $\mu\text{F}$ , nF and pF.

4.3

EASY

Reference sheet R.4

An electric dipole is:

✓ Two equal and opposite charges separated by a small distance

Correct. Two equal and opposite charges a small distance apart, with the dipole moment  $p = qd$  pointing from the negative charge to the positive one.

✗ A single charge on a conductor

Wrong. A dipole needs two charges.

✗ Two equal charges of the same sign

Wrong. They must be OPPOSITE, or the net effect cancels differently.

✗ A conductor with no net charge

Wrong. A neutral conductor is not a dipole, though it can be polarised into one.

✗ An insulator between capacitor plates

Wrong. That is a dielectric, though dielectrics work BY forming dipoles.

Two equal and opposite charges a small distance apart, with dipole moment  $p = qd$  pointing from the negative to the positive. The polarised atom is the everyday example: an applied field pulls the electron cloud one way and the nucleus the other, creating exactly this arrangement, which is what polarisation in an insulator means.

4.4

EASY

Reference sheet R.1

Inserting a dielectric between capacitor plates always increases the capacitance.

Answer: true

True.  $C = \epsilon_r$  times  $C_0$ , and the relative permittivity  $\epsilon_r$  is always greater than 1 for a real material, being exactly 1 only for vacuum. The mechanism is polarisation: the dielectric's induced dipoles partly cancel the field between the plates, so the same charge sits at a lower voltage, and  $C = Q/V$  rises. Note it works the OPPOSITE way for the Coulomb force, which a dielectric divides by the same factor.

4.5

EASY

Reference sheet R.1

Capacitance is  $C = \frac{Q}{V}$ , the charge stored per  $\{\{1\}\}$  applied.

Answer:

Per volt. It is a ratio, and crucially a FIXED one: raise the voltage and you push in proportionally more charge, so  $Q/V$  does not move. Capacitance is set by geometry, never by what you store in it.

4.6

EASY

Reference sheet R.4

Capacitance is defined as the charge stored per volt applied, and is measured in farads.

Answer: true

True, and the farad is a coulomb per volt. It is an enormous unit in practice: the capacitors in this course run from pF to  $\mu\text{F}$ , and a 1 F capacitor would be a remarkable object. Seeing an answer of 4.77 F should therefore fail the step 7 believability check instantly.

4.7

EASY

Reference sheet R.4

A dipole is two equal and  $\{\{1\}\}$  charges separated by a small distance.

Answer:

Opposite. A polarised atom is the everyday case: an applied field pulls the electron cloud one way and the nucleus the other, producing exactly this arrangement. Its dipole moment is  $p = qd$ , pointing from the negative charge to the positive one.

4.8

EASY

Reference sheet R.4

One farad is equal to:

✓ One coulomb per volt

Correct. A coulomb per volt, read straight off the definition  $C = Q/V$ .

✗ One volt per coulomb

Wrong, and inverted.

✗ One coulomb per second

Wrong. That is the ampere.

✗ One joule per coulomb

Wrong. That is the volt.

✗ One volt per amp

Wrong. That is the ohm.

A coulomb per volt, read straight off  $C = Q/V$ . The other four options are all real units of other things, which is why they make good distractors: a coulomb per second is the AMPERE, a joule per coulomb is the VOLT, and a volt per amp is the OHM.

4.9

EASY

Lecture slides, module 2, slide 11

The dipole moment is  $p = qd$ , where  $q$  is a charge in coulombs and  $d$  a separation in metres. Its unit is therefore the coulomb metre.

Answer:

Coulomb metre, written C m. The unit of any product is the product of the units, so a charge in coulombs times a distance in metres gives coulomb metres. This is the class drill the slide sets, and it is worth doing rather than skipping: reading a unit off a formula is the fastest check that you have the formula the right way up.

4.10

MEDIUM

Reference sheet R.1

The torque on an electric dipole in a uniform field is  $\tau = pE \sin \theta$ . At what angle is the torque zero?

✓ Zero degrees, with the dipole aligned along the field

Correct. Aligned means  $\sin \theta = 0$ , so no torque.

✗ Ninety degrees, with the dipole across the field

Wrong.  $90^\circ$  is where the torque is at its MAXIMUM.

✗ Forty-five degrees

Wrong.  $\sin 45$  is about 0.71, so the torque is substantial.

✗ The torque is never zero in a field

Wrong. It is zero at alignment, which is why dipoles line up.

✗ One hundred and eighty degrees only

Wrong.  $\sin 180$  is also zero, but so is  $\sin 0$ , and 0 is the stable one.

$\sin 0 = 0$ , so the torque vanishes when the dipole lies along the field. That is the position the field twists it INTO: a dipole placed at an angle rotates until it is aligned, and then stops. The torque is greatest at  $90^\circ$ , where  $\sin$  is 1, because that is when the field has the most turning to do.

4.11

MEDIUM

Reference sheet R.1

The energy of a dipole in a field is  $U = \text{minus } pE \cos \theta$ , and it is at a minimum when the dipole is aligned with the field.

Answer: true

True. At  $\theta = 0$ ,  $\cos 0 = 1$ , so  $U = \text{minus } pE$ , the most negative value available and therefore the minimum. Systems settle into their lowest energy, which is exactly why a dipole turns to align itself. The torque formula and the energy formula are telling the same story from two directions.

4.12

MEDIUM

Reference sheet R.2

Asked twice. Also set as: Reference sheet R.1.

### How do capacitors combine, compared with resistors?

✓ Backwards: capacitors ADD in parallel, while resistors add in series

Correct, and remembering one and flipping is the reliable way to keep them apart.

✗ The same way as resistors in both cases

Wrong. They behave oppositely, which is the whole difficulty.

✗ Capacitors add in series, and resistors add in parallel too

Wrong on both halves.

✗ Capacitors always add, whichever way they are connected

Wrong. In series they combine reciprocally.

✗ Capacitors never simply add

Wrong. In parallel they add straightforwardly.

Capacitors ADD in parallel:  $C = C_1 + C_2$ . In series they combine reciprocally, giving less than the smallest. Resistors are the exact opposite: they add in series and combine reciprocally in parallel. The hook is to learn ONE of them properly and flip it for the other. Physically, putting capacitors side by side is just making one bigger plate, so the capacitances add.

4.13

MEDIUM

Reference sheet R.1

**Capacitors in series combine as 1 over C equivalent = 1 over C1 + 1 over C2, which always gives a total LESS than the {{1}} capacitor in the combination.**

Answer:

Less than the smallest. That is a free check on every series answer: work it out, and if your total is not below the smallest capacitor you have forgotten to flip the reciprocal back. Test 1 question 2 uses it, where 1 and 5  $\mu\text{F}$  in series give 0.833, below the 1.

4.14

MEDIUM

Reference sheet R.1

Which of these is NOT an equivalent expression for the energy stored in a capacitor?

✓  $W = QV$

Correct as the odd one out.  $QV$  has no half, so it is double the true energy.

✗  $W = \text{one half } CV \text{ squared}$

Wrong choice: this IS a valid form, the one used most often.

✗  $W = \text{one half } QV$

Wrong choice: this is valid, and follows from  $C = Q/V$ .

✗  $W = \frac{Q^2}{2C}$

Wrong choice: this is valid, and is the form to use when you know  $Q$  rather than  $V$ .

✗ All four are equivalent

Wrong.  $QV$  is not among the valid forms.

The three valid forms all carry a factor of one half:  $W = \text{one half } CV \text{ squared} = \text{one half } QV = \frac{Q^2}{2C}$ .  $QV$  without the half is

wrong, and the half is there for a physical reason: the voltage climbed from zero to  $V$  as the capacitor charged, so the average voltage the charge crossed was  $V/2$ , not  $V$ . Forgetting it doubles the answer, and Test 1 question 15 offers exactly that wrong value.

4.15

MEDIUM

Reference sheet R.4

Match each quantity to what it describes.

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| Permittivity, symbol epsilon | How readily a material permits an ELECTRIC field               |
| Permeability, symbol mu      | How readily a material permits a MAGNETIC field                |
| Polarisation                 | Induced dipoles forming in an insulator under an applied field |

The two names are deliberately parallel, and so are their roles:  $\epsilon_0$  appears in every electrostatic formula and  $\mu_0$  in every magnetic one. Their product is what gives the speed of light. Polarisation is the mechanism by which a dielectric raises capacitance.

4.16

MEDIUM

Reference sheet R.2

Capacitance is charge per volt, and it is fixed by the capacitor's  $\{\{1\}\}$ .

Answer:

Geometry. For a parallel plate that means the area and the separation, plus the dielectric between the plates.

4.17

MEDIUM

Reference sheet R.2

The half in one half  $CV$  squared is there because the voltage rose from zero as the capacitor was charged.

Answer: true

True. The charge that arrives first is moved across almost no potential difference, so the average voltage over the whole charging process is half the final value.

4.18

MEDIUM

Reference sheet R.1

Asked twice. Also set as: Reference sheet R.2.

### How do capacitors combine compared with resistors?

✓ Capacitors ADD in parallel, which is backwards from resistors, which add in series

Correct, and this is the pair most often confused under time pressure.

✗ Capacitors add in series, exactly like resistors

Wrong. Series capacitors combine by reciprocals, giving less than the smallest.

✗ Both add in parallel

Wrong. Resistors in parallel take reciprocals.

✗ Both use reciprocals in both arrangements

Wrong. Each uses the reciprocal rule in one arrangement only.

✗ Capacitors cannot be combined at all

Wrong. Combining them is routine, and Test 1 question 2 does exactly that.

Capacitors add in parallel and take reciprocals in series; resistors do the opposite. Remember ONE of them and flip it for the other. The physical reason is that putting capacitors side by side effectively enlarges the plate area, and a larger area means more capacitance.

4.19

MEDIUM

Reference sheet R.2

### Capacitors add when they are in parallel, which is backwards from resistors.

Answer:

Parallel. Resistors add in series, so learning one rule and flipping it for the other is safer than trying to hold both.

A metal sphere mounted on an insulating rod carries a charge of 4.5 nC when its potential is 180 V higher than its surroundings. What is the capacitance of the capacitor formed by the sphere and its surroundings?

✓ 25 pF

Correct.  $C = Q/V = 4.5 \times 10^{-9} / 180 = 2.5 \times 10^{-11} \text{ F} = 25 \text{ pF}$ .

✗ 12 pF

Wrong, and about half: this comes from doubling the potential difference.

✗ 8.1 pF

Wrong. No route to this value from 4.5 nC and 180 V.

✗ 30 pF

Wrong. This would need 5.4 nC at the same potential.

✗ 6.3 pF

Wrong. No route to this value from these numbers.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Capacitance is not about how much charge something is holding at this moment; it is about how much charge it holds PER VOLT. A big capacitance means you can pile on a lot of charge without the voltage climbing much. Even an isolated metal sphere has a capacitance, because raising its charge raises its potential above its surroundings, and the ratio of the two is fixed by its size and shape alone.

**GIVEN**

- $Q = 4.5 \text{ nC} = 4.5 \times 10^{-9} \text{ C}$
- $V = 180 \text{ V}$  (the potential above the surroundings)

**Find:** The capacitance  $C$ , in picofarads (pF)

**FORMULA**

$$C = \frac{Q}{V}$$

**Why this one.** This is the definition of capacitance itself, so it needs no derivation and no geometry. The question gives one charge and one voltage and asks for capacitance, which is exactly the ratio. The sphere and the insulating rod are scene-setting: they explain WHY there is a capacitance to speak of, but they add no numbers to the calculation.

**WHAT EACH SYMBOL MEANS**

- $C$  = the capacitance, in farads (F)
- $Q$  = the charge stored, in coulombs
- $V$  = the potential difference that charge produces, in volts
- $1 \text{ pF} = 10^{-12} \text{ F}$ , and  $1 \text{ nF} = 10^{-9} \text{ F}$

**THE WORKING, LINE BY LINE**

1. Convert the charge:  $4.5 \text{ nC} = 4.5 \times 10^{-9} \text{ C}$
2.  $C = Q \div V$
3.  $C = (4.5 \times 10^{-9}) \div 180$
4. Front numbers:  $4.5 \div 180 = 0.025$
5.  $C = 0.025 \times 10^{-9} \text{ F}$
6. Move to picofarads:  $0.025 \times 10^{-9} = 25 \times 10^{-12}$
7.  $C = 25 \text{ pF}$

**CHECK IT**

Every option is in picofarads, so the conversion from  $10^{-9}$  to  $10^{-12}$  is compulsory and is where the mark is won. The rule: to go from nano to pico, multiply by 1000. And 25 pF is a believable capacitance for a small isolated sphere, being about what a short length of coaxial cable has.

Capacitance is charge per unit potential difference:  $C = Q/V$ . The geometry is a distraction, because the sphere and its surroundings are simply being treated as a capacitor.  $C = 4.5 \times 10^{-9} / 180 = 2.5 \times 10^{-11} \text{ F} = 25 \text{ pF}$ . Remember  $1 \text{ pF} = 10^{-12} \text{ F}$ .

**What is the capacitance of a capacitor having a stored energy of 2.25 mJ after being charged with a 30-volt source?**

**X** 10  $\mu\text{F}$

Wrong, and it is the classic slip: forgetting the  $1/2$  and computing  $U/V^2$  doubled, or equivalently dropping the factor of two.

**✓** 5  $\mu\text{F}$

Correct.  $C = 2U/V^2 = 4.5 \times 10^{-3} / 900 = 5 \mu\text{F}$ .

**X** 25  $\mu\text{F}$

Wrong. No route to this value from 2.25 mJ at 30 V.

**X** 13  $\mu\text{F}$

Wrong. No route to this value from these numbers.

**X** 2.5  $\mu\text{F}$

Wrong, and exactly half the answer: this comes from using  $U/V^2$  without the factor of two.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Charging a capacitor is not free. The first charge crosses easily, but each new charge has to be pushed against the ones already sitting there, so the work builds up as the voltage rises. Averaging that over the whole process is where the half comes from in the energy formula. That stored energy is what a camera flash releases in a millisecond, and what makes a large charged capacitor genuinely dangerous long after the power is off.

**GIVEN**

- $U = 2.25 \text{ mJ} = 2.25 \times 10^{-3} \text{ J}$
- $V = 30 \text{ V}$

**Find:** The capacitance  $C$ , in microfarads

**FORMULA**

$$U = \frac{1}{2}CV^2, \text{ so } C = \frac{2U}{V^2}$$

**Why this one.** An energy and a voltage are given and a capacitance is wanted, and this is the formula joining exactly those three. Rearranging it moves the half across as a doubling on top, and the  $V^2$  down to the bottom. The other energy forms,  $U = \frac{1}{2}QV$  and  $U = Q^2 / 2C$ , would need the charge, which is not given.

**WHAT EACH SYMBOL MEANS**

- $U$  = the energy stored, in joules
- $C$  = the capacitance, in farads
- $V$  = the voltage the capacitor was charged to, in volts, and it is SQUARED
- $1 \text{ mJ} = 10^{-3} \text{ J}$  and  $1 \mu\text{F} = 10^{-6} \text{ F}$

**THE WORKING, LINE BY LINE**

1. Convert the energy:  $2.25 \text{ mJ} = 2.25 \times 10^{-3} \text{ J}$
2. Rearrange:  $C = (2 \times U) \div V^2$
3. Top:  $2 \times 2.25 \times 10^{-3} = 4.5 \times 10^{-3}$
4. Bottom:  $V^2 = 30^2 = 900$
5.  $C = (4.5 \times 10^{-3}) \div 900$
6. Front numbers:  $4.5 \div 900 = 0.005$
7.  $C = 0.005 \times 10^{-3} = 5 \times 10^{-6} \text{ F}$
8.  $C = 5 \mu\text{F}$

**CHECK IT**

Forgetting to double gives  $2.5 \mu\text{F}$  and forgetting to square the 30 gives a wildly different number; both wrong routes are sitting in the option list waiting. Verify forwards in one line:  $\frac{1}{2} \times 5 \times 10^{-6} \times 900 = 2.25 \times 10^{-3} \text{ J}$ , which is the energy given. Substituting your answer back into the original formula takes seconds and catches nearly every rearrangement error.

Energy stored is  $U = \frac{1}{2} C V^2$ , so rearranging for the capacitance gives  $C = 2U / V^2$ . Substituting:  $C = 2 \times 2.25 \times 10^{-3} / 30^2 = 4.5 \times 10^{-3} / 900 = 5 \times 10^{-6} \text{ F} = 5 \mu\text{F}$ . The half and the SQUARE on the voltage are where marks are lost here.

**4.22**

MEDIUM

Reference sheet R.1

**The capacitance of an isolated sphere is  $C = 4 \pi \epsilon_0 R$ , so a larger sphere has a larger capacitance.**

Answer: **true**

True.  $R$  is the only variable in it, so capacitance rises directly with radius. Notice what is absent: no  $Q$  and no  $V$ , because they cancelled when  $C = Q/V$  was formed. That cancellation is the clearest proof in the course that capacitance depends on geometry alone.

4.23

MEDIUM

Reference sheet R.1

For  $C = \epsilon_r \frac{\epsilon_0 A}{d}$ , which change **INCREASES** the capacitance?

✓ Moving the plates closer together

Correct. Smaller  $d$ , larger  $C$ .

✗ Moving the plates further apart

Wrong, and the opposite.

✗ Reducing the plate area

Wrong.  $A$  is on top, so less area means less capacitance.

✗ Replacing the dielectric with vacuum

Wrong. Vacuum has  $\epsilon_r = 1$ , the smallest possible, so this reduces it.

✗ Raising the applied voltage

Wrong, and it is the classic misconception: no  $V$  appears in the formula at all.

$d$  is in the denominator, so a smaller gap gives a bigger capacitance. That is why real capacitors use very thin insulating layers. Area helps too, since  $A$  is on top, and so does a dielectric, since  $\epsilon_r$  is at least 1. Voltage does nothing at all: it is not in the formula.

4.24

MEDIUM

Reference sheet R.1

**Capacitors in parallel add directly,  $C = C_1 + C_2$ , which is backwards from how resistors behave in parallel.**

Answer: **true**

True. Capacitors add in PARALLEL and combine reciprocally in series; resistors do exactly the opposite. Physically, putting capacitors side by side is just building one larger plate, so the capacitances sum. The reliable way to remember is to learn one of the four cases properly and flip it for the other.

4.25

MEDIUM

Lecture slides, module 5, slide 30

**Permittivity, symbol  $\epsilon$ , describes how readily a material permits an electric field, while permeability, symbol  $\mu$ , describes how readily it permits a magnetic field.**

Answer:

A magnetic field. The two names are deliberately parallel and so are their roles:  $\epsilon_0$  appears in every electrostatic formula,  $\mu_0$  in every magnetic one. Their product is what gives the speed of light, which is how the two subjects turned out to be one.

Determine the value of the capacitance if the radius of the spherical shell is  $R = 100 \text{ cm}$ . Take  $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ .

✓  $1.1 \times 10^{-10} \text{ F}$

Correct.  $4 \pi$  times  $8.85 \times 10^{-12} \times 1 \text{ metre}$ .

✗  $8.85 \times 10^{-12} \text{ F}$

Wrong. That is  $\epsilon_0$  itself, with the  $4 \pi R$  never applied.

✗  $1.1 \times 10^{-8} \text{ F}$

Wrong by a factor of 100, which is what leaving  $R$  as 100 cm instead of 1 m does.

✗  $3.5 \times 10^{-11} \text{ F}$

Wrong. This uses  $\pi$  rather than  $4 \pi$ .

✗  $1.1 \times 10^{-12} \text{ F}$

Wrong by a factor of 100 the other way.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Capacitance is not only a property of two facing plates. Any isolated conductor has one, because putting charge on it raises its potential, and capacitance is simply the ratio of the two. For a sphere the answer depends on nothing but its radius: a bigger sphere spreads the same charge over more surface, so its potential rises less, so its capacitance is larger.

**GIVEN**

- $R = 100 \text{ cm} = 1.0 \text{ m}$
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
- $\pi = 3.142$

**Find:** The capacitance  $C$  of the spherical shell, in farads

**FORMULA**

$$C = 4\pi\epsilon_0 R$$

**Why this one.** It follows from the two things already known about a sphere: its potential is  $V = kQ \div R$ , and capacitance is  $C = Q \div V$ . Divide one by the other and the  $Q$  cancels, leaving  $C = R \div k$ , which is the same as  $4\pi\epsilon_0 R$  because  $k = 1 \div (4\pi\epsilon_0)$ . There is no second plate and no gap, so the parallel plate formula cannot apply.

**WHAT EACH SYMBOL MEANS**

- $C$  = the capacitance, in farads
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
- $R$  = the radius of the sphere, in METRES
- $4\pi = 12.566$ , a pure number

**THE WORKING, LINE BY LINE**

1. Convert the radius:  $100 \text{ cm} = 1.0 \text{ m}$
2.  $C = 4 \times \pi \times \epsilon_0 \times R$
3.  $4\pi = 12.566$
4.  $C = 12.566 \times 8.85 \times 10^{-12} \times 1.0$
5. Front numbers:  $12.566 \times 8.85 = 111.2$
6.  $C = 111.2 \times 10^{-12} = 1.112 \times 10^{-10} \text{ F}$
7.  $C \approx 1.1 \times 10^{-10} \text{ F}$ , which is 111 pF

**CHECK IT**

A one metre sphere, which is enormous, manages only about a hundred picofarads. That is worth remembering as a feel for the size of things: capacitance is hard to come by from geometry alone, which is exactly why real capacitors use large rolled-up areas and gaps measured in microns. The option  $8.85 \times 10^{-12}$  is what you get by forgetting the  $4\pi$  entirely.

Convert first: 100 cm is 1.00 m, not 100. Then  $C = 4 \pi \epsilon_0 R = 4 \times 3.1416 \times 8.85 \times 10^{-12} \times 1.00 = 1.11 \times 10^{-10} \text{ F}$ , which the slide gives as  $1.1 \times 10^{-10} \text{ F}$ . A whole metre of sphere yields only about a tenth of a nF, which is why real capacitors use closely spaced plates rather than spheres.

A parallel plate capacitor has capacitance  $8.85 \times 10^{-11}$  F. What is the capacitance if a dielectric material with dielectric constant of 3 is inserted between the plates?

✓  $2.7 \times 10^{-10}$  F

Correct. Three times the original, written in standard form.

✗  $2.95 \times 10^{-11}$  F

Wrong. This DIVIDES by 3, but a dielectric always increases capacitance, never reduces it.

✗  $8.85 \times 10^{-11}$  F

Wrong. That is the original capacitance, unchanged, but inserting a dielectric must change it.

✗  $2.7 \times 10^{-11}$  F

Wrong by a factor of ten:  $3 \times 8.85$  is 26.55, so the exponent moves from -11 to -10.

✗  $1.77 \times 10^{-10}$  F

Wrong. This multiplies by 2 rather than by the dielectric constant of 3 that the question gives.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A dielectric is an insulator slipped between the plates, and it improves a capacitor without any change of shape. Its molecules turn to face the field and set up a small field of their own pointing the other way, which partly cancels the original. Less net field means less voltage for the same charge, and less voltage for the same charge means more capacitance. The dielectric constant is simply the factor by which it helps.

#### GIVEN

- $C_0 = 8.85 \times 10^{-11}$  F, the capacitance with nothing between the plates
- $K = 3$ , the dielectric constant of the material inserted

**Find:** The new capacitance  $C$  with the dielectric in place

#### FORMULA

$$C = KC_0$$

**Why this one.** The geometry is untouched, since neither the plate area nor the gap changes, so the only new ingredient is the material. Its whole effect is one multiplying factor. This is quicker and safer than rebuilding the capacitance from  $\epsilon_0 A \div d$ , and it works whatever the shape of the capacitor.

#### WHAT EACH SYMBOL MEANS

- $C$  = the capacitance with the dielectric, in farads
- $K$  = the dielectric constant, sometimes written  $\epsilon_r$ , a pure number with no unit
- $C_0$  = the original capacitance in vacuum or air, in farads

#### THE WORKING, LINE BY LINE

1.  $C = K \times C_0$
2.  $C = 3 \times 8.85 \times 10^{-11}$
3. Front numbers:  $3 \times 8.85 = 26.55$
4.  $C = 26.55 \times 10^{-11}$
5. Tidy into standard form:  $C = 2.655 \times 10^{-10}$  F
6.  $C \approx 2.7 \times 10^{-10}$  F

#### CHECK IT

A dielectric can only INCREASE capacitance, because  $K$  is never less than 1 for any real material. So the answer must be bigger than  $8.85 \times 10^{-11}$ , which instantly kills the options offering  $2.95 \times 10^{-11}$  and  $2.7 \times 10^{-11}$ , both of which are smaller. The remaining care is tidying  $26.55 \times 10^{-11}$  into  $2.655 \times 10^{-10}$ , which is the same number.

A dielectric multiplies the capacitance:  $C = \epsilon_r$  times  $C_0 = 3 \times 8.85 \times 10^{-11} = 2.655 \times 10^{-10}$  F, which the slide rounds to  $2.7 \times 10^{-10}$  F. Note the exponent shifts from -11 to -10 because  $3 \times 8.85 = 26.55$ , which is 2.655 in standard form with the exponent raised by one.

4.28

HARD

Reference sheet R.1

**A capacitor is charged to a higher voltage. What happens to its capacitance?**

✓ Nothing, because capacitance is fixed by geometry rather than by Q or V

Correct. Geometry sets it; Q and V simply move together.

✗ It increases, because  $C = Q/V$  and Q has increased

Wrong. Q rises, but V rises with it, so the ratio holds.

✗ It decreases, because V is in the denominator

Wrong, for the same reason: the numerator rises too.

✗ It doubles

Wrong. Nothing about the capacitor has changed.

✗ It becomes zero

Wrong. Capacitance is a property of the object.

$C = Q/V$  is charge per volt, but the RATIO is set by the geometry: the plate area, the separation and the dielectric. Raising V raises Q in exact proportion, so the ratio does not move. For a parallel plate,  $C = \epsilon_r \epsilon_0 A/d$  contains no Q and no V at all.

4.29

HARD

Reference sheet R.1

**Why does the energy stored in a capacitor carry a factor of one half,  $W = \text{one half } CV \text{ squared}$ ?**

✓ Because the voltage climbed from zero as the capacitor charged, so the average voltage was half the final one

Correct, and it is why the identical half appears in the inductor formula.

✗ Because only half the charge is stored on each plate

Wrong. Both plates carry the full magnitude Q, one positive and one negative.

✗ Because half the energy is always lost as heat

Wrong. The formula is for stored energy, not for losses.

✗ Because the two plates share the energy equally

Wrong. The energy is in the field between them, not split between the plates.

✗ It is an arbitrary constant with no reason behind it

Wrong. It follows directly from the voltage rising linearly with charge.

The first charge to arrive crosses no potential difference; the last crosses the full V. The average is  $V/2$ , so the work done is Q times  $V/2$ , that is one half QV, which becomes one half CV squared. The same reasoning gives the half in one half LI squared for an inductor.

Determine the total capacitance of the capacitors given in Figure 1

✗ 5.1  $\mu\text{F}$

Wrong. No arrangement of 1, 5 and 8 gives 5.1.

✗ 3.43  $\mu\text{F}$

Wrong. This is all three in series:  $1/(1/1 + 1/5 + 1/8) = 0.784$ , or a mis-stated version of it. Series can never exceed the smallest capacitor, so it cannot be right here.

✗ 0.8  $\mu\text{F}$

Wrong, and it is the near miss: 0.833  $\mu\text{F}$  is C1 and C2 in series, the FIRST step only, with C3 forgotten.

✓ 8.8  $\mu\text{F}$

Correct. C1 in series with C2 gives  $5/6 = 0.833 \mu\text{F}$ , and in parallel with C3 that is  $0.833 + 8 = 8.8 \mu\text{F}$ .

✗ 14.0  $\mu\text{F}$

Wrong. This is all three added as though every one were in parallel:  $1 + 5 + 8 = 14$ .

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Capacitors combine the OPPOSITE way round to resistors, and the reason is physical rather than arbitrary. Put two capacitors side by side across the same two points and you have simply built one capacitor with more plate area, so the capacitance adds. String them one after the other and the charge has to cross both gaps in turn, which is like building one capacitor with a wider gap, so the capacitance falls. Wide gap, weak capacitor.

#### GIVEN

- $C_1 = 1 \mu\text{F}$  and  $C_2 = 5 \mu\text{F}$ , drawn one above the other in the SAME branch
- $C_3 = 8 \mu\text{F}$ , drawn in its own branch beside them
- Both branches connect the same top wire to the same bottom wire

**Find:** The total capacitance of the network, in  $\mu\text{F}$

#### FORMULA

$$1/C_s = 1/C_1 + 1/C_2, \text{ then } C = C_s + C_3$$

**Why this one.** Read the picture before reaching for a formula.  $C_1$  and  $C_2$  sit one after the other with no junction between them, which is the definition of SERIES, so their reciprocals add. That combined branch and  $C_3$  both run from the top wire to the bottom wire, sharing the same two ends, which is the definition of PARALLEL, so those simply add. Series first, parallel last: work from the inside of the picture outwards.

#### WHAT EACH SYMBOL MEANS

- $C_1, C_2$  = the two capacitors in series, in microfarads
- $C_s$  = their combined series value, always SMALLER than either of them
- $C_3$  = the capacitor in parallel with that branch
- $C$  = the total capacitance seen at the terminals

#### THE WORKING, LINE BY LINE

1. Series pair first:  $1/C_s = 1/C_1 + 1/C_2$
2.  $1/C_s = 1/1 + 1/5$
3. Common denominator:  $1/C_s = 5/5 + 1/5 = 6/5$
4. Turn it up the other way:  $C_s = 5/6 = 0.833 \mu\text{F}$
5. Now the parallel step:  $C = C_s + C_3$
6.  $C = 0.833 + 8$
7.  $C = 8.83 \approx 8.8 \mu\text{F}$

#### CHECK IT

Two guards, both instant. A series combination must come out SMALLER than the smallest capacitor in it, and 0.833 is duly under 1. And adding anything in parallel can only INCREASE the total, so the final answer must exceed 8. That single fact kills 5.1, 3.43 and 0.8 without any arithmetic at all. The remaining trap, 14.0, is what you get by adding all three as if they were parallel.

Read the picture before reaching for a formula.  $C_1$  and  $C_2$  sit one above the other in the SAME branch, so they are in series; that branch and  $C_3$  span the same two terminals, so the pair is in parallel with  $C_3$ . Series first:  $1/C_s = 1/1 + 1/5 = 6/5$ , so  $C_s = 5/6 = 0.833 \mu\text{F}$ . Then parallel adds:  $C = 0.833 + 8 = 8.83$ , which is  $8.8 \mu\text{F}$ . The clue that parallel comes last is that the answer must exceed 8, since adding a capacitor in parallel can only increase the total.

The area of each plate of a parallel plate capacitor is  $4 \times 10^{-2} \text{ m}^2$ . If the thickness of the dielectric placed in between the plates is  $10^{-3} \text{ m}$  and the relative permittivity of the dielectric is 7, then the capacitance of the capacitor is?

✗  $0.2478 \times 10^{-9} \text{ F}$

Wrong by a factor of ten, which is what you get from mishandling the  $10^{-2}$  area or the  $10^{-3}$  thickness.

✗  $2.478 \times 10^9 \text{ F}$

Wrong sign on the exponent.  $2.478 \times 10^9 \text{ F}$  is larger than any capacitor ever built.

✗  $4.77 \text{ F}$

Wrong. This drops the permittivity of free space entirely.

✓  $2.478 \times 10^{-9} \text{ F}$

Correct.  $C = \epsilon_0 \epsilon_r A / d = 2.478 \times 10^{-9} \text{ F}$ .

✗  $0.2478 \times 10^9 \text{ F}$

Wrong on both the digits and the sign of the exponent.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A parallel plate capacitor stores more charge when the plates are big, when the gap is small, and when the space between them is filled with an insulator rather than air. The insulator is called a dielectric, and it works because its molecules line up against the field and partly cancel it, which lowers the voltage for the same charge, and less voltage for the same charge means more capacitance. Its relative permittivity is simply the factor by which it improves things.

### GIVEN

- $A = 4 \times 10^{-2} \text{ m}^2$  (area of each plate)
- $d = 10^{-3} \text{ m}$  (thickness of the dielectric, which is the plate separation)
- $\epsilon_r = 7$  (relative permittivity of the dielectric, a pure number)
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$

**Find:** The capacitance  $C$ , in farads

### FORMULA

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

**Why this one.** Plate area and plate separation are given, which is the signature of the parallel plate formula, and a dielectric is named, which supplies the  $\epsilon_r$  factor. Without a dielectric the formula would be just  $\epsilon_0 A \div d$ ; the dielectric multiplies it by 7. Note that the thickness of the dielectric IS the gap, because the dielectric fills the gap completely.

### WHAT EACH SYMBOL MEANS

- $C$  = the capacitance, in farads (F)
- $\epsilon_0$  = the permittivity of free space,  $8.85 \times 10^{-12} \text{ F m}^{-1}$
- $\epsilon_r$  = the relative permittivity, how many times better than vacuum, no unit
- $A$  = the area of ONE plate, in  $\text{m}^2$
- $d$  = the separation of the plates, in metres

### THE WORKING, LINE BY LINE

1. Top, part one:  $\epsilon_0 \times \epsilon_r = 8.85 \times 10^{-12} \times 7 = 61.95 \times 10^{-12} = 6.195 \times 10^{-11}$
2. Top, part two:  $\times A = 6.195 \times 10^{-11} \times 4 \times 10^{-2}$
3. Front numbers:  $6.195 \times 4 = 24.78$
4. Powers of ten:  $10^{-11} \times 10^{-2} = 10^{-13}$
5. Top =  $24.78 \times 10^{-13} = 2.478 \times 10^{-12}$
6. Divide by  $d$ :  $C = (2.478 \times 10^{-12}) \div 10^{-3}$
7. Powers of ten:  $10^{-12} \div 10^{-3} = 10^{-9}$
8.  $C = 2.478 \times 10^{-9} \text{ F}$  (that is 2.478 nF)

### CHECK IT

The options are the same digits with the sign of the exponent flipped, so the exponent is the whole mark. A capacitance of  $2.478 \times 10^{-9} \text{ F}$  would be larger than every capacitor ever built; real ones live in the picofarad to microfarad range, so a negative exponent is the only physically possible answer. That reasoning alone eliminates two options and the 4.77 F.

$C = (\epsilon_0 \times \epsilon_r \times A) / d$ . Substituting:  $(8.854 \times 10^{-12} \times 7 \times 4 \times 10^{-2}) / 10^{-3}$ . Numerator:  $8.854 \times 10^{-12} \times 7 = 6.198 \times 10^{-11}$ , times  $4 \times 10^{-2} = 2.479 \times 10^{-12}$ . Dividing by  $10^{-3}$  gives  $2.478 \times 10^{-9} \text{ F}$ , that is about 2.5 nF. A capacitance in nF is the sanity check: a positive power of ten here would be an impossibly large capacitor.

**An insulated conducting sphere of shell radius  $R$  carries charge  $Q$ , and its outer surface is an equipotential. Determine the potential of the outer surface and the capacitance of the shell.**

✓  $V = Q/(4 \pi \epsilon_0 R)$  and  $C = 4 \pi \epsilon_0 R$

Correct, and the  $Q$  cancelling is the point: what remains depends only on the geometry.

✗  $V = Q/(4 \pi \epsilon_0 R^2)$  and  $C = 4 \pi \epsilon_0 R^2$

Wrong. Potential uses a plain  $R$ , not  $R^2$ .  $R^2$  belongs to the FIELD.

✗  $V = QR/(4 \pi \epsilon_0)$  and  $C = 4 \pi \epsilon_0 / R$

Wrong.  $R$  is in the denominator of the potential, not the numerator.

✗  $V = Q/(4 \pi \epsilon_0 R)$  and  $C = Q / (4 \pi \epsilon_0 R)$

Wrong. This forgets to divide, so it never cancels the  $Q$ , and a capacitance cannot depend on the charge stored.

✗  $V = 4 \pi \epsilon_0 QR$  and  $C = Q/R$

Wrong on both, and it inverts the roles of the constants.

For a conducting sphere all the charge sits on the surface and behaves, from outside, like a point charge at the centre. So the potential at the surface is  $V = Q/(4 \pi \epsilon_0 R)$ , which is just  $kQ/r$  with  $r = R$ . Then capacitance is  $C = Q/V$ , and the  $Q$  cancels:  $C = Q$  divided by  $Q/(4 \pi \epsilon_0 R) = 4 \pi \epsilon_0 R$ . Notice what is left. The charge has vanished, and the capacitance depends on the RADIUS alone, which is the whole lesson: capacitance is fixed by geometry. A larger sphere has a larger capacitance.

## 5 Current, Resistance and Networks

**Speed tip.** Resistivity questions hide the area: a wire of radius  $r$  has  $A = \pi r^2$ , not  $r$ . Halving the radius quarters the area and quadruples the resistance.

5.1

EASY

Reference sheet R.4

**A point where two or more conductors meet is a junction, also called a node. A closed conducting path is a  $\{\{1\}\}$ .**

Answer:

A loop. The three words matter because Kirchhoff's two laws are stated in terms of them: the CURRENT law applies at a junction and conserves charge, while the VOLTAGE law applies around a loop and conserves energy. A branch is a single element such as a resistor or source.

Given a circuit where R1, R2 and R3 are in series with a voltage source of 48 V, such that  $R1 = 7\ \Omega$ ,  $R2 = 6\ \Omega$  and  $R3 = 3\ \Omega$ , calculate the current that flows through R3?

✓ 3 A

Correct.  $R_{\text{total}} = 16\ \text{ohms}$ , so  $I = 48/16 = 3\ \text{A}$ , and that same current flows through R3.

✗ 6 A

Wrong. This is  $48/8$ , which no combination of these resistors gives.

✗ 4 A

Wrong. This is  $48/12$ , that is the sum of two of the three resistors.

✗ 12 A

Wrong, and it is the trap:  $48/3$  uses R3 alone, ignoring that current is set by the WHOLE series loop.

✗ 5 A

Wrong. No route to 5 A from 48 V and 16 ohms.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

In a series circuit there is only one path, so every electron that leaves the source must pass through every component in turn. That means the **CURRENT** is the same everywhere, while the voltage divides itself among the resistors in proportion to their size. Parallel circuits are the mirror image: same voltage everywhere, current divides. Knowing which quantity is shared is most of circuit analysis.

**GIVEN**

- $R_1 = 7\ \Omega$ ,  $R_2 = 6\ \Omega$ ,  $R_3 = 3\ \Omega$ , all in **SERIES**
- $V = 48\ \text{V}$

**Find:** The current through  $R_3$ , in amperes

**FORMULA**

$$R = R_1 + R_2 + R_3, \text{ then } I = \frac{V}{R}$$

**Why this one.** Series resistances add directly, because the current meets them one after the other and each takes its share of the voltage. Once the total resistance is known, Ohm's law gives the single current that flows round the whole loop, and that same current is what passes through  $R_3$ .

**WHAT EACH SYMBOL MEANS**

- $R$  = the total series resistance, in ohms
- $V$  = the source voltage, in volts
- $I$  = the current, the **SAME** through every component in a series loop

**THE WORKING, LINE BY LINE**

1. Total resistance:  $R = 7 + 6 + 3$
2.  $R = 16\ \Omega$
3. Ohm's law for the whole loop:  $I = V \div R$
4.  $I = 48 \div 16$
5.  $I = 3\ \text{A}$
6. Series means one path, so the current through  $R_3$  is that same  $3\ \text{A}$

**CHECK IT**

The question singles out  $R_3$ , which invites you to do something extra with the  $3\ \Omega$ , and there is nothing extra to do. In series the answer is identical for  $R_1$ ,  $R_2$  or  $R_3$ . Verify by adding the individual voltages:  $3 \times 7 = 21\ \text{V}$ ,  $3 \times 6 = 18\ \text{V}$ ,  $3 \times 3 = 9\ \text{V}$ , and  $21 + 18 + 9 = 48\ \text{V}$ , the full supply. The voltages account for themselves exactly.

In series the resistances add and the **SAME** current flows through every component, so the answer for  $R_3$  is the answer for the whole loop.  $R = 7 + 6 + 3 = 16\ \text{ohms}$ , and  $I = V/R = 48/16 = 3\ \text{A}$ . Naming  $R_3$  specifically is the trap: it invites you to divide 48 by 3.

**5.3****EASY**

Reference sheet R.1

**Electric current is charge per unit time,  $I = \frac{\Delta Q}{\Delta t}$ , and one ampere is one coulomb per  $\{1\}$ .**

$$\Delta t$$

Answer:

One second.  $1\ \text{A} = 1\ \text{C/s}$ , straight from the definition. It is why the CD-player question can go from a current and a time to a charge in one multiplication.

5.4

EASY

Reference sheet R.1

**Ohm's law is  $V = IR$ , and one ohm is one volt per amp.**

Answer: **true**

True on both. Cover the quantity you want and the formula rearranges itself: cover  $V$  and you see  $IR$ , cover  $I$  and you see  $V$  over  $R$ . The unit follows from the same relation, since  $R = V/I$  gives volts per amp.

5.5

EASY

Reference sheet R.1

**Resistors in series simply add, and the same current flows through every one of them.**

Answer: **true**

True on both, and the two facts are linked: there is only one path, so the current has nowhere else to go, and each resistor adds its own opposition to that single path. Test 1 question 12 depends on it, where naming  $R_3$  invites you to divide 48 by 3 when the answer is the whole loop's current.

5.6

EASY

Reference sheet R.4

**The ampere is the unit of current, and it equals one coulomb per second.**

Answer: **true**

True, and it is simply the definition  $I = \Delta Q$  over  $\Delta t$  read as units. The slides note that a 1 A current is fairly strong, and that phones and music players run on small fractions of an amp.

5.7

EASY

Reference sheet R.4

**In circuit language, what is a BRANCH?**

✓ A single element such as a resistor or a source

Correct. A branch is one element, such as a single resistor or a single source.

✗ A point where two or more conductors meet

Wrong. That is a junction or node.

✗ A closed conducting path

Wrong. That is a loop.

✗ Any path of zero resistance

Wrong. That is a short circuit.

✗ The whole circuit

Wrong. A circuit contains many branches.

A branch is one element. A junction, or node, is a meeting point, and a loop is a closed path. The three words are the vocabulary Kirchhoff's laws are written in: the current law works at junctions and the voltage law around loops.

5.8

EASY

Reference sheet R.4

**A material is called **ohmic** when it obeys Ohm's law, with its resistance constant at a fixed temperature.**

Answer:

Ohmic. Metal wires and ordinary resistors are ohmic; semiconductors are the important exception the slides name, because their resistance falls as they warm. The temperature qualification matters: even a metal stops being ohmic if you let it heat up.

Which of the following situations has the greatest electric current? (a) the flow of 6 C in 3 s, (b) the flow of 3 C in 1 s, (c) a person walking 8 m in 4 s.

✓ Situation (b), at 3 A

Correct. 3 C in 1 s is 3 A, beating the 2 A of situation (a).

✗ Situation (a), at 2 A

Wrong.  $\frac{6 \text{ C}}{3 \text{ s}}$  is 2 A, which is real current but less than 3 A.

✗ Situation (c), because the person is moving fastest

Wrong, and it is the trap. Speed is not current. A person is electrically neutral, so their net charge is essentially zero and the current is zero however fast they walk.

✗ All three carry the same current

Wrong. The three come to 2 A, 3 A and zero.

✗ Situation (a), at 6 A

Wrong. 6 C is the charge, not the current; it must be divided by the 3 seconds.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Current is the rate at which charge flows past a point, which is charge divided by time. The word RATE is what matters: a large charge moving slowly can carry less current than a small charge moving quickly. And current is about CHARGE, so anything that is not charge, however fast it moves, carries no current at all.

**GIVEN**

- Situation (a): 6 C in 3 s
- Situation (b): 3 C in 1 s
- Situation (c): a person walking 8 m in 4 s

**Find:** Which situation has the greatest electric current

**FORMULA**

$$I = \frac{Q}{t}$$

**Why this one.** Comparison questions are answered by computing each candidate on the same footing rather than by guessing from the raw numbers. Situation (a) has the larger charge, which invites the wrong answer, so the division must actually be done.

**WHAT EACH SYMBOL MEANS**

- $I$  = the current, in amperes, where  $1 \text{ A} = 1 \text{ C s}^{-1}$
- $Q$  = the charge that flows, in coulombs
- $t$  = the time it takes, in seconds

**THE WORKING, LINE BY LINE**

1. Situation (a):  $I = 6 \div 3 = 2 \text{ A}$
2. Situation (b):  $I = 3 \div 1 = 3 \text{ A}$
3. Situation (c): a walking person moves mass, not charge, so  $I = 0$
4. Compare: 3 A is the largest
5. The answer is situation (b), at 3 A

**CHECK IT**

Situation (a) has twice the charge of (b) and still loses, which is the whole point of the question: current is a rate, so the time matters as much as the charge. Situation (c) is there to test whether you noticed that metres per second is a speed and not a current. It would only carry current if the walker were charged.

Current is charge per time,  $I = \Delta Q$  over  $\Delta t$ . (a) gives 6 C over 3 s = 2 A. (b) gives 3 C over 1 s = 3 A. (c) is zero, because a walking person carries essentially no NET charge, so moving them from place to place transports no charge at all. The greatest is therefore (b) at 3 A, and the smallest is (c) at zero.

**A potential difference of 24 V is applied to a 150 ohm resistor. How much current flows through the resistor?**

✓ 0.16 A

Correct. 24 divided by 150.

✗ 6.25 A

Wrong, and upside down: this is 150 over 24.

✗ 3600 A

Wrong. This multiplies instead of dividing, which would be V times R.

✗ 1.6 A

Wrong by a factor of ten.

✗ 0.016 A

Wrong by a factor of ten the other way.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Ohm's law is the most-used relation in the whole of electricity, and it is worth seeing it as a statement about push and opposition. The voltage is the push, the resistance is the opposition, and the current is what results. Push harder and more flows; oppose more and less flows.

#### GIVEN

- $V = 24 \text{ V}$
- $R = 150 \text{ } \Omega$

**Find:** The current  $I$  through the resistor, in amperes

#### FORMULA

$$I = \frac{V}{R}$$

**Why this one.** A voltage and a resistance are given and a current is wanted, which is Ohm's law arranged with  $I$  on the left. There is only one component, so no combining of resistances is needed first.

#### WHAT EACH SYMBOL MEANS

- $I$  = the current, in amperes
- $V$  = the potential difference across the resistor, in volts
- $R$  = the resistance, in ohms

#### THE WORKING, LINE BY LINE

1.  $I = V \div R$
2.  $I = 24 \div 150$
3.  $I = 0.16 \text{ A}$

#### CHECK IT

The resistance in ohms is much bigger than the voltage in volts, so the current must come out well BELOW one amp. That single glance rules out 6.25 A and 3600 A before any arithmetic. Among the survivors, 0.016 differs from 0.16 by a factor of ten, so count the decimal places carefully:  $24 \div 150$  is a bit more than  $24 \div 240$ , which is 0.1.

Ohm's law, rearranged for the current:  $I = V \text{ over } R = 24 \text{ over } 150 = 0.16 \text{ A}$ . Cover the quantity you want in  $V = IR$  and the formula rearranges itself.

5.11

MEDIUM

Reference sheet R.1

Resistance is  $R = \frac{\rho L}{A}$ . Two copper wires carry the same current. Wire X is long and thin, wire Y is short and fat. Which has the greater resistance, and why?

✓ X, because resistance rises with length and falls with cross-sectional area

Correct on both halves of the formula.

✗ Y, because a fatter wire has more material to push through

Wrong, and backwards: more cross-section gives current more room, so less resistance.

✗ They are equal, since both are copper

Wrong. Same material means same  $\rho$ , but resistance also depends on shape.

✗ X, because copper resistivity depends on the shape

Right answer, wrong reason: resistivity does NOT depend on shape, only the resistance does.

✗ Y, because resistance rises with area

Wrong. Resistance falls as area rises.

L is on top and A underneath, so long raises resistance and wide lowers it. The hook is long and thin resists, short and fat does not, and the water analogy makes it obvious: a long narrow hose fights the flow, a short wide one barely does. Note that  $\rho$ , the resistivity, is a property of the MATERIAL alone and is the same for both wires because both are copper.

5.12

MEDIUM

Reference sheet R.4

Resistivity, symbol  $\rho$ , is a material's intrinsic opposition to current, measured in ohm  $\{\{1\}\}$ .

Answer:

Ohm metres. Read the unit off  $R = \rho L$  over A: rearranged,  $\rho = RA$  over L, which is ohms times metres squared divided by metres, giving ohm metres. Insulators have large resistivities and conductors small ones.

5.13

MEDIUM

Reference sheet R.4

A short circuit is a path of zero resistance placed across part of a circuit. What does it do to the current?

✓ It carries all the current, and the equivalent resistance of that parallel section becomes zero

Correct, and it follows directly from the parallel rule.

✗ It carries none of the current

Wrong, and exactly backwards: it takes all of it.

✗ It splits the current equally with the other branches

Wrong. Current divides in inverse proportion to resistance, so zero resistance takes everything.

✗ It stops the current entirely

Wrong. It increases the current, which is the hazard.

✗ It doubles the total resistance

Wrong. It reduces the resistance to zero.

Current takes the easiest path, and a parallel combination is always less than or equal to the smallest resistance in it, so a zero-resistance branch drags the whole section to zero. Everything in parallel with the short is bypassed and carries nothing. That is why a short circuit is dangerous: with almost no resistance left, the current rises until something melts.

5.14

MEDIUM

Reference sheet R.2

For  $V = IR$ ,  $\{\{1\}\}$  the one you want and the formula rearranges itself.

Answer:

Cover it. Hiding  $V$  leaves  $I$  times  $R$ ; hiding  $I$  leaves  $V$  over  $R$ ; hiding  $R$  leaves  $V$  over  $I$ . It is a memory aid for rearranging under pressure, not a physical law.

5.15

MEDIUM

Reference sheet R.1

A wire is made longer and thinner. What happens to its resistance?

✓ It rises, because  $R = \frac{\rho L}{A}$  means long and thin resists

Correct on both counts, and both changes push the same way here.

✗ It falls, because a longer wire carries more current

Wrong. Length is in the numerator, so more length means more resistance.

✗ It is unchanged, because resistivity is a property of the material

Wrong. Resistivity is fixed by the material, but RESISTANCE also depends on shape.

✗ It falls, because a smaller area means less material to resist

Wrong, and reversed: a smaller cross-section squeezes the current and raises the resistance.

✗ It rises only if the material is a semiconductor

Wrong. The geometry argument holds for any ohmic material.

$R = \rho L/A$ . Length is on top, so a longer wire resists more; area is underneath, so a thinner wire resists more too. Long and thin resists; short and fat does not. Resistivity  $\rho$  is the material property and does not change with shape.

5.16

MEDIUM

Reference sheet R.2

The hook for  $R = \frac{\rho L}{A}$  is that long and  $\{\{1\}\}$  resists, while short and fat does not.

Answer:

Thin. Length on top and area underneath, so both being unfavourable raises the resistance.

5.17

MEDIUM

Reference sheet R.2

The equivalent resistance of a parallel combination is always less than the smallest resistor in it.

Answer: **true**

True, and it is the fastest sanity check in circuit work. Every extra parallel branch gives the current another route, so the combination always conducts better than the best single branch.

5.18

MEDIUM

Reference sheet R.1

**What does each of Kirchhoff's laws express?**

✓ The current law conserves charge at junctions, and the voltage law conserves energy around loops

Correct. Charge cannot pile up at a junction, and a charge going once round a loop must gain and lose equal energy.

✗ The current law conserves energy, and the voltage law conserves charge

Wrong, and swapped.

✗ Both conserve charge

Wrong. Only the current law is about charge.

✗ Both conserve energy

Wrong. Only the voltage law is about energy.

✗ Neither is a conservation law; both are experimental rules with no deeper basis

Wrong. Both follow from conservation principles.

KCL is charge, KVL is energy. Current in equals current out at a junction because charge cannot pile up there; the voltages around a closed loop sum to zero because a charge returning to its starting point must have gained and lost equal energy.

5.19

MEDIUM

Reference sheet R.2

**Kirchhoff's current law is about charge and his voltage law is about {{1}}.**

Answer:

Energy. Junctions conserve charge, loops conserve energy: two words that keep the pair straight.

**A  $10\ \Omega$  resistor is connected to a battery of internal resistance  $5\ \Omega$ . The p.d. across the terminals of the battery is  $2\ \text{V}$ . Estimate the current flowing the resistor after a time of  $2\ \text{seconds}$ .**

✓ **0.2 A**

Correct.  $I = V/R = 2/10 = 0.2\ \text{A}$ , using the terminal p.d. across the external resistor.

✗ **0.1 A**

Wrong. This is  $2/(10 + 5 + 5)$  or similar double counting of the internal resistance.

✗ **0.3 A**

Wrong. No route to  $0.3\ \text{A}$  from  $2\ \text{V}$  and  $10\ \text{ohms}$ .

✗ **0.4 A**

Wrong. This would need  $4\ \text{V}$  across the resistor.

✗ **0.5 A**

Wrong. This uses the internal resistance as though it were the external one:  $2/5$  wrongly halved, or  $2\ \text{V}$  across  $4\ \text{ohms}$ .

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

The terminal p.d. of a cell is not measured inside the cell, it is what appears at the terminals, and that is the same voltage that sits across whatever is connected to them. So the moment a question gives you the terminal p.d. and the external resistor, Ohm's law on that resistor alone gives the current. The internal resistance has already had its effect: it is the reason the terminal p.d. is only 2 V instead of the full emf.

**GIVEN**

- $R = 10\ \Omega$  (the external resistor)
- $r = 5\ \Omega$  (internal resistance of the battery)
- $V = 2\ \text{V}$  (the p.d. ACROSS THE TERMINALS)
- $t = 2\ \text{s}$  (the time given in the question)

**Find:** The current  $I$  flowing through the  $10\ \Omega$  resistor, in amperes

**FORMULA**

$$I = \frac{V}{R}$$

**Why this one.** The voltage given is the terminal p.d., and the terminal p.d. is precisely what is across the external resistor, so the current follows from Ohm's law on that resistor by itself. The  $5\ \Omega$  is a distractor: dividing by 15 would answer a different question, the one where the EMF was 2 V rather than the terminal p.d. Reading which voltage you have been handed is the whole difficulty.

**WHAT EACH SYMBOL MEANS**

- $I$  = the current, in amperes
- $V$  = the terminal p.d., in volts, equal to the p.d. across  $R$
- $R$  = the external resistance, in ohms

**THE WORKING, LINE BY LINE**

1.  $I = V \div R$
2.  $I = 2 \div 10$
3.  $I = 0.2\ \text{A}$

**CHECK IT**

The 2 seconds never enters the calculation. In a steady d.c. circuit the current is the same at every instant, so a time can only matter if the question asks for charge ( $Q = It$ ) or energy. It asks for current, so the time is there to be ignored. Notice too that dividing by 15 would give 0.133 A, which is not among the options, further confirming the reading.

The terminal p.d. is the voltage that appears ACROSS the external resistor, so Ohm's law needs nothing else:  $I = V/R = 2/10 = 0.2\ \text{A}$ . Two pieces of the stem are deliberate distractions: the internal resistance is already accounted for by quoting the terminal p.d. rather than the emf, and the time of 2 seconds does not enter a steady-current calculation at all.

**5.21**

MEDIUM

Reference sheet R.1

Resistance from resistivity is  $R = \frac{\rho L}{A}$ , so a wire that is long and thin has a **higher** resistance than one that is

**short and fat.**

Answer:

Higher.  $L$  is on top so length raises it, and  $A$  is underneath so width lowers it. The water-hose picture makes it stick: a long narrow hose fights the flow, a short wide one hardly does.

5.22

MEDIUM

Reference sheet R.1

**Resistors in parallel combine reciprocally, and the answer is always  $\frac{1}{n}$  than the smallest resistor in the group.**

Answer:

Less. Adding another path can only make it easier for current to flow, never harder. It is a free check on every parallel answer: if your total is not below the smallest, you forgot to flip the reciprocal back at the end.

5.23

MEDIUM

Reference sheet R.1

**For exactly two resistors in parallel,  $R_{\text{equivalent}} = \frac{R_1 R_2}{R_1 + R_2}$ , product over sum.**

Answer: **true**

True, and it is worth memorising because two-resistor parallel groups appear constantly. It is only a shortcut for TWO: with three or more you must go back to the reciprocal sum. Checking it on 100 and 200 ohms gives  $\frac{20000}{300} = 66.7$  ohms, duly below the smaller.

5.24

MEDIUM

Reference sheet R.1

**In the voltage divider  $V_1 = \frac{R_1}{R_{\text{total}}} V$ , the resistor on top is the  $\frac{1}{n}$  one.**

Answer:

The WANTED one, the resistor whose voltage you are after. That is the opposite of the current divider, where the OTHER resistor goes on top. Getting them the wrong way round is the single commonest circuit error, and both wrong answers usually appear as options.

5.25

MEDIUM

Reference sheet R.1

**Kirchhoff's current law says the currents into a junction equal the currents out of it, which is conservation of  $\frac{1}{n}$ .**

Answer:

Charge. Nothing accumulates at a junction, so whatever arrives must leave. The partner law, the voltage law around a loop, is conservation of ENERGY. KCL is charge, KVL is energy, and keeping the pair straight is what makes the two-loop problems tractable.

5.26

MEDIUM

Reference sheet R.1

**Kirchhoff's voltage law says the sum of the potential differences around any closed loop is zero, and it expresses conservation of energy.**

Answer: **true**

True. Go all the way round a loop and you arrive back where you started, so the total change in potential must be nothing, exactly as walking a closed path on a hillside returns you to your starting height. The rises across sources balance the drops across resistors.

5.27

MEDIUM

Reference sheet R.4

**A semiconductor sits between a conductor and an insulator, and unlike a metal its resistance  $\frac{1}{n}$  as it heats.**

Answer:

Falls. Heating a metal makes its atoms jiggle and collide with electrons, raising resistance; heating a semiconductor frees more charge carriers, which wins. That backwards behaviour is why semiconductors are the standing exception to Ohm's law in this course.

**A short circuit has zero resistance, so it carries all the current and bypasses everything in parallel with it.**

Answer: **true**

True. A parallel combination can never exceed its smallest member, so a zero-resistance branch pulls the whole section to zero and everything beside it is bypassed. With almost no resistance left the current climbs until something fails, which is why a short circuit is a hazard rather than a curiosity.

The disk drive in a portable CD player is connected to a battery that supplies it with a current of 0.22 A. How many electrons pass through the disk drive in 4.5 s? Recall that the charge of an electron has magnitude  $e = 1.60 \times 10^{-19} \text{ C}$ .

✓  $6.2 \times 10^{18}$  electrons

Correct. 0.99 C divided by  $1.60 \times 10^{-19} \text{ C}$  per electron.

✗  $0.99 \times 10^{18}$  electrons

Wrong. 0.99 is the CHARGE in coulombs, the intermediate answer, not the number of electrons.

✗  $1.6 \times 10^{19}$  electrons

Wrong. This inverts the electronic charge instead of dividing by it.

✗  $3.1 \times 10^{18}$  electrons

Wrong, and about half: this comes from using  $3.2 \times 10^{-19}$  for the electronic charge.

✗  $9.9 \times 10^{19}$  electrons

Wrong by a factor of ten, a slip in the exponent when dividing.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Current is a flow of electrons, but it is measured in coulombs per second rather than electrons per second, because an electron carries an absurdly small charge. Converting between the two is always a two-step job: first find how many coulombs flowed, then ask how many electrons that many coulombs is. Trying to do it in one step is where the error creeps in.

#### GIVEN

- $I = 0.22 \text{ A}$
- $t = 4.5 \text{ s}$
- $e = 1.60 \times 10^{-19} \text{ C}$ , the charge on one electron

**Find:** The number of electrons  $n$  that pass through the drive

#### FORMULA

$$Q = It, \text{ then } n = Q / e$$

**Why this one.** Two formulas because there are two conversions. The first turns a current and a time into a charge, which is just the definition of current rearranged. The second turns that charge into a count, using the fact that charge comes in fixed lumps of size  $e$ . Neither can be skipped.

#### WHAT EACH SYMBOL MEANS

- $Q$  = the total charge that flowed, in coulombs
- $I$  = the current, in amperes
- $t$  = the time, in seconds
- $n$  = the number of electrons, a pure count
- $e$  = the elementary charge,  $1.60 \times 10^{-19} \text{ C}$

#### THE WORKING, LINE BY LINE

1. Step one, the charge:  $Q = I \times t$
2.  $Q = 0.22 \times 4.5 = 0.99 \text{ C}$
3. Step two, the count:  $n = Q \div e$
4.  $n = 0.99 \div (1.60 \times 10^{-19})$
5. Front numbers:  $0.99 \div 1.60 = 0.619$
6. Dividing by  $10^{-19}$  multiplies by  $10^{19}$ :  $n = 0.619 \times 10^{19}$
7.  $n = 6.19 \times 10^{18} \approx 6.2 \times 10^{18}$  electrons

#### CHECK IT

The option  $0.99 \times 10^{18}$  is a trap for stopping after step one and then decorating the charge with a power of ten. Notice the number  $6.2 \times 10^{18}$ : it is the same figure that appears in the charge-quantisation question, because  $6.24 \times 10^{18}$  electrons is what makes one coulomb. Since  $0.99 \text{ C}$  is very nearly one coulomb, the count had to come out near that value.

Two steps. First the charge:  $\Delta Q = I \Delta t = 0.22 \times 4.5 = 0.99 \text{ C}$ . Then the count:  $N = \Delta Q \text{ over } e = 0.99 / 1.60 \times 10^{-19} = 6.2 \times 10^{18}$  electrons. Worth memorising, because Test 1 question 10 hands you  $6.2 \times 10^{18}$  electrons and asks for the charge, which is this same calculation run backwards to  $0.99 \text{ C}$ .

**A battery with an emf of 1.5 V delivers a current of 0.44 A to a flashlight bulb for 64 s. Find the charge that passes through the circuit and the work done by the battery.**

✓ 28 C and 42 J

Correct. 28 C from  $I$  times  $t$ , then 42 J from  $Q$  times emf.

✗ 28 C and 19 J

Wrong on the work: this divides by the emf instead of multiplying.

✗ 0.44 C and 0.66 J

Wrong. This forgets to multiply by the 64 seconds.

✗ 96 C and 144 J

Wrong. This uses a current or a time larger than the ones given.

✗ 42 C and 28 J

Wrong, and it swaps the two answers over.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A battery does work on every coulomb it pushes round the circuit, and the emf says how much: 1.5 V means 1.5 joules given to each coulomb. So finding the total work is again a two-step job. First find how many coulombs went round, then multiply by the joules each one received. The energy did not come from nowhere; it came out of the chemicals inside the cell.

#### GIVEN

- $\text{emf} = 1.5 \text{ V}$
- $I = 0.44 \text{ A}$
- $t = 64 \text{ s}$

**Find:** The charge that passes, in coulombs, and the work done, in joules

#### FORMULA

$$Q = It, \text{ then } W = QV$$

**Why this one.** The current and the time give the charge, because current is charge per second. The charge and the voltage then give the energy, because voltage is energy per coulomb. Each formula is a definition read the other way round, which is why they chain together so neatly.

#### WHAT EACH SYMBOL MEANS

- $Q$  = the charge that circulates, in coulombs
- $I$  = the current, in amperes
- $t$  = the time, in seconds
- $W$  = the work done by the battery, in joules
- $V$  = the emf, the energy given per coulomb, in volts

#### THE WORKING, LINE BY LINE

1. Charge first:  $Q = I \times t$
2.  $Q = 0.44 \times 64 = 28.16 \approx 28 \text{ C}$
3. Work next:  $W = Q \times V$
4.  $W = 28.16 \times 1.5$
5.  $W = 42.24 \approx 42 \text{ J}$

#### CHECK IT

One option offers the same two numbers the other way round, 42 C and 28 J, so read your own labels before choosing. A quick guard: the emf is 1.5, which is more than 1, so the work in joules must be LARGER than the charge in coulombs. 42 is larger than 28, so the pairing is the right way round.

Charge first:  $\Delta Q = I \Delta t = 0.44 \times 64 = 28 \text{ C}$ . Then work:  $W = \Delta Q \text{ times emf} = 28 \times 1.5 = 42 \text{ J}$ . The insight the slide draws is that the more charge a battery moves, and the greater its emf, the more work it does, which is why a 12 V car battery delivering several amps does far more work than this 1.5 V flashlight cell.

**A 20 V source is applied across  $R_1 = 15$  kilohms and  $R_2 = 10$  kilohms in series. What is the voltage across  $R_2$ ?**

✓ 8.0 V

Correct. Ten twenty-fifths of 20 V.

✗ 12.0 V

Wrong, and it is the voltage across  $R_1$ : this puts the OTHER resistor on top, which is the current divider's rule, not the voltage divider's.

✗ 10.0 V

Wrong. 10 is the resistance in kilohms, not the voltage.

✗ 20.0 V

Wrong. That is the whole source voltage, which is shared between the two resistors.

✗ 1.33 V

Wrong. This divides rather than taking the ratio of the resistances.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

When resistors sit in series they share the supply voltage between them, and they share it in proportion to their resistance. The bigger resistor takes the bigger share, because the same current passes through both and  $V = IR$ . This is called a voltage divider, and it is how a circuit produces a smaller voltage from a larger one without a transformer.

#### GIVEN

- $V = 20 \text{ V}$
- $R_1 = 15 \text{ k}\Omega$
- $R_2 = 10 \text{ k}\Omega$ , in series with  $R_1$

**Find:** The voltage across  $R_2$ , in volts

#### FORMULA

$$V_2 = \frac{V R_2}{(R_1 + R_2)}$$

**Why this one.** The divider rule is the same physics as finding the current and multiplying by  $R_2$ , but written as one line so there is only one chance to slip. Read it as a fraction:  $R_2$  over the total is the SHARE that  $R_2$  takes, and that share of the supply voltage is what appears across it.

#### WHAT EACH SYMBOL MEANS

- $V_2$  = the voltage across  $R_2$ , in volts
- $V$  = the total supply voltage, in volts
- $R_2$  = the resistance you want the voltage across
- $R_1 + R_2$  = the total series resistance
- $1 \text{ k}\Omega = 1000 \Omega$ , but the k cancels here so it can stay

#### THE WORKING, LINE BY LINE

1. Total resistance:  $R_1 + R_2 = 15 + 10 = 25 \text{ k}\Omega$
2. The share taken by  $R_2$ :  $10 \div 25 = 0.4$
3.  $V_2 = 20 \times 0.4$
4.  $V_2 = 8.0 \text{ V}$

#### CHECK IT

$R_2$  is the smaller resistor, so it must take the SMALLER share, meaning under half of 20 V. 8.0 V is under 10, so it leans the right way; the option 12.0 V is the share belonging to  $R_1$ , offered for anyone who put the wrong resistance on top. Check the pair adds up:  $8 + 12 = 20 \text{ V}$ , the full supply, with nothing unaccounted for. Note the kilohms never had to be converted, because the same unit appears on top and bottom and cancels.

The voltage divider puts the WANTED resistor on top:  $V_2 = R_2 \text{ over } R \text{ total, times } V = 10 \text{ over } 25, \text{ times } 20 = 8.0 \text{ V}$ . Note the total resistance is 25 kilohms, and that the kilohms cancel in the ratio so there is no need to convert them.

5.32

MEDIUM

Lecture slides, module 3, slide 66

A 20 mA source feeds  $R_1 = 100$  ohms in parallel with  $R_2 = 200$  ohms. What is the value of the current in  $R_1$ ?

✓ 13.3 mA

Correct. Two thirds of 20 mA.

✗ 6.7 mA

Wrong. 6.7 mA is the current in  $R_2$ , the larger resistor's smaller share.

✗ 20 mA

Wrong. 20 mA is the total from the source, before dividing.

✗ 26.7 mA

Wrong, and impossible: a branch cannot carry more than the total entering the junction.

✗ 10 mA

Wrong. The split would only be equal if the two resistors were equal.

Current divider, other resistor on top:  $I_1 = R_2 \text{ over } R_1 + R_2$ , times  $I_{\text{total}} = 200 \text{ over } 300$ , times 20 = 13.3 mA.  $R_1$  is the smaller resistor so it takes the larger share, and the remaining 6.7 mA flows through  $R_2$ .

5.33

HARD

Reference sheet R.4

What happens to the resistance of a semiconductor as it heats, and what is doping?

✓ Its resistance FALLS as it heats, and doping is adding impurities to make it conduct

Correct on both, and the falling resistance is what makes it an exception to Ohm's law.

✗ Its resistance RISES as it heats, and doping is adding impurities to make it conduct

Wrong on the temperature. That is how a METAL behaves.

✗ Its resistance falls as it heats, and doping is removing impurities to purify it

Wrong on doping. It ADDS impurities on purpose.

✗ Its resistance is unaffected by temperature

Wrong. Temperature is precisely what changes it.

✗ Its resistance rises as it heats, and doping is a heat treatment

Wrong on both halves: the resistance falls rather than rises, and doping is the deliberate addition of impurities, not a heat treatment.

A semiconductor sits between conductor and insulator, and it behaves BACKWARDS from a metal: heating a metal wire raises its resistance, because the atoms jiggle more and collide with electrons, but heating a semiconductor FREES more charge carriers and its resistance falls. Doping means deliberately adding impurities, such as one arsenic atom per million silicon atoms, which frees electrons and makes the silicon conduct. That is an n-type semiconductor, n for the negative electrons carrying the current.

5.34

HARD

Reference sheet R.1

**You combine three resistors in parallel and get a value LARGER than the smallest of them. What has happened?**

✓ A mistake: parallel resistance is always less than the smallest, so the reciprocal was not flipped back

Correct, and this is a free check worth running on every parallel answer.

✗ Nothing: parallel resistance can be larger than the smallest resistor

Wrong. It is impossible.

✗ Nothing, provided the resistors are not equal

Wrong. The rule holds whether or not the resistors are equal.

✗ The resistors must have been in series instead

Wrong. Series is where resistances add; the question specifies parallel.

✗ The supply voltage was too low

Wrong. The combination does not depend on the supply.

Adding a parallel path always gives the current somewhere else to go, so the combination must conduct better than any single branch, which means a resistance below the smallest. Getting a larger value almost always means stopping at  $1/R$  without inverting to get  $R$ .

5.35

HARD

Reference sheet R.1

**In the current divider for two resistors, the resistor that appears on top of the fraction is the OTHER one, not the one whose current you want.**

Answer: **true**

True, and it is the mirror image of the voltage divider.  $I_1 = \frac{R_2}{R_1 + R_2}$ , times  $I$  total. The physical sense is that current prefers

the easier path, so the SMALLER resistor takes the LARGER share, which only works out if the other resistance sits on top.

A circuit consists of three resistors connected in series to a 24.0 V battery. The current in the circuit is 0.0320 A. Given that  $R_1 = 250.0$  ohms and  $R_2 = 150.0$  ohms, find the resistance of  $R_3$ .

✓ 350 ohms

Correct. 750 total – the 400 already accounted for.

✗ 750 ohms

Wrong. 750 ohms is the total of all three, not  $R_3$  alone.

✗ 400 ohms

Wrong. 400 ohms is  $R_1 + R_2$ , the part you subtract.

✗ 250 ohms

Wrong. That is  $R_1$  repeated.

✗ 1150 ohms

Wrong. This adds 750 to  $R_2$  rather than subtracting.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

In a series circuit the same current passes through every component, because there is only one path for it to take. That makes the whole string behave like one big resistor whose value is the simple total. So if you know the supply voltage and the current, Ohm's law hands you that total, and anything missing from the string can then be found by subtraction.

#### GIVEN

- $V = 24.0 \text{ V}$
- $I = 0.0320 \text{ A}$
- $R_1 = 250.0 \Omega$
- $R_2 = 150.0 \Omega$
- All three resistors are in SERIES

**Find:** The resistance of  $R_3$ , in ohms

#### FORMULA

$$R = \frac{V}{I}, \text{ then } R_3 = R - R_1 - R_2$$

**Why this one.** Working backwards is the trick here. Ohm's law applied to the whole circuit gives the total resistance, because in series the current through the whole loop is the current the battery supplies. Once the total is known, the unknown resistor is whatever is left after removing the two you were told.

#### WHAT EACH SYMBOL MEANS

- $R$  = the total resistance of the series string, in ohms
- $V$  = the battery voltage, in volts
- $I$  = the current, the SAME everywhere in a series loop
- $R_1, R_2, R_3$  = the three individual resistances

#### THE WORKING, LINE BY LINE

1. Total resistance from Ohm's law:  $R = V \div I$
2.  $R = 24.0 \div 0.0320$
3.  $R = 750 \Omega$
4. Subtract the known ones:  $R_3 = 750 - 250 - 150$
5.  $750 - 250 = 500$
6.  $R_3 = 500 - 150 = 350 \Omega$

#### CHECK IT

The 750 is deliberately offered as an option, so do not stop at the total. Verify forwards in one line:  $250 + 150 + 350 = 750$ , and  $24 \div 750 = 0.032 \text{ A}$ , which is the current given. That closes the loop and proves the answer.

First the whole circuit:  $R_{\text{equivalent}} = \text{emf over } I = 24.0 \text{ over } 0.0320 = 750 \text{ ohms}$ . In series the resistances simply add, so  $R_3 = 750 - 250 - 150 = 350 \text{ ohms}$ . Checking with the potential differences:  $V_1 = 0.0320 \times 250 = 8.00 \text{ V}$ ,  $V_2 = 0.0320 \times 150 = 4.80 \text{ V}$ ,  $V_3 = 0.0320 \times 350 = 11.2 \text{ V}$ , and  $8.00 + 4.80 + 11.2 = 24.0 \text{ V}$ , the battery voltage exactly.

Consider a circuit with three resistors,  $R_1 = 250.0$  ohms,  $R_2 = 150.0$  ohms and  $R_3 = 350.0$  ohms, connected in parallel with a  $24.0$  V battery. Find the total current supplied by the battery.

✓ 0.325 A

Correct, and it equals the three branch currents added together.

✗ 0.0320 A

Wrong. 0.0320 A is the current when these same resistors are in SERIES, which is the neighbouring example.

✗ 0.0960 A

Wrong. That is the branch current through  $R_1$  alone,  $\frac{24}{250}$ .

✗ 0.160 A

Wrong. That is the branch current through  $R_2$  alone,  $\frac{24}{150}$ .

✗ 0.0686 A

Wrong. That is the branch current through  $R_3$  alone,  $\frac{24}{350}$ .

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

In parallel, every resistor is connected straight across the battery, so every one of them gets the FULL supply voltage. They do not share it. What they share is the current, each taking as much as its own resistance allows, and the battery has to supply the total. That is why adding more parallel branches makes the circuit draw more current, not less.

**GIVEN**

- $R_1 = 250.0\ \Omega$ ,  $R_2 = 150.0\ \Omega$ ,  $R_3 = 350.0\ \Omega$ , all in PARALLEL
- $V = 24.0\ \text{V}$  across all of them

**Find:** The total current supplied by the battery, in amperes

**FORMULA**

$$I = I_1 + I_2 + I_3, \text{ where each } I = V / R$$

**Why this one.** There are two valid routes and this is the safer one. You could combine the resistances with reciprocals first and then divide once, but that means three reciprocals and a final flip, which is four chances to slip. Finding each branch current separately and adding uses only division and addition, and it also shows which branch carries most, which is often the next part of the question.

**WHAT EACH SYMBOL MEANS**

- $I$  = the total current drawn from the battery, in amperes
- $I_1, I_2, I_3$  = the currents in the three branches
- $V = 24.0\ \text{V}$ , the same across EVERY branch in parallel
- $R_1, R_2, R_3$  = the branch resistances, in ohms

**THE WORKING, LINE BY LINE**

1. Branch one:  $I_1 = 24.0 \div 250.0 = 0.0960\ \text{A}$
2. Branch two:  $I_2 = 24.0 \div 150.0 = 0.1600\ \text{A}$
3. Branch three:  $I_3 = 24.0 \div 350.0 = 0.0686\ \text{A}$
4. Add them:  $0.0960 + 0.1600 = 0.2560$
5.  $0.2560 + 0.0686 = 0.3246$
6.  $I \approx 0.325\ \text{A}$

**CHECK IT**

Every individual branch current is offered as a wrong option, so make sure you added rather than stopping at one of them. Two guards on the result: the total must be LARGER than the biggest single branch current, and here 0.325 comfortably exceeds 0.160. And compare with the series version of the same three resistors, which drew only 0.032 A, ten times less. Parallel always draws more.

1 over R equivalent =  $1/250 + 1/150 + 1/350 = 0.01352$ , so R equivalent = 73.96 ohms, and  $I = 24.0 \text{ over } 73.96 = 0.325\ \text{A}$ . Two free checks: 73.96 ohms is LESS than the smallest resistor, 150, as a parallel combination must be. And the branch currents  $0.0960 + 0.160 + 0.0686$  add back to 0.325 A, as they must.

The emf of the battery is 12 V and each resistor has a resistance of 450 ohms. Two of the resistors are in series with one another, and that pair is in parallel with the third. Find the current supplied by the battery.

✓ 0.040 A

Correct. 12 V across an equivalent 300 ohms.

✗ 0.0267 A

Wrong. This uses 450 ohms as the whole circuit.

✗ 0.0089 A

Wrong. This treats all three as being in series, giving 1350 ohms.

✗ 0.080 A

Wrong. This halves the resistance once too often.

✗ 0.027 A

Wrong, and a near miss on the 450 ohm error.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Most real circuits are neither purely series nor purely parallel, but a mixture, and the way through is always the same: find a group that is clearly one or the other, replace it with a single equivalent resistor, and repeat until only one resistor is left. Work from the inside out, and never try to see the whole circuit in one go.

**GIVEN**

- emf = 12 V
- Each resistor is 450  $\Omega$
- Two of them are in series with each other
- That pair is in parallel with the third

**Find:** The current supplied by the battery, in amperes

**FORMULA**

$$R = \frac{(R_1 R_2)}{(R_1 + R_2)}$$

**Why this one.** The series pair goes first, because it is the innermost group: two resistors one after the other simply add, giving 900  $\Omega$ . That 900  $\Omega$  branch and the remaining 450  $\Omega$  branch span the same two points, so they are in parallel, and the product-over-sum form is the quickest way to combine exactly two resistors. Ohm's law then finishes it.

**WHAT EACH SYMBOL MEANS**

- $R_1, R_2$  = the two BRANCHES being combined in parallel, here 900  $\Omega$  and 450  $\Omega$
- $R$  = their combined resistance, always SMALLER than either branch
- $I$  = the current from the battery, in amperes

**THE WORKING, LINE BY LINE**

1. Series pair first:  $450 + 450 = 900 \Omega$
2. Now parallel with the third:  $R = (900 \times 450) \div (900 + 450)$
3. Top:  $900 \times 450 = 405000$
4. Bottom:  $900 + 450 = 1350$
5.  $R = 405000 \div 1350 = 300 \Omega$
6. Ohm's law:  $I = V \div R$
7.  $I = 12 \div 300 = 0.040 \text{ A}$

**CHECK IT**

The parallel result must be smaller than both branches, and 300 is duly under 450 and under 900. Two wrong routes appear in the options: treating all three as one series string gives 1350  $\Omega$  and a current of 0.0089 A, while treating all three as parallel gives 150  $\Omega$  and 0.080 A. The correct answer sits between them, which is exactly where a mixed circuit should land.

Work outwards from the innermost group. The two in series give  $R + R = 2R = 900$  ohms. That 900 in parallel with the remaining 450 gives 1 over R equivalent =  $1/450 + 1/900 = 3/900$ , so R equivalent = 300 ohms, which is two thirds of 450. Then  $I = 12 \text{ over } 300 = 0.040 \text{ A}$ . The insight is that three 450 ohm resistors together come to only 300 ohms, less than one of them, because the parallel step dominates.

A 20 V source is applied across  $R_1 = 470\ \Omega$  and  $R_2 = 330\ \Omega$  in series. Using the voltage divider rule to find  $V_1$  and  $V_2$ , find the total power dissipated.

✓ 0.5 W

Correct.  $0.29 + 0.21$ .

✗ 0.29 W

Wrong. 0.29 W is the power in  $R_1$  alone.

✗ 0.21 W

Wrong. 0.21 W is the power in  $R_2$  alone.

✗ 1.0 W

Wrong. This is double the answer, from adding each power twice.

✗ 0.08 W

Wrong. This uses the wrong voltage across each resistor.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Power is the rate at which a resistor turns electrical energy into heat, and it has three forms,  $P = VI$ ,  $P = I^2R$  and  $P = V^2 \div R$ . They are the same equation with Ohm's law substituted in, so all three always agree. Choose whichever one matches what you have been given, since that avoids an intermediate step and an intermediate chance to slip.

**GIVEN**

- $V = 20 \text{ V}$
- $R_1 = 470 \text{ } \Omega$
- $R_2 = 330 \text{ } \Omega$ , in series

**Find:** The total power dissipated, in watts

**FORMULA**

$$P = \frac{V^2}{R}$$

**Why this one.** The supply voltage and the total resistance are both known, so this form gets there in one step. There is a shortcut worth seeing: the TOTAL power can be found from the total resistance directly, without ever computing  $V_1$  and  $V_2$  separately, because the two resistors share the same current and the powers simply add.

**WHAT EACH SYMBOL MEANS**

- $P$  = the total power dissipated as heat, in watts
- $V$  = the supply voltage, in volts, and it is SQUARED
- $R$  = the total series resistance, in ohms

**THE WORKING, LINE BY LINE**

1. Total resistance:  $R = 470 + 330 = 800 \text{ } \Omega$
2.  $P = V^2 \div R$
3.  $V^2 = 20^2 = 400$
4.  $P = 400 \div 800$
5.  $P = 0.5 \text{ W}$

**CHECK IT**

Verify the long way in two lines to prove the shortcut. The current is  $I = 20 \div 800 = 0.025 \text{ A}$ , so  $V_1 = 0.025 \times 470 = 11.75 \text{ V}$  and  $V_2 = 0.025 \times 330 = 8.25 \text{ V}$ , which correctly add to 20 V. Then  $P_1 = 11.75 \times 0.025 = 0.294 \text{ W}$  and  $P_2 = 8.25 \times 0.025 = 0.206 \text{ W}$ , and those add to 0.5 W. Both of those partial powers are sitting in the option list, so make sure you gave the TOTAL.

The total resistance is 800 ohms.  $V_1 = 470 \text{ over } 800$ , times 20 = 11.75 V, and  $V_2 = 330 \text{ over } 800$ , times 20 = 8.25 V, and those two add back to 20 V as a check. Then  $P = V^2 \text{ over } R$  for each:  $P_1 = 11.75 \text{ squared over } 470 = 0.29 \text{ W}$ ,  $P_2 = 8.25 \text{ squared over } 330 = 0.21 \text{ W}$ . Total 0.5 W.

**R1 is a 2.2 kilohm resistor in parallel with R2, which is 4.7 kilohms. If the total current into the resistors is 8.0 mA, what is the current in R1?**

✓ 5.45 mA

Correct, and it is the larger share because R1 is the smaller resistor.

✗ 2.55 mA

Wrong. 2.55 mA is the current in R2, found by putting R1 on top instead.

✗ 4.0 mA

Wrong. The current only splits equally when the resistors are equal.

✗ 8.0 mA

Wrong. 8.0 mA is the total entering the pair, before it divides.

✗ 1.70 mA

Wrong. No route to this from these numbers.

The current divider puts the OTHER resistor on top, which is the opposite of the voltage divider:  $I_1 = R_2 \text{ over } R_1 + R_2, \text{ times } I_{\text{total}} = 4.7 \text{ over } 6.9, \text{ times } 8.0 = 5.45 \text{ mA}$ . The remaining 2.55 mA goes through R2, and the two add back to 8.0 mA. Notice the LARGER resistor carries the SMALLER current, which is the sanity check for every current divider.

**A 20 mA source feeds  $R_1 = 100$  ohms in parallel with  $R_2 = 200$  ohms. Find the voltage across  $R_2$ .**

✓ 1.33 V

Correct, whichever of the two routes you take.

✗ 4.0 V

Wrong. This puts the whole 20 mA through the 200 ohm resistor, but the current divides.

✗ 0.67 V

Wrong. This uses the 100 ohm resistor with the wrong branch current.

✗ 0 V

Wrong. Only a short circuit of zero resistance would have no voltage across it.

✗ 2.0 V

Wrong. This uses 100 ohms with the whole source current.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

A current source is the mirror image of a voltage source: it insists on pushing a fixed current out, and lets the voltage settle wherever the circuit requires. So the reasoning runs the other way round from usual. Instead of starting with a voltage and finding the current, you start with the current, find the resistance it faces, and multiply.

**GIVEN**

- $I = 20 \text{ mA} = 0.020 \text{ A}$ , from a current source
- $R_1 = 100 \text{ } \Omega$  and  $R_2 = 200 \text{ } \Omega$ , in PARALLEL

**Find:** The voltage across  $R_2$ , in volts

**FORMULA**

$$R = \frac{(R_1 R_2)}{(R_1 + R_2)}$$

**Why this one.** The whole 20 mA has to pass through the parallel combination, so the voltage that appears across it is the source current times the combined resistance. And because the two resistors are in parallel, that same voltage sits across BOTH of them, so once you have it for the pair you have it for  $R_2$  without any further work.

**WHAT EACH SYMBOL MEANS**

- $R$  = the combined parallel resistance, in ohms
- $R_1, R_2$  = the two resistors, in ohms
- $I$  = the source current, in amperes
- $V$  = the voltage across the pair, which is the voltage across each

**THE WORKING, LINE BY LINE**

1. Convert the current:  $20 \text{ mA} = 0.020 \text{ A}$
2. Combine in parallel:  $R = (100 \times 200) \div (100 + 200)$
3. Top:  $100 \times 200 = 20000$
4. Bottom:  $100 + 200 = 300$
5.  $R = 20000 \div 300 = 66.67 \text{ } \Omega$
6.  $V = I \times R = 0.020 \times 66.67$
7.  $V = 1.33 \text{ V}$

**CHECK IT**

In parallel the voltage is common, so the answer for  $R_2$  is identical to the answer for  $R_1$ ; if you found yourself splitting the current first, you took a longer road to the same place. Guard the parallel step:  $66.67 \text{ } \Omega$  is smaller than both 100 and 200, as any parallel combination must be. Using  $300 \text{ } \Omega$  instead, by adding as if in series, would give 6 V, and using only  $R_2$  would give 4 V, which is duly offered as a trap.

Two routes, and both give the same answer. Either combine first:  $R_{\text{equivalent}} = 100 \times 200 \text{ over } 300 = 66.7 \text{ ohms}$ , so  $V = 20 \text{ mA} \times 66.7 = 1.33 \text{ V}$ . Or use the branch current from the previous question: the 6.7 mA through  $R_2$  gives  $V = 6.7 \text{ mA} \times 200 = 1.33 \text{ V}$ . Parallel components share the same voltage, so this is the voltage across  $R_1$  as well.

A 10 A source feeds an 8 ohm resistor carrying  $I_1$  in parallel with a 4 ohm resistor carrying  $I_2$ . Using Kirchhoff's laws, find  $I_1$  and the voltage across the pair.

✓  $I_1 = 3.33 \text{ A}$  and  $V = 26.67 \text{ V}$

Correct, and both branches agree on the voltage.

✗  $I_1 = 6.67 \text{ A}$  and  $V = 26.67 \text{ V}$

Wrong. 6.67 A is  $I_2$ , the current in the 4 ohm resistor.

✗  $I_1 = 5.00 \text{ A}$  and  $V = 40 \text{ V}$

Wrong. The current only splits equally when the resistances are equal.

✗  $I_1 = 3.33 \text{ A}$  and  $V = 13.3 \text{ V}$

Wrong on the voltage: 13.3 V would need a 4 ohm resistor carrying  $I_1$ .

✗  $I_1 = 10 \text{ A}$  and  $V = 80 \text{ V}$

Wrong. The whole 10 A cannot flow through one branch when there are two paths.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Kirchhoff's two laws are just conservation restated for circuits. The junction law says charge cannot pile up at a junction, so whatever flows in must flow out. The loop law says energy cannot be created going round a loop, so the voltage rises and drops must cancel. For two resistors fed by a current source, both laws are visible at once: the currents split at the junction, and both resistors share one voltage.

#### GIVEN

- Source current: 10 A total
- $R_1 = 8\ \Omega$  carrying  $I_1$
- $R_2 = 4\ \Omega$  carrying  $I_2$
- The two resistors are in parallel

**Find:** The current  $I_1$  in the  $8\ \Omega$  resistor, and the voltage  $V$  across the pair

#### FORMULA

$$R = \frac{(R_1 R_2)}{(R_1 + R_2)}$$

**Why this one.** Find the voltage first and the currents come free. The 10 A must cross the parallel combination, so  $V = IR$  gives the shared voltage in one step, and then each branch current is that voltage divided by its own resistance. Trying to split the current first means using the current divider rule, which is easy to get upside down.

#### WHAT EACH SYMBOL MEANS

- $R$  = the combined parallel resistance, in ohms
- $V$  = the voltage across the pair, common to both branches
- $I_1$  = the current in the  $8\ \Omega$  branch, in amperes
- $I_2$  = the current in the  $4\ \Omega$  branch, in amperes

#### THE WORKING, LINE BY LINE

1. Combine:  $R = (8 \times 4) \div (8 + 4) = 32 \div 12 = 2.667\ \Omega$
2. Shared voltage:  $V = 10 \times 2.667$
3.  $V = 26.67\ \text{V}$
4. Branch one:  $I_1 = V \div 8 = 26.67 \div 8$
5.  $I_1 = 3.33\ \text{A}$
6. Branch two, for the check:  $I_2 = 26.67 \div 4 = 6.67\ \text{A}$

#### CHECK IT

Kirchhoff's junction law is the check, and it costs one line:  $I_1 + I_2 = 3.33 + 6.67 = 10\ \text{A}$ , which is exactly what the source supplies, so nothing has been lost or invented. A second guard by eye: the SMALLER resistor must take the BIGGER current, so the  $4\ \Omega$  branch should carry more than the  $8\ \Omega$  branch, and 6.67 duly exceeds 3.33. The option offering 6.67 A for  $I_1$  has the two branches swapped.

Both resistors sit across the same two nodes, so by the voltage law  $I_1 \times 8 = I_2 \times 4$ , which gives  $I_2 = 2 I_1$ : the smaller resistor takes twice the current. By the current law  $10 = I_1 + I_2 = I_1 + 2 I_1 = 3 I_1$ , so  $I_1 = 3.33\ \text{A}$  and  $I_2 = 6.67\ \text{A}$ . The voltage is then  $3.33 \times 8 = 26.67\ \text{V}$ , and checking on the other branch,  $6.67 \times 4 = 26.67\ \text{V}$  too.

A 50 V source drives a circuit with a 10 ohm resistor carrying  $I_1$ , an 8 ohm resistor carrying  $I_2$ , and a 6 ohm and 4 ohm resistor in series carrying  $I_3$ . Traversing anti-clockwise, find  $I_3$ .

✓ 1.538 A

Correct. 50 volts divided by 32.5 ohms.

✗ 1.25 A

Wrong. 1.25 is the RATIO of  $I_2$  to  $I_3$ , not a current in amps.

✗ 2.5 A

Wrong. This drops one of the two loop equations.

✗ 5.0 A

Wrong. This puts 50 V across the 10 ohm resistor alone.

✗ 0.65 A

Wrong. This inverts the final division.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

When a circuit has more than one loop, Kirchhoff's laws are the general tool, but a circuit that is only series and parallel underneath can be unpicked more quickly by reduction. Collapse the parallel part into one resistor, find the total current, then expand back out to recover the individual branch currents. Reduce, solve, expand.

**GIVEN**

- $V = 50 \text{ V}$
- $R_1 = 10 \text{ } \Omega$ , carrying  $I_1$ , in series with the rest
- $R_2 = 8 \text{ } \Omega$ , carrying  $I_2$
- $6 \text{ } \Omega$  and  $4 \text{ } \Omega$  in series carrying  $I_3$ , forming a branch parallel to  $R_2$

**Find:** The current  $I_3$ , in amperes

**FORMULA**

$$R_p = \frac{(R_2 R_3)}{(R_2 + R_3)}$$

**Why this one.** The  $6 \text{ } \Omega$  and  $4 \text{ } \Omega$  sit one after the other in the same branch, so they add to a single  $10 \text{ } \Omega$  before anything else happens. That branch and the  $8 \text{ } \Omega$  branch span the same two nodes, so they combine in parallel. What is left is the  $10 \text{ } \Omega$  of  $R_1$  in series with that combination, which is a single loop Ohm's law can finish.

**WHAT EACH SYMBOL MEANS**

- $R_3 = 6 + 4 = 10 \text{ } \Omega$ , the series branch treated as one resistor
- $R_p$  = the parallel combination of  $R_2$  and  $R_3$ , in ohms
- $I_1$  = the total current from the source, in amperes
- $I_3$  = the current in the  $6 \text{ } \Omega$  and  $4 \text{ } \Omega$  branch, which is what we want

**THE WORKING, LINE BY LINE**

1. Collapse the series branch:  $R_3 = 6 + 4 = 10 \text{ } \Omega$
2. Parallel it with  $R_2$ :  $R_p = (8 \times 10) \div (8 + 10) = 80 \div 18 = 4.444 \text{ } \Omega$
3. Total round the loop:  $R = R_1 + R_p = 10 + 4.444 = 14.444 \text{ } \Omega$
4. Total current:  $I_1 = 50 \div 14.444 = 3.462 \text{ A}$
5. Voltage across the parallel part:  $V_p = I_1 \times R_p = 3.462 \times 4.444$
6.  $V_p = 15.385 \text{ V}$
7. Expand back out:  $I_3 = V_p \div R_3 = 15.385 \div 10$
8.  $I_3 = 1.538 \text{ A}$

**CHECK IT**

Recover the other branch and test the junction:  $I_2 = 15.385 \div 8 = 1.923 \text{ A}$ , and  $I_2 + I_3 = 1.923 + 1.538 = 3.461 \text{ A}$ , which matches the total current  $I_1$  to three decimals. Sense check as well: the  $10 \text{ } \Omega$  branch has more resistance than the  $8 \text{ } \Omega$  branch, so it must carry LESS current, and  $1.538$  is duly below  $1.923$ .

Three equations, three unknowns. First loop, voltage law: minus  $50 + 10 I_1 + 8 I_2 = 0$ . Second loop: minus  $8 I_2 + 6 I_3 + 4 I_3 = 0$ , which gives  $8 I_2 = 10 I_3$ , so  $I_2 = 1.25 I_3$ . Current law at the junction:  $I_1 = I_2 + I_3$ . Substituting into the first loop gives  $18 I_2 + 10 I_3 = 50$ , then replacing  $I_2$  gives  $1.25 I_3 \times 18 + 10 I_3 = 50$ , that is  $22.5 I_3 + 10 I_3 = 32.5 I_3 = 50$ , so  $I_3 = 50 \text{ over } 32.5 = 1.538 \text{ A}$ .

---

## 6 EMF and Internal Resistance

**Speed tip.** Terminal voltage is always LESS than the emf, by exactly  $Ir$ . If your answer exceeds the emf you added where you should have subtracted.

A cell has an e.m.f. of 9 V and an internal resistance of 1  $\Omega$ . If the current flowing in the circuit is 2 A, what is the terminal voltage of the cell?

✓ 7 V

Correct.  $V = E - Ir = 9 - 2 = 7$  V.

✗ 5 V

Wrong. This subtracts twice the internal drop, as if  $r$  were 2 ohms.

✗ 9 V

Wrong. 9 V is the emf, which is only the terminal voltage when no current flows.

✗ 11 V

Wrong, and it ADDS the internal drop. Terminal voltage can never exceed the emf.

✗ 4 V

Wrong. No route to 4 V from a 9 V cell dropping 2 V internally.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

The terminal voltage is what a voltmeter across the battery would actually read while current is flowing. It is always less than the emf, because part of the push is spent driving current through the cell's own internal resistance. Open the circuit so no current flows and the loss vanishes, which is why a flat battery can still read its full voltage until you put it under load.

### GIVEN

- $E = 9 \text{ V}$  (the emf)
- $r = 1 \Omega$  (the internal resistance)
- $I = 2 \text{ A}$  (the current in the circuit)

**Find:** The terminal voltage  $V$  of the cell, in volts

### FORMULA

$$V = E - Ir$$

**Why this one.** The emf is given and the terminal voltage is wanted, so this is the subtraction form, not  $E = I(R + r)$ . The lost volts are  $Ir$ , and they are taken away because they are consumed inside the cell before anything reaches the terminals. Watching the sign is the whole question.

### WHAT EACH SYMBOL MEANS

- $V$  = the terminal voltage, what the external circuit gets, in volts
- $E$  = the emf, the total push, in volts
- $I$  = the current, in amperes
- $r$  = the internal resistance, in ohms
- $Ir$  = the volts lost inside the cell

### THE WORKING, LINE BY LINE

1. Lost volts:  $I \times r = 2 \times 1 = 2 \text{ V}$
2.  $V = E - Ir$
3.  $V = 9 - 2$
4.  $V = 7 \text{ V}$

### CHECK IT

The answer must be below 9 V, always. One of the options is 11 V, which is the trap for anyone who added instead of subtracting, and 9 V is the trap for anyone who forgot the internal resistance entirely. Both are impossible: a cell cannot deliver more than it generates.

Terminal voltage is the emf – what is lost inside the cell:  $V = E - Ir = 9 - (2 \times 1) = 7 \text{ V}$ . This is the mirror image of the previous question: there the terminal p.d. was given and the emf wanted, here the emf is given and the terminal p.d. wanted. The terminal voltage is always the **SMALLER** of the two.

6.2

MEDIUM

Reference sheet R.1

**Why do a car's headlights dim at the moment the engine is started?**

- ✓ The starter draws a large current, so the internal drop  $Ir$  grows and the terminal voltage  $V = E - Ir$  falls

Correct. It is  $I$  that changes, and  $Ir$  with it.

- ✗ The emf of the battery falls

Wrong. The emf is the chemical property of the cell and does not drop like that.

- ✗ The internal resistance of the battery rises sharply

Wrong. The internal resistance is roughly constant; the current is what surges.

- ✗ The headlights are switched off automatically

Wrong. Nothing switches them off.

- ✗ The alternator reverses polarity

Wrong. No polarity reversal is involved.

$V = E - Ir$ . The emf  $E$  and the internal resistance  $r$  stay much the same, but the starter motor pulls a very large current  $I$ , so the internal drop  $Ir$  becomes large and the terminal voltage available to the lamps falls. It is the everyday illustration of internal resistance.

6.3

MEDIUM

Reference sheet R.2

**The terminal voltage is the emf – the internal drop,  $V = E$  minus  $\{\{1\}\}$ .**

Answer:

$Ir$ , with the lower case  $r$  for the INTERNAL resistance. Capital  $R$  is the external resistance, and  $IR$  is the useful voltage delivered to the circuit.

In a circuit, the external resistor has a value of  $5\ \Omega$ . The internal resistance of the cell is  $0.5\ \Omega$ . If the current flowing is  $1\ \text{A}$ . Calculate the e.m.f of the cell.

✓ 5.5 V

Correct.  $E = I(R + r) = 1 \times 5.5 = 5.5\ \text{V}$ .

✗ 4.5 V

Wrong, and it SUBTRACTS the internal resistance instead of adding it, which would make the emf smaller than the terminal p.d.

✗ 3.5 V

Wrong. No route to 3.5 V from these numbers.

✗ 6.5 V

Wrong. This would need an internal resistance of 1.5 ohms.

✗ 7.5 V

Wrong. This would need an internal resistance of 2.5 ohms.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A cell is not a perfect source. The chemicals inside it have resistance of their own, so some of the push the cell generates is used up driving current through the cell itself, before any of it reaches the circuit outside. The full push is the emf; what is left over at the terminals is the terminal p.d. This is why a torch battery that works fine dims when you add a second bulb: more current means more of the emf is lost inside.

#### GIVEN

- $R = 5\ \Omega$  (the external resistor)
- $r = 0.5\ \Omega$  (the internal resistance of the cell)
- $I = 1\ \text{A}$  (the current flowing)

**Find:** The emf  $E$  of the cell, in volts

#### FORMULA

$$E = I(R + r)$$

**Why this one.** The emf has to push the current through EVERYTHING in the loop, and the current meets the outside resistor and the inside of the cell one after the other. Resistances met one after the other simply add, so the total the emf works against is  $R + r$ . This is just Ohm's law applied to the whole circuit rather than to one component.

#### WHAT EACH SYMBOL MEANS

- $E$  = the electromotive force, the full push of the cell, in volts
- $I$  = the current in the circuit, in amperes
- $R$  = the external or load resistance, in ohms
- $r$  = the internal resistance of the cell, in ohms

#### THE WORKING, LINE BY LINE

1. Add the two resistances first:  $R + r = 5 + 0.5 = 5.5\ \Omega$
2.  $E = I \times (R + r)$
3.  $E = 1 \times 5.5$
4.  $E = 5.5\ \text{V}$

#### CHECK IT

The emf must be LARGER than the terminal p.d., which here is  $IR = 1 \times 5 = 5\ \text{V}$ . It is, by exactly the  $0.5\ \text{V}$  lost inside the cell. If your emf ever comes out below  $IR$ , you subtracted where you should have added.

The emf drives the current through BOTH the external resistor and the cell's own internal resistance, so  $E = I(R + r) = 1 \times (5 + 0.5) = 5.5 \text{ V}$ . Equivalently  $E = IR + Ir$ , that is 5 V delivered to the circuit + 0.5 V lost inside the cell. The emf is always LARGER than the terminal p.d., which is the check.

6.5

MEDIUM

Reference sheet R.1

**For a source with internal resistance,  $V = E - Ir$ , so the terminal voltage is always **less** than the emf whenever current flows.**

Answer:

Less. The internal resistance eats  $Ir$  volts before anything reaches the terminals. It is why headlights dip when the starter motor draws a huge current, and it is a free check: any answer where the emf comes out below the terminal p.d. is wrong.

6.6

MEDIUM

Reference sheet R.4

**emf is the energy per unit charge supplied by a source, and despite the name electromotive FORCE it is measured in volts.**

Answer: **true**

True, and the name is a historical accident worth knowing about: emf is not a force at all, and it carries the unit of potential difference. Energy per unit charge is exactly what a volt measures, which is why  $E$  and  $V$  can be added and subtracted in  $V = E - Ir$ .

6.7

MEDIUM

Reference sheet R.4

**Internal resistance is the resistance of the source **itself**, and it is what causes the terminal voltage to fall below the emf.**

Answer:

Of the source itself. A battery is not a perfect supply: its chemistry and construction resist the current passing through it, and that resistance sits in series with everything outside. Capital  $R$  is the external resistance and small  $r$  the internal one, and swapping them fails every emf question.

6.8

MEDIUM

Reference sheet R.4

**The terminal voltage of a source is:**

✓ The voltage across it while current flows, always less than the emf

Correct. What you would measure at the terminals with current flowing, always below the emf by exactly  $Ir$ .

✗ The same as the emf under all conditions

Wrong while current flows, though it becomes true at zero current.

✗ The voltage across the internal resistance alone

Wrong. That is the internal DROP, the part you lose.

✗ Always greater than the emf

Wrong, and impossible: that would mean the internal resistance added energy.

✗ The voltage measured with the circuit disconnected

Wrong as a definition of terminal voltage, though disconnecting is indeed how you measure the emf.

What you would actually measure at the terminals with current flowing, and it is always below the emf by exactly  $Ir$ . There is an edge case worth knowing: with NO current flowing,  $Ir$  is zero and the terminal voltage does equal the emf, which is how an emf is measured in practice.

**While a battery is delivering current, the voltage measured at its terminals is lower than its emf.**

Answer: **true**

True. The current has to pass through the source's own internal resistance first, and that costs  $Ir$  volts, so what reaches the terminals is  $E - Ir$ . Draw a bigger current and you lose more, which is why a torch dims as its cells age and their internal resistance climbs.

In a circuit, the external resistor has value 5 ohms. The internal resistance of the cell is 0.5 ohms. If the current flowing is 1 A, calculate the emf of the cell.

✓ 5.5 V

Correct. 1 A driven through 5.5 ohms in total.

✗ 4.5 V

Wrong. This SUBTRACTS the internal resistance instead of adding it, but internal resistance costs voltage, it does not supply it.

✗ 5.0 V

Wrong. This uses only the external 5 ohms and forgets the cell has resistance of its own.

✗ 0.5 V

Wrong. That is the internal resistance alone, in ohms rather than volts.

✗ 6.5 V

Wrong. This adds a whole ohm too many.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A real cell is a perfect source of emf with a small resistance built in, and that internal resistance is not a defect to be ignored but the reason a battery's voltage sags under load. The emf must drive the current through the outside circuit AND through the cell itself, so the two resistances are met one after the other and simply add.

#### GIVEN

- $R = 5 \, \Omega$ , the external resistor
- $r = 0.5 \, \Omega$ , the internal resistance
- $I = 1 \, \text{A}$

**Find:** The emf  $E$  of the cell, in volts

#### FORMULA

$$E = I(R + r)$$

**Why this one.** This is Ohm's law applied to the complete loop rather than to one component. The current is the same everywhere in a single loop, so the total opposition it meets is  $R + r$ , and the emf is what pushes it through both.

#### WHAT EACH SYMBOL MEANS

- $E$  = the emf, the full push generated by the cell, in volts
- $I$  = the current in the loop, in amperes
- $R$  = the external or load resistance, in ohms
- $r$  = the internal resistance of the cell, in ohms

#### THE WORKING, LINE BY LINE

1. Add the resistances:  $R + r = 5 + 0.5 = 5.5 \, \Omega$
2.  $E = I \times (R + r)$
3.  $E = 1 \times 5.5$
4.  $E = 5.5 \, \text{V}$

#### CHECK IT

The emf must exceed the terminal p.d., which here is  $IR = 1 \times 5 = 5 \, \text{V}$ . It does, by exactly the 0.5 V lost inside. The option 5.0 V is that terminal p.d. offered as a trap for anyone who forgot the cell's own resistance. This same question was set again in test 2, which tells you the examiner likes it.

The emf has to drive the current through BOTH resistances, the external one and the cell's own.  $E = I(R + r) = 1 \times (5 + 0.5) = 5.5 \text{ V}$ . This slide is worth knowing by name: Test 2 question 3 is this question, word for word and number for number, so the examiner set the test straight from the deck.

A wire of 4 ohms is connected to a battery of emf 6 V and internal resistance  $r$ . The current flowing is 1 A. Calculate  $r$ .

✓ 2 ohms

Correct. The total resistance must be 6 ohms, and 4 of them are external.

✗ 6 ohms

Wrong. 6 is the TOTAL resistance, or the emf in volts, not the internal part.

✗ 1.5 ohms

Wrong. This divides 6 by 4 instead of subtracting.

✗ 10 ohms

Wrong. This ADDS 6 and 4 rather than subtracting.

✗ 0.5 ohms

Wrong. No route to 0.5 from these numbers.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Every question in this family is the same equation,  $E = I(R + r)$ , read with a different unknown. Here the emf and the current are known and the internal resistance is the mystery, so the algebra runs backwards, but the physics has not changed at all. Recognising the family saves you from learning four separate formulas.

#### GIVEN

- $R = 4 \Omega$ , the wire
- $E = 6 \text{ V}$
- $I = 1 \text{ A}$

**Find:** The internal resistance  $r$  of the battery, in ohms

#### FORMULA

$$E = I(R + r)$$

**Why this one.** One equation with one unknown, so it can simply be rearranged. Divide by the current to get the total resistance, then subtract the part you already know. That two-move pattern, total then subtract, is the same one used to find a missing resistor in a series string.

#### WHAT EACH SYMBOL MEANS

- $E$  = the emf, in volts
- $I$  = the current, in amperes
- $R$  = the external resistance, in ohms
- $r$  = the internal resistance, the unknown

#### THE WORKING, LINE BY LINE

1. Start from:  $E = I(R + r)$
2. Divide both sides by  $I$ :  $R + r = E \div I$
3.  $R + r = 6 \div 1 = 6 \Omega$
4. Subtract the known part:  $r = 6 - 4$
5.  $r = 2 \Omega$

#### CHECK IT

Substitute back:  $E = 1 \times (4 + 2) = 6 \text{ V}$ , which is the emf given, so the answer stands. The option  $6 \Omega$  is the TOTAL resistance offered as a trap for stopping one line early, and  $10 \Omega$  is what adding gives instead of subtracting.

Start from  $E = I(R + r)$  and rearrange in symbols before substituting:  $R + r = E/I$ , so  $r = E/I - R$ . Then  $r = 6/1 - 4 = 6 - 4 = 2$  ohms. Check it forwards: 1 A through  $4 + 2 = 6$  ohms needs exactly 6 V, which is the emf given.

A cell of unknown e.m.f  $E$ , and internal resistance  $2\ \Omega$  is connected to a  $5\ \Omega$  resistor. If the terminal p.d. is 1 volt, calculate the value of " $E$ " if a current of " $I$ " flows through the cell.

✓ 1.4 V

Correct.  $I = 1/5 = 0.2\text{ A}$ , then  $E = I(R + r) = 0.2 \times 7 = 1.4\text{ V}$ .

✗ 1.2 V

Wrong. This adds only half the internal drop, as if  $r$  were 1 ohm.

✗ 1.3 V

Wrong. No consistent route gives 1.3 V here.

✗ 1 V

Wrong, and it is the trap: 1 V is the TERMINAL p.d. given in the stem, not the emf. The emf must exceed it by the internal drop.

✗ 1.5 V

Wrong. This would need a current of 0.25 A, that is a 4 ohm external resistor.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

This is the same cell idea as before, read from the other end. You are told what the outside world actually receives, the terminal p.d., and asked to work back to the full push the cell was generating. The lost volts, the part burned inside the cell, are exactly  $Ir$ . So emf = what got out + what was lost inside.

#### GIVEN

- $r = 2\ \Omega$  (internal resistance)
- $R = 5\ \Omega$  (external resistor)
- $V = 1\text{ V}$  (terminal p.d., the voltage actually across the  $5\ \Omega$ )

**Find:** The emf  $E$  of the cell, in volts

#### FORMULA

$$E = V + Ir$$

**Why this one.** The terminal p.d. is given, not the current, so the circuit has to be entered through Ohm's law on the external resistor first: the terminal p.d. is what sits across  $R$ , so  $I = V \div R$ . Once the current is known, the volts lost inside are  $Ir$ , and adding them back to  $V$  rebuilds the emf.

#### WHAT EACH SYMBOL MEANS

- $E$  = the emf, the total push generated, in volts
- $V$  = the terminal p.d., what the outside circuit receives, in volts
- $I$  = the current, in amperes, found from  $V$  and  $R$
- $r$  = the internal resistance, in ohms
- $Ir$  = the lost volts, the part used up inside the cell

#### THE WORKING, LINE BY LINE

1. The terminal p.d. sits across  $R$ , so find  $I$  from Ohm's law:
2.  $I = V \div R = 1 \div 5 = 0.2\text{ A}$
3. Lost volts inside the cell:  $Ir = 0.2 \times 2 = 0.4\text{ V}$
4.  $E = V + Ir = 1 + 0.4$
5.  $E = 1.4\text{ V}$

#### CHECK IT

Cross-check with the other form:  $E = I(R + r) = 0.2 \times (5 + 2) = 0.2 \times 7 = 1.4\text{ V}$ . Two independent routes to the same number is the strongest check there is, and it costs one line.

Two steps. First the current, from the terminal p.d. across the external resistor:  $I = V/R = 1/5 = 0.2$  A. Then the emf, which must also push that current through the internal resistance:  $E = I(R + r) = 0.2 \times (5 + 2) = 0.2 \times 7 = 1.4$  V. Equivalently  $E = V + Ir = 1 + 0.2 \times 2 = 1.4$  V.

A cell of unknown emf  $E$  and internal resistance  $2\ \Omega$  is connected to a  $5\ \Omega$  resistance. If the terminal p.d.  $V$  is  $1\ \text{V}$ , calculate  $E$ .

✓  $1.4\ \text{V}$

Correct.  $1\ \text{V}$  at the terminals +  $0.4\ \text{V}$  lost inside the cell.

✗  $1.0\ \text{V}$

Wrong, and it is the trap:  $1\ \text{V}$  is the terminal p.d. copied straight out of the question. The emf must always be LARGER.

✗  $1.2\ \text{V}$

Wrong. This uses an internal drop of  $0.2\ \text{V}$  rather than  $0.4\ \text{V}$ .

✗  $1.5\ \text{V}$

Wrong. This would need an internal resistance of  $2.5\ \Omega$ .

✗  $1.3\ \text{V}$

Wrong. A near miss on the internal drop.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Read from the other end, the same cell idea asks you to rebuild the emf from what actually reached the outside world. The terminal p.d. is what got out; the volts lost inside are  $Ir$ ; add them and you have the full push. What makes it a step harder is that the current is not given, so it has to be found first.

### GIVEN

- $r = 2 \, \Omega$
- $R = 5 \, \Omega$
- $V = 1 \, \text{V}$ , the terminal p.d.

**Find:** The emf  $E$  of the cell, in volts

### FORMULA

$$E = V + Ir$$

**Why this one.** The terminal p.d. is the voltage across the external resistor, so Ohm's law on  $R$  alone gives the current. Only then can the lost volts be worked out. Jumping straight to  $E = I(R + r)$  is not wrong, but it needs the same current first, so nothing is saved.

### WHAT EACH SYMBOL MEANS

- $E$  = the emf, in volts
- $V$  = the terminal p.d., what the outside circuit receives, in volts
- $I$  = the current, found from  $V$  and  $R$
- $r$  = the internal resistance, in ohms
- $Ir$  = the volts lost inside the cell

### THE WORKING, LINE BY LINE

1. The terminal p.d. sits across  $R$ , so:  $I = V \div R$
2.  $I = 1 \div 5 = 0.2 \, \text{A}$
3. Volts lost inside:  $Ir = 0.2 \times 2 = 0.4 \, \text{V}$
4.  $E = V + Ir = 1 + 0.4$
5.  $E = 1.4 \, \text{V}$

### CHECK IT

Cross-check by the other route:  $E = I(R + r) = 0.2 \times 7 = 1.4 \, \text{V}$ . Agreement from two independent formulas is worth the one line it costs. Sanity: the emf must be bigger than the  $1 \, \text{V}$  that got out, and  $1.4$  is. This question was also set in test 2, word for word.

Two steps, and the first is the one people miss. The terminal p.d. is the voltage across the EXTERNAL resistor, so it gives you the current:  $I = V/R = 1/5 = 0.2 \, \text{A}$ . The drop inside the cell is then  $Ir = 0.2 \times 2 = 0.4 \, \text{V}$ , and the emf is the terminal voltage + that internal loss:  $E = 1 + 0.4 = 1.4 \, \text{V}$ . Equivalently  $E = I(R + r) = 0.2 \times 7 = 1.4 \, \text{V}$ . This is Test 2 question 4, verbatim from this slide.

The terminal p.d. of a battery is 12 V when an external resistance of 20 ohms is connected, and 13.5 V when an external resistance of 45 ohms is connected. Calculate the emf of the battery.

✓ 15 V

Correct.  $r$  works out at 5 ohms and  $E$  at 15 V.

✗ 13.5 V

Wrong. 13.5 V is the higher terminal reading, still below the emf.

✗ 25.5 V

Wrong. This adds the two terminal voltages, which have no reason to be added.

✗ 12 V

Wrong. 12 V is the lower terminal reading.

✗ 14 V

Wrong. A near miss that does not satisfy both readings at once.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

This is the standard experiment for measuring what is inside a cell without opening it. Take two readings with different loads and you get two equations; the two unknowns, the emf and the internal resistance, then fall out together. It works because the emf is a fixed property of the cell while the terminal p.d. changes with the load, and the difference between them is always  $Ir$ .

### GIVEN

- First reading:  $V = 12 \text{ V}$  with  $R = 20 \Omega$
- Second reading:  $V = 13.5 \text{ V}$  with  $R = 45 \Omega$

**Find:** The emf  $E$  of the battery, in volts

### FORMULA

$$E = V + Ir, \text{ where } I = V / R$$

**Why this one.** One reading is not enough, because it gives one equation with two unknowns. A second reading gives a second equation, and since  $E$  and  $r$  are the same cell in both, setting the two expressions for  $E$  equal eliminates  $E$  and leaves  $r$  alone. This is simultaneous equations doing exactly what they are for.

### WHAT EACH SYMBOL MEANS

- $E$  = the emf, the same in both readings
- $r$  = the internal resistance, also the same in both
- $V$  = the terminal p.d. for that reading, in volts
- $R$  = the external resistance for that reading, in ohms
- $I$  = the current for that reading, found as  $V \div R$

### THE WORKING, LINE BY LINE

1. First reading, current:  $I_1 = 12 \div 20 = 0.6 \text{ A}$
2. First equation:  $E = 12 + 0.6r$
3. Second reading, current:  $I_2 = 13.5 \div 45 = 0.3 \text{ A}$
4. Second equation:  $E = 13.5 + 0.3r$
5. The cell is the same, so set them equal:  $12 + 0.6r = 13.5 + 0.3r$
6. Collect the  $r$  terms:  $0.6r - 0.3r = 13.5 - 12$
7.  $0.3r = 1.5$ , so  $r = 5 \Omega$
8. Put  $r$  back into either equation:  $E = 12 + 0.6 \times 5$
9.  $E = 12 + 3 = 15 \text{ V}$

### CHECK IT

Test  $r = 5$  in the OTHER equation, the one you did not use to finish:  $E = 13.5 + 0.3 \times 5 = 13.5 + 1.5 = 15 \text{ V}$ . Both agree, so the pair is consistent. Sense check: the emf must exceed both terminal readings, and 15 is above 13.5 and above 12. Notice also the physical story, that the larger load resistance drew less current, so less was lost inside and the terminal reading climbed closer to the true emf.

Two readings give two equations, and two unknowns,  $E$  and  $r$ , need exactly that. First currents:  $I_1 = 12/20 = 0.6 \text{ A}$  and  $I_2 = 13.5/45 = 0.3 \text{ A}$ . Then  $V = E - Ir$  twice over:  $12 = E - 0.6r$  and  $13.5 = E - 0.3r$ . Subtracting the first from the second gives  $1.5 = 0.3r$ , so  $r = 5 \text{ ohms}$ . Substituting back:  $E = 12 + 0.6 \times 5 = 15 \text{ V}$ . Notice the pattern the question is teaching: a LARGER external resistance draws LESS current, so less is lost inside the cell and the terminal voltage climbs closer to the emf.

## 7 Magnetic Force on Charges and Wires

**Speed tip.** The magnetic force never changes the speed, only the direction. Kinetic energy is unchanged and the path is a circle of radius  $mv/qB$ .

7.1

EASY

Reference sheet R.4

**Magnetic flux density, the quantity everyone simply calls the magnetic field, is measured in  $\{1\}$ .**

Answer:

The tesla, symbol T. Keep it apart from the weber, which is the unit of magnetic FLUX rather than flux density, and one weber is one tesla square metre. A tesla is a large unit: the fields in this course run from 0.03 T up to 0.6 T, and a fridge magnet is only a few thousandths.

7.2

EASY

Reference sheet R.1

**What is the mass of an electron?**

✓  $9.11 \times 10^{-31}$  kg

Correct.  $9.11 \times 10^{-31}$  kg.

✗  $1.67 \times 10^{-27}$  kg

Wrong. That is the PROTON mass, which is far larger.

✗  $1.602 \times 10^{-19}$  kg

Wrong. That is the elementary charge, in coulombs not kilograms.

✗  $9.11 \times 10^{31}$  kg

Wrong sign on the exponent, which would make an electron heavier than a star.

✗  $8.85 \times 10^{-12}$  kg

Wrong. That is  $\epsilon_0$ .

The electron mass is  $9.11 \times 10^{-31}$  kg, about  $1836 \times$  smaller than the proton's  $1.67 \times 10^{-27}$  kg. It appears in the radius of a circular path,  $r = mv/(qB)$ , and in the cyclotron frequency.

7.3

EASY

Reference sheet R.1

**What is the mass of a proton?**✓  $1.67 \times 10^{-27} \text{ kg}$ Correct.  $1.67 \times 10^{-27} \text{ kg}$ .✗  $9.11 \times 10^{-31} \text{ kg}$ 

Wrong. That is the electron mass.

✗  $1.602 \times 10^{-19} \text{ kg}$ 

Wrong. That is a charge, not a mass.

✗  $1.67 \times 10^{27} \text{ kg}$ 

Wrong sign on the exponent.

✗  $4 \pi \times 10^{-7} \text{ kg}$ Wrong. That is  $\mu_0$ .

The proton mass is  $1.67 \times 10^{-27} \text{ kg}$ . Keep the three small constants apart by their exponents: charge at 10 to the  $-19 \text{ C}$ , proton at 10 to the  $-27 \text{ kg}$ , electron at 10 to the  $-31 \text{ kg}$ .

7.4

EASY

Reference sheet R.4

**Which pairing is correct?**

✓ The tesla measures magnetic flux DENSITY, and the weber measures magnetic FLUX

Correct. Tesla goes with B, the density; weber goes with  $\Phi$ , the flux.

✗ The tesla measures magnetic flux, and the weber measures flux density

Wrong, and exactly reversed.

✗ Both measure magnetic flux

Wrong. They measure related but different quantities.

✗ The tesla measures current and the weber measures field

Wrong. Current is in amperes.

✗ The weber measures inductance

Wrong. Inductance is in henries.

B, the flux density, is in tesla.  $\Phi$ , the flux, is in webers, and  $\Phi = BA \cos \theta$  ties them: one weber is one tesla square metre. The distinction matters on Test 2 question 11, whose answer is that a changing FLUX induces an emf, not a changing field.

When BOTH an electric and a magnetic field are present, the force on a moving charge is:

✓  $F = q(E + v \text{ cross } B)$

Correct, and this is the full Lorentz force.

✗  $F = qE$  only, since magnetic fields do no work

Wrong. Doing no work is not the same as exerting no force; a magnetic force bends the path.

✗  $F = qvB \sin \theta$  only

Wrong. That drops the electric contribution entirely.

✗  $F = q(E \text{ cross } B)$

Wrong. E and B are not crossed with each other here. That combination is the Poynting vector, a different quantity.

✗  $F = qE$  times  $qvB$

Wrong. Forces add; they do not multiply.

The two contributions simply add: the electric part  $qE$ , which acts whether or not the charge is moving, plus the magnetic part  $qv \text{ cross } B$ , which needs motion. Note that the magnetic part doing no WORK does not mean it exerts no FORCE: it steers without changing the speed.

The period of a charge circling in a magnetic field is  $T = \frac{2 \pi m}{qB}$ , which like the frequency is independent of the

{{1}}.

Answer:

The speed. Mass, charge and field all appear in the formula and all change the period; speed does not appear and does not. It matters for the helical path, where the pitch is  $v \cos \theta$  times  $T$ : the period is fixed by the field and the particle, and only the along-field velocity decides how far it advances in that time.

A straight wire of length  $L$  carries current  $I$  in a magnetic field  $B$ . The force on it is  $F = BIL \sin \theta$ . At what angle is this force greatest?

✓  $90^\circ$ , with the wire across the field

Correct. Perpendicular gives  $\sin = 1$ .

✗  $0^\circ$ , with the wire along the field

Wrong, and it is the ZERO-force case.

✗  $45^\circ$

Wrong.  $\sin 45$  is about 0.71, so substantial but not maximal.

✗  $180^\circ$

Wrong.  $\sin 180 = 0$ , so no force, same as along the field.

✗ The angle makes no difference

Wrong. The sine factor is precisely the angle's effect.

$\sin 90 = 1$ , the largest the sine can be, so the force is maximum with the wire perpendicular to the field. Along the field,  $\sin 0 = 0$  and the wire feels nothing at all, which is the same rule as for a single moving charge: only the component of motion ACROSS the field is deflected.

7.8

MEDIUM

Reference sheet R.1

The magnetic dipole moment of a current loop is  $m = NIA$ , where  $N$  is the number of turns,  $I$  the current and  $A$  the area the loop encloses.

Answer:

The area, in square metres, giving  $m$  the unit  $A\ m^2$ . Test 2 question 7 works it backwards, giving a torque and asking for  $m$ . Note that the  $m$  here is a MAGNETIC MOMENT and not the mass that appears in  $r = \frac{mv}{qB}$ , which is the most confusable

pair of symbols in the topic.

7.9

MEDIUM

Reference sheet R.1

The torque on a current loop is  $\tau = mB \sin \theta = NIAB \sin \theta$ . When is it zero?

✓ When the magnetic moment  $m$  lies along  $B$

Correct, and it is the stable resting position.

✗ When  $m$  is perpendicular to  $B$

Wrong. Perpendicular is where the torque is at its MAXIMUM.

✗ When the current is at its maximum

Wrong. More current gives more torque, not less.

✗ Only when the current is zero

Wrong. Zero current is one way, but alignment gives zero torque with the current still flowing.

✗ When the loop has more than one turn

Wrong. More turns increase the torque.

$\sin 0 = 0$ , so aligned means no torque. It is the exact counterpart of the ELECTRIC dipole, whose torque  $pE \sin \theta$  also vanishes on alignment. Both are the position the field is trying to turn the object into, which is why a compass needle settles pointing along the field and then stops.

7.10

MEDIUM

Reference sheet R.4

Asked twice. Also set as: Reference sheet R.4.

The cyclotron frequency is the number of revolutions per second of a charge in a magnetic field, and it is independent of the particle's speed.

Answer: **true**

True on both counts.  $f = qB$  over  $2\pi m$  contains the charge, the field and the mass, and nothing about how fast the particle is going. A faster particle circles on a wider radius and arrives back at the same moment.

7.11

MEDIUM

Reference sheet R.1

The electron mass is  $9.11 \times 10^{-31}$  kg.

Answer:

Minus 31. The proton, at  $10^{-27}$ , is about two thousand times heavier.

7.12

MEDIUM

Reference sheet R.1

The proton mass is  $1.67 \times 10^{-27}$  kg.

Answer:

Minus 27, against the electron's - 31.

7.13

MEDIUM

Reference sheet R.2

A charge moving **ALONG** a magnetic field feels no force, because  $\sin 0$  equals 0.

Answer:

Zero. The angle is measured between the velocity and the field, so moving along the field kills the force entirely.

7.14

MEDIUM

Reference sheet R.2

A magnetic field can increase the speed of a charged particle.

Answer: **false**

It cannot. The magnetic force is always perpendicular to the velocity, so it does no work and cannot change the speed. It changes direction only.

7.15

MEDIUM

Reference sheet R.2

The cyclotron frequency is independent of the particle's mass.

Answer:

Speed. The charge, the mass and the field all appear in  $f = qB/(2\pi m)$ ; the speed does not.

7.16

MEDIUM

Reference sheet R.1

The magnetic force is  $F = qvB \sin \theta$ , so it is zero either when  $v = 0$  or when the charge travels **parallel to** the field.

Answer:

Along the field, where  $\theta = 0$  and  $\sin 0 = 0$ . Those are the two escapes: stand still, or travel straight down the field lines. A stationary charge in the strongest magnet on earth feels nothing at all, which is the sharpest difference between a magnetic force and an electric one.

7.17

MEDIUM

Reference sheet R.1

In  $r = \frac{mv}{qB}$ , a faster particle travels a wider circle but takes exactly the same time to go round.

Answer: **true**

True, and it is the cyclotron principle. The radius is proportional to  $v$ , so doubling the speed doubles the circle, but the period  $T = 2\pi m / qB$  contains no  $v$  at all, so the longer path is covered in the same time. That is what lets a fixed-frequency accelerator keep pushing particles as they speed up.

The force on a current-carrying conductor is  $F = BIL \sin \theta$ , which is at a maximum when the wire is at **90 degrees** to the field.

Answer:

90°, where  $\sin$  is 1. Along the field,  $\sin 0 = 0$  and the wire feels nothing, exactly as a single charge moving along the field feels nothing. Only the component ACROSS the field is deflected, whether it is one charge or a wire full of them.

A coil of 50 turns, each enclosing  $4 \times 10^{-4} \text{ m}^2$ , carries 2 A. What is its magnetic dipole moment?

✓ 0.04 A m<sup>2</sup>

Correct.  $50 \times 2 \times 4 \times 10^{-4}$ .

✗ 0.0008 A m<sup>2</sup>

Wrong. This omits the 50 turns.

✗ 400 A m<sup>2</sup>

Wrong. This drops the  $10^{-4}$  entirely.

✗ 0.02 A m<sup>2</sup>

Wrong. This omits the current.

✗ 100 A m<sup>2</sup>

Wrong. That is N times I, with the area forgotten.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

A current loop behaves like a tiny bar magnet, and its magnetic moment measures how strongly. Three things make it stronger: more current, more area enclosed, and more turns, because each turn adds its own contribution on top of the others. That is why a compass needle can be replaced by a coil, and why motors use many turns rather than one.

**GIVEN**

- $N = 50$  turns
- $A = 4 \times 10^{-4} \text{ m}^2$  per turn
- $I = 2 \text{ A}$

**Find:** The magnetic dipole moment  $m$ , in  $\text{A m}^2$

**FORMULA**

$$m = NIA$$

**Why this one.** The unit gives the formula away:  $\text{A m}^2$  is amperes times metres squared, so it must be current times area, and the turns multiply because each one contributes equally. No field appears in the question, and none is needed, because the moment is a property of the loop itself and exists before any field is applied.

**WHAT EACH SYMBOL MEANS**

- $m$  = the magnetic dipole moment, in  $\text{A m}^2$
- $N$  = the number of turns
- $I$  = the current, in amperes
- $A$  = the area enclosed by ONE turn, in  $\text{m}^2$

**THE WORKING, LINE BY LINE**

1.  $m = N \times I \times A$
2.  $m = 50 \times 2 \times 4 \times 10^{-4}$
3.  $50 \times 2 = 100$
4.  $100 \times 4 = 400$
5.  $m = 400 \times 10^{-4}$
6.  $m = 0.04 \text{ A m}^2$

**CHECK IT**

Reconstruct the unit as the check: amperes  $\times$  metres squared =  $\text{A m}^2$ , which matches what was asked for, so the multiplication was the right operation. The option 0.0008 is what you get by forgetting the 50 turns, and the turns are exactly why a coil is used instead of a single loop.

$m = NIA = 50 \times 2 \times 4 \times 10^{-4} = 100 \times 4 \times 10^{-4} = 0.04 \text{ A m}^2$ . All three multiply, so more turns, more current or more area each raise it. Note this  $m$  is a MAGNETIC MOMENT, not the mass that appears in  $r = \frac{mv}{qB}$ .

**7.20**

MEDIUM

Reference sheet R.1

The torque on a current loop is  $\tau = NIAB \sin \theta$ , and it falls to zero when the magnetic moment lies **parallel** to the field.

Answer:

Along the field, where  $\sin \theta = 0$ . That is the resting position the field is turning the loop towards, and it is exactly the same behaviour as an electric dipole in an electric field. A compass needle settling on north and then stopping is the everyday case.

7.21

MEDIUM

Reference sheet R.4

Asked twice. Also set as: Reference sheet R.4.

**The cyclotron frequency is the number of revolutions per second of a charge in a magnetic field, and it is independent of the particle's  $\{1\}$ .**

Answer:

Its speed.  $f = qB$  over  $2\pi m$  contains the charge, the field and the mass, and nothing about how fast the particle moves.

Charge, mass and field all DO change it, which is worth noting because three of the four candidate answers are genuinely in the formula.

7.22

HARD

Reference sheet R.1

**The cyclotron frequency is  $f = \frac{qB}{2\pi m}$ . A proton is injected twice into the same field, the second time at double the speed. What happens to its frequency of revolution?**

✓ Unchanged, because there is no  $v$  in the formula

Correct, and the absence of  $v$  from the formula is the whole point.

✗ It doubles

Wrong. Doubling the speed doubles the RADIUS, not the frequency.

✗ It halves

Wrong. Nothing halves here.

✗ It quadruples

Wrong. No quantity here scales as a square.

✗ It falls by the square root of two

Wrong. No square root appears in this relationship.

There is no  $v$  in  $f = qB$  over  $2\pi m$ , so the frequency is independent of speed, and so is the period  $T = 2\pi m$  over  $qB$ . A faster proton simply travels a wider circle, since  $r = mv$  over  $qB$ , and covers that longer path in exactly the same time. That is the cyclotron principle: it is what allows a fixed-frequency accelerating voltage to keep pushing particles that are getting faster.

7.23

HARD

Reference sheet R.1

**In  $F = qvB \sin \theta$ , name TWO situations in which a charged particle in a magnetic field feels no force at all.**

✓ When it is stationary, and when it moves along the field direction

Correct. Either no motion, or motion parallel to the field.

✗ When it is stationary, and when it moves at  $90^\circ$  to the field

Wrong on the second:  $90^\circ$  is where the force is at its MAXIMUM.

✗ When it moves fast, and when the field is strong

Wrong. Both of those increase the force.

✗ When the charge is negative, and when the field is uniform

Wrong. A negative charge feels a force in the opposite direction, not no force.

✗ There is no situation in which the force vanishes

Wrong. Both cases above give exactly zero.

Two ways to make the product zero. If  $v = 0$  the whole expression is zero, so a stationary charge feels nothing. If  $\theta = 0$  the particle moves ALONG the field and  $\sin 0 = 0$ , so again nothing. The force is greatest at  $90^\circ$ , where the sine is 1.

### What happens to the kinetic energy of a charged particle moving through a magnetic field alone?

- ✓ It stays constant, because a magnetic field steers but never speeds up

Correct. Magnetic fields steer, they do not accelerate in the sense of changing speed.

- ✗ It increases steadily

Wrong. That would require work to be done, which a perpendicular force cannot do.

- ✗ It decreases steadily

Wrong, for the same reason.

- ✗ It oscillates between zero and a maximum

Wrong. The speed is constant throughout.

- ✗ It depends on the sign of the charge

Wrong. The sign changes which way it curves, not the speed.

The magnetic force is always perpendicular to the velocity, and a force at right angles to the motion does no work. So the speed, and with it the kinetic energy, is unchanged: the field only changes the DIRECTION. That is why the path is a circle at constant speed.

### What is notable about the cyclotron frequency $f = qB / (2 \pi m)$ ?

- ✓ It contains no  $v$ , so the frequency does not depend on the particle's speed

Correct, and the cancellation with the radius is why.

- ✗ It rises in proportion to the speed

Wrong. Speed cancels out entirely.

- ✗ It falls in proportion to the speed

Wrong, for the same reason.

- ✗ It depends on the radius but not on the field

Wrong.  $B$  is in the formula and  $r$  is not.

- ✗ It applies only to electrons

Wrong. It holds for any charged particle.

No  $v$  appears in the formula. A faster particle travels a larger circle, since  $r = mv/(qB)$ , and the two effects cancel exactly, so it goes round in the same time. That is the cyclotron principle, and it is what makes the machine work with a fixed driving frequency.

A proton experiences a magnetic force of  $3.2 \times 10^{-14}$  N while moving through a magnetic field of 0.2 T. If the angle between the velocity vector and the magnetic field is  $60^\circ$ , calculate the speed of the proton. Take: Charge of proton: ( $q = 1.6 \times 10^{-19}$  C).

✓  $1.155 \times 10^6$  m/s

Correct.  $v = F/(qB \sin 60) = 3.2 \times 10^{-14} / 2.771 \times 10^{-20} = 1.155 \times 10^6$  m/s.

✗  $6.155 \times 10^6$  m/s

Wrong. No route to this value; the leading digit cannot be 6 with this force and field.

✗  $4.155 \times 10^6$  m/s

Wrong. No route to this value from these numbers.

✗  $3.155 \times 10^6$  m/s

Wrong. No route to this value from these numbers.

✗  $9.155 \times 10^6$  m/s

Wrong. No route to this value from these numbers.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A magnetic field only pushes on a charge that is MOVING, and only on the part of that motion which cuts across the field. Move exactly along the field lines and there is no force at all. Move exactly across them and the force is at its largest. Every angle in between is handled by  $\sin \theta$ , which is 0 when you move along the field and 1 when you move across it. That is the opposite of the flux formula, which uses  $\cos \theta$ , and mixing them up is the commonest way to lose this mark.

### GIVEN

- $F = 3.2 \times 10^{-14} \text{ N}$  (the magnetic force felt by the proton)
- $B = 0.2 \text{ T}$  (the field strength, in teslas)
- $\theta = 60^\circ$  (the angle between the velocity and the field)
- $q = 1.6 \times 10^{-19} \text{ C}$  (the charge on a proton, given in the question)

**Find:** The speed  $v$  of the proton, in  $\text{m s}^{-1}$

### FORMULA

$$F = qvB \sin \theta, \text{ so } v = \frac{F}{(qB \sin \theta)}$$

**Why this one.** The question names a moving charge, a field, an angle and a force, which is exactly the four quantities this formula ties together. It is not the wire formula  $F = BIL$ , because nothing here carries a current along a length; it is a single particle. Rearranging for  $v$  means everything except  $F$  moves to the bottom.

### WHAT EACH SYMBOL MEANS

- $F$  = the magnetic force, in newtons
- $q$  = the charge on the particle, in coulombs
- $v$  = the speed of the particle, in  $\text{m s}^{-1}$ , which is what we want
- $B$  = the magnetic flux density, in teslas
- $\theta$  = the angle between  $v$  and  $B$ , taking SINE not cosine

### THE WORKING, LINE BY LINE

1. Rearrange:  $v = F \div (q B \sin \theta)$
2.  $\sin 60^\circ = 0.866$
3. Bottom, step one:  $q \times B = 1.6 \times 10^{-19} \times 0.2 = 3.2 \times 10^{-20}$
4. Bottom, step two:  $3.2 \times 10^{-20} \times 0.866 = 2.771 \times 10^{-20}$
5.  $v = (3.2 \times 10^{-14}) \div (2.771 \times 10^{-20})$
6. Front numbers:  $3.2 \div 2.771 = 1.155$
7. Powers of ten:  $10^{-14} \div 10^{-20} = 10^6$
8.  $v = 1.155 \times 10^6 \text{ m s}^{-1}$

### CHECK IT

$\sin 60^\circ$  is less than 1, so dividing by it makes the answer BIGGER than it would be at  $90^\circ$ . At  $90^\circ$  you would get  $1.0 \times 10^6$ , and  $1.155 \times 10^6$  is duly larger. An answer smaller than the perpendicular case means you multiplied by the sine instead of dividing.

The magnetic force is  $F = qvB \sin(\theta)$ , so  $v = F / (qB \sin \theta)$ . The sine is what makes this more than a plug-in:  $\sin 60 = 0.866$ , not 1.  $v = 3.2 \times 10^{-14} / (1.6 \times 10^{-19} \times 0.2 \times 0.866) = 3.2 \times 10^{-14} / 2.771 \times 10^{-20} = 1.155 \times 10^6 \text{ m/s}$ . Forgetting the sine would give  $1.0 \times 10^6$ , which is not offered, so the sine is exactly what is being tested.

A rectangular current loop placed in a uniform magnetic field experiences a torque of  $4.8 \times 10^{-3} \text{ Nm}$ . If the magnetic field strength is  $0.6 \text{ T}$  and the angle between the loop's magnetic moment and the field is  $30^\circ$ , calculate the magnetic dipole moment of the loop.

✓  $1.6 \times 10^{-2} \text{ Am}^2$

Correct.  $m = \tau / (B \sin 30) = 4.8 \times 10^{-3} / 0.3 = 1.6 \times 10^{-2} \text{ Am}^2$ .

✗  $13 \times 10^{-2} \text{ Am}^2$

Wrong. No route to this value from these numbers.

✗  $20 \times 10^{-2} \text{ Am}^2$

Wrong. No route to this value from these numbers.

✗  $3 \times 10^{-2} \text{ Am}^2$

Wrong. This uses  $\sin 30$  as  $0.25$ , or divides by  $1.6$  rather than  $0.3$ .

✗  $5 \times 10^{-2} \text{ Am}^2$

Wrong. This drops the sine altogether and then mishandles the division.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

A current loop sitting in a magnetic field behaves exactly like a small compass needle. It has a magnetic moment, an arrow saying which way it is facing and how strongly, and the field tries to twist that arrow into line with itself. That twist is the torque. When the loop already lies along the field there is nothing left to twist, so the torque is zero; when it sits across the field the twist is greatest. This is precisely how an electric motor turns.

**GIVEN**

- $\tau = 4.8 \times 10^{-3} \text{ N m}$  (the torque on the loop)
- $B = 0.6 \text{ T}$  (the field strength)
- $\theta = 30^\circ$  (the angle between the magnetic moment and the field)

**Find:** The magnetic dipole moment  $m$  of the loop, in  $\text{A m}^2$

**FORMULA**

$$\tau = mB \sin \theta, \text{ so } m = \frac{\tau}{(B \sin \theta)}$$

**Why this one.** A torque, a field and an angle are given and a moment is wanted, and this formula is the only one linking those four. It takes SINE because the twist is greatest when the moment lies across the field and vanishes when it lies along it, which is exactly what sine does:  $\sin 90^\circ = 1$  and  $\sin 0^\circ = 0$ .

**WHAT EACH SYMBOL MEANS**

- $\tau$  = the torque, in newton metres (N m)
- $m$  = the magnetic dipole moment, in ampere metres squared ( $\text{A m}^2$ )
- $B$  = the magnetic flux density, in teslas
- $\theta$  = the angle between the moment and the field

**THE WORKING, LINE BY LINE**

1. Rearrange:  $m = \tau \div (B \sin \theta)$
2.  $\sin 30^\circ = 0.5$  (worth knowing by heart, no calculator needed)
3. Bottom:  $B \times \sin \theta = 0.6 \times 0.5 = 0.3$
4.  $m = (4.8 \times 10^{-3}) \div 0.3$
5. Front numbers:  $4.8 \div 0.3 = 16$
6.  $m = 16 \times 10^{-3} = 1.6 \times 10^{-2} \text{ A m}^2$

**CHECK IT**

$16 \times 10^{-3}$  and  $1.6 \times 10^{-2}$  are the same number written two ways, so recognise your answer even when the options move the decimal point. If the angle had been  $90^\circ$  the moment needed would be only half as large, since a square-on loop feels the full twist from a smaller moment.

Torque on a magnetic dipole is  $\tau = mB \sin(\theta)$ , so  $m = \tau / (B \sin \theta)$ . With  $\sin 30 = 0.5$ :  $m = 4.8 \times 10^{-3} / (0.6 \times 0.5) = 4.8 \times 10^{-3} / 0.3 = 1.6 \times 10^{-2} \text{ A m}^2$ . Note the contrast with the proton question: there the angle was  $60$  and the sine  $0.866$ , here it is  $30$  and the sine exactly  $0.5$ , so read the angle before reaching for a value.

An electron of mass  $9.1 \times 10^{-31}$  kg and charge  $1.6 \times 10^{-19}$  enters a magnetic field of 0.03 T perpendicularly with velocity  $4 \times 10^6$  m/s. Find the radius of its circular path.

✓  $7.58 \times 10^{-4}$  m

Correct.  $r = mv/(qB) = 3.64 \times 10^{-24} / 4.8 \times 10^{-21} = 7.58 \times 10^{-4}$  m.

✗  $7.58 \times 10^{-6}$  m

Wrong exponent, off by 100. Check the subtraction: -24 minus -21 is -3, then the leading 3.64/4.8 being less than one shifts it to -4.

✗  $7.58 \times 10^{-2}$  m

Wrong exponent, off by 100 the other way.

✗  $7.58 \times 10^{-8}$  m

Wrong exponent, off by 10000.

✗  $7.58 \times 10^{-1}$  m

Wrong exponent, and a radius of 0.758 m would be larger than the apparatus.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

A magnetic force always acts at right angles to the direction of travel. A force that is permanently sideways can never speed a particle up or slow it down, it can only bend it, so the particle keeps its speed and curls into a circle forever. That is the principle behind the cyclotron and behind the way a mass spectrometer separates isotopes: heavier particles have more momentum, so they curve more gently and land further along.

**GIVEN**

- $m = 9.1 \times 10^{-31}$  kg (the mass of the electron)
- $q = 1.6 \times 10^{-19}$  C (the charge on the electron)
- $B = 0.03$  T
- $v = 4 \times 10^6$  m s<sup>-1</sup>, entering PERPENDICULAR to the field

**Find:** The radius  $r$  of the circular path, in metres

**FORMULA**

$$r = \frac{mv}{qB}$$

**Why this one.** It comes from setting the magnetic force equal to the centripetal force the circle requires:  $qvB = mv^2 \div r$ . Cancel one  $v$  from both sides and rearrange, and  $r = mv \div qB$  drops out. Because the particle enters perpendicular,  $\sin \theta = 1$  and the angle disappears from the problem entirely. The top line,  $mv$ , is just momentum, which is the useful way to remember it: more momentum means a wider curve.

**WHAT EACH SYMBOL MEANS**

- $r$  = the radius of the circular path, in metres
- $m$  = the mass of the particle, in kilograms
- $v$  = its speed, in m s<sup>-1</sup>
- $q$  = its charge, in coulombs
- $B$  = the flux density, in teslas

**THE WORKING, LINE BY LINE**

1. Top:  $m \times v = 9.1 \times 10^{-31} \times 4 \times 10^6$
2. Front numbers:  $9.1 \times 4 = 36.4$
3. Powers of ten:  $10^{-31} \times 10^6 = 10^{-25}$
4. Top =  $36.4 \times 10^{-25} = 3.64 \times 10^{-24}$
5. Bottom:  $q \times B = 1.6 \times 10^{-19} \times 0.03 = 4.8 \times 10^{-21}$
6.  $r = (3.64 \times 10^{-24}) \div (4.8 \times 10^{-21})$
7. Front numbers:  $3.64 \div 4.8 = 0.758$
8. Powers of ten:  $10^{-24} \div 10^{-21} = 10^{-3}$
9.  $r = 0.758 \times 10^{-3} = 7.58 \times 10^{-4}$  m

**CHECK IT**

Every option in this question is 7.58 with a different power of ten, so the whole mark rides on the exponent arithmetic and not on the multiplication. Do the powers of ten on their own line, as above, and the trap disappears. Sanity:  $7.58 \times 10^{-4}$  m is under a millimetre, which is believable for a light electron in a real laboratory field.

A charge moving perpendicular to a field goes in a circle, with the magnetic force supplying the centripetal force:  $qvB = mv^2/r$ , so  $r = mv/(qB)$ . Substituting:  $r = (9.1 \times 10^{-31} \times 4 \times 10^6) / (1.6 \times 10^{-19} \times 0.03) = 3.64 \times 10^{-24} / 4.8 \times 10^{-21} = 7.58 \times 10^{-4}$  m. Every option shares the digits 7.58, so the ONLY thing being tested is the power of ten.

**Tutorial question 1.** A proton enters a magnetic field of 0.5 T perpendicular to the field at  $2 \times 10^6$  m/s. Calculate the magnetic force and the radius of its path. Take the proton mass as  $1.67 \times 10^{-27}$  kg and its charge as  $1.6 \times 10^{-19}$  C.

✓  $F = 1.6 \times 10^{-13}$  N and  $r = 4.2 \times 10^{-2}$  m

Correct on both.

✗  $F = 1.6 \times 10^{-13}$  N and  $r = 4.2 \times 10^{-4}$  m

Wrong on the radius by a factor of 100, a slip when dividing the exponents.

✗  $F = 8.0 \times 10^{-14}$  N and  $r = 4.2 \times 10^{-2}$  m

Wrong on the force, from halving it as though the sine were 0.5 rather than 1.

✗  $F = 1.6 \times 10^{-19}$  N and  $r = 2.1 \times 10^{-2}$  m

Wrong.  $1.6 \times 10^{-19}$  is the proton's CHARGE copied out as though it were a force.

✗  $F = 3.2 \times 10^{-13}$  N and  $r = 8.4 \times 10^{-2}$  m

Wrong on both, from doubling the force and the radius together.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A charged particle fired across a magnetic field is pushed sideways, and because the push stays sideways as the particle turns, it never speeds up or slows down. It just curls into a circle. Two questions follow naturally: how hard is the push, and how tight is the circle. The first is the force formula, the second comes from setting that force equal to what a circle requires.

### GIVEN

- $B = 0.5 \text{ T}$
- $v = 2 \times 10^6 \text{ m s}^{-1}$ , entering PERPENDICULAR to the field
- $m = 1.67 \times 10^{-27} \text{ kg}$ , the proton mass
- $q = 1.6 \times 10^{-19} \text{ C}$

**Find:** The magnetic force  $F$ , in newtons, and the radius  $r$  of the path, in metres

### FORMULA

$$F = qvB, \text{ then } r = mv / (qB)$$

**Why this one.** Perpendicular entry means  $\theta = 90^\circ$  and  $\sin 90^\circ = 1$ , so the angle drops out of the force formula entirely. The radius formula comes from  $qvB = mv^2 / r$ : the magnetic force IS the centripetal force, since it is the only force acting. Cancel one  $v$  and rearrange.

### WHAT EACH SYMBOL MEANS

- $F$  = the magnetic force, in newtons
- $q$  = the charge on the proton, in coulombs
- $v$  = its speed, in  $\text{m s}^{-1}$
- $B$  = the flux density, in teslas
- $m$  = the proton mass, in kilograms
- $r$  = the radius of the circular path, in metres

### THE WORKING, LINE BY LINE

1. Force first:  $F = q \times v \times B$
2.  $F = 1.6 \times 10^{-19} \times 2 \times 10^6 \times 0.5$
3. Front numbers:  $1.6 \times 2 \times 0.5 = 1.6$
4. Powers of ten:  $10^{-19} \times 10^6 = 10^{-13}$
5.  $F = 1.6 \times 10^{-13} \text{ N}$
6. Radius next:  $r = (m \times v) \div (q \times B)$
7. Top:  $1.67 \times 10^{-27} \times 2 \times 10^6 = 3.34 \times 10^{-21}$
8. Bottom:  $1.6 \times 10^{-19} \times 0.5 = 8.0 \times 10^{-20}$
9.  $r = (3.34 \times 10^{-21}) \div (8.0 \times 10^{-20})$
10. Front numbers:  $3.34 \div 8.0 = 0.4175$
11. Powers of ten:  $10^{-21} \div 10^{-20} = 10^{-1}$
12.  $r = 0.4175 \times 10^{-1} = 4.2 \times 10^{-2} \text{ m}$ , about 4 cm

### CHECK IT

Cross-check the two answers against each other using the third relation,  $F = mv^2 / r$ :  $(1.67 \times 10^{-27} \times (2 \times 10^6)^2) \div 0.0418 = (1.67 \times 10^{-27} \times 4 \times 10^{12}) \div 0.0418 = 6.68 \times 10^{-15} \div 0.0418 = 1.6 \times 10^{-13} \text{ N}$ , matching the force found first. A 4 cm circle is also a believable laboratory size, whereas the option offering  $4.2 \times 10^{-4} \text{ m}$  would be a quarter of a millimetre.

Perpendicular means  $\theta$  is  $90^\circ$  and  $\sin 90 = 1$ , so the sine quietly disappears. Force:  $F = qvB = 1.6 \times 10^{-19} \times 2 \times 10^6 \times 0.5 = 1.6 \times 10^{-13} \text{ N}$ . Radius:  $r = mv \text{ over } qB = (1.67 \times 10^{-27} \times 2 \times 10^6) / (1.6 \times 10^{-19} \times 0.5) = 3.34 \times 10^{-21} / 8.0 \times 10^{-20} = 4.2 \times 10^{-2} \text{ m}$ , which is about 4 cm. A believable size for a proton curving in half a tesla, which is the step 7 check.

**A proton moves with a velocity of  $3.5 \times 10^6$  m/s in a magnetic field of 0.25 T at an angle of  $60^\circ$ . Which component of the velocity sets the RADIUS of the helix, and which sets its PITCH?**

✓  $v \sin 60$  sets the radius, and  $v \cos 60$  sets the pitch

Correct. Sine for the radius, cosine for the pitch.

✗  $v \cos 60$  sets the radius, and  $v \sin 60$  sets the pitch

Wrong, and it is the trap, since these two are the easiest pair in the course to swap.

✗ The full  $v$  sets both

Wrong. Using the whole speed for both ignores that only the perpendicular part is deflected.

✗  $v \sin 60$  sets both

Wrong. If the pitch used sine too, the path would never advance along the field.

✗  $v \tan 60$  sets the radius, and  $v \sin 60$  sets the pitch

Wrong. A tangent does not appear anywhere in this topic.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

When a particle enters a field at an angle rather than square on, split its velocity into two parts and treat them separately. The part **ACROSS** the field feels the magnetic force and curls into a circle. The part **ALONG** the field feels no force at all, since a charge moving parallel to the field is not cut by it, so it just carries on at constant speed. A circle plus a steady drift along the axis is a helix, like a stretched spring.

**GIVEN**

- $v = 3.5 \times 10^6 \text{ m s}^{-1}$
- $B = 0.25 \text{ T}$
- $\theta = 60^\circ$  between the velocity and the field

**Find:** Which velocity component sets the **RADIUS**, and which sets the **PITCH**

**FORMULA**

$$r = \frac{m v \sin \theta}{(qB)}$$

**Why this one.** The formula itself answers the question. Only the perpendicular component appears in it, so only  $v \sin \theta$  can affect the radius. The pitch, which is how far the helix advances along the field in one full turn, is set by the parallel component  $v \cos \theta$  multiplied by the time for one revolution, because that is the part doing the advancing.

**WHAT EACH SYMBOL MEANS**

- $v \sin \theta$  = the component **ACROSS** the field, which is bent into the circle
- $v \cos \theta$  = the component **ALONG** the field, which is never bent
- $r$  = the radius of the helix, in metres
- the pitch = the advance along the field per complete turn, in metres

**THE WORKING, LINE BY LINE**

1. Split the velocity at the angle to the **FIELD**:
2. Across the field:  $v \sin 60^\circ = 3.5 \times 10^6 \times 0.866 = 3.03 \times 10^6 \text{ m s}^{-1}$
3. Along the field:  $v \cos 60^\circ = 3.5 \times 10^6 \times 0.5 = 1.75 \times 10^6 \text{ m s}^{-1}$
4. The across part feels the force  $qvB$  and curls, so it sets the **RADIUS**
5. The along part feels no force and drifts, so it sets the **PITCH**
6. Answer:  $v \sin 60$  sets the radius,  $v \cos 60$  sets the pitch

**CHECK IT**

Test the two extremes, which settles it without memorising. At  $\theta = 90^\circ$ , straight across the field,  $\sin 90 = 1$  and  $\cos 90 = 0$ , so the radius is at its largest and the pitch is zero: a flat circle, which is right. At  $\theta = 0^\circ$ , straight along the field,  $\sin 0 = 0$  so there is no circle at all and the particle flies straight: infinite pitch, zero radius, also right. Only the sine-for-radius pairing survives both tests.

Split the velocity into the part **ACROSS** the field and the part **ALONG** it. The perpendicular part,  $v \sin \theta$ , is what the magnetic force acts on, so it drives the circular motion and fixes the radius:  $r = mv \sin \theta$  over  $qB$ . The parallel part,  $v \cos \theta$ , feels no force at all, because  $\sin 0$  is zero along the field, so the proton simply drifts steadily in that direction, and that drift over one period is the pitch:  $p = v \cos \theta$  times  $T$ . Circle across the field, drift along it, and the two together trace a helix.

## 8 Sources of a Magnetic Field

**Speed tip.** The loop centre formula has NO  $\pi$  in the denominator. The straight wire and the toroid both do. That difference alone identifies the formula.

8.1

EASY

Reference sheet R.4

**An Amperian loop is an {{1}} closed path, chosen to make Ampere's law easy.**

Answer:

Imaginary, exactly like the Gaussian surface in electrostatics. It is a path you draw on the diagram, not a wire, and you choose its shape to match the symmetry: a circle around a straight wire, a rectangle threading a solenoid.

8.2

EASY

Reference sheet R.4

**What is a solenoid, and what is a toroid?**

✓ A solenoid is a long coil producing a uniform field inside; a toroid is a solenoid bent into a ring

Correct on both.

✗ A solenoid is a single loop; a toroid is a straight wire

Wrong. A solenoid is many turns, not one.

✗ A solenoid is a ring; a toroid is a long coil

Wrong, and reversed.

✗ Both are single loops of different radius

Wrong. Both are multi-turn coils.

✗ A solenoid carries AC and a toroid carries DC

Wrong. Neither is defined by the kind of current.

A long coil, and the same coil closed into a doughnut. The uniform interior field is what makes a solenoid useful, and bending it into a ring confines the field entirely within the loop, which is why toroids are used where stray fields would be a nuisance. Note their formulas differ in exactly the way the previous question describes.

8.3

EASY

Reference sheet R.1

**What is the value of the permeability of free space,  $\mu_0$ ?**

✓  $4 \pi \times 10^{-7} \text{ H m}^{-1}$

Correct.  $4 \pi \times 10^{-7} \text{ H m}^{-1}$ .

✗  $8.85 \times 10^{-12} \text{ H m}^{-1}$

Wrong. That is  $\epsilon_0$ , the electric counterpart.

✗  $8.99 \times 10^9 \text{ H m}^{-1}$

Wrong. That is the Coulomb constant.

✗  $4 \pi \times 10^7 \text{ H m}^{-1}$

Wrong sign on the exponent.

✗  $1.602 \times 10^{-19} \text{ H m}^{-1}$

Wrong. That is the elementary charge.

$\mu_0 = 4 \pi \times 10^{-7} \text{ H m}^{-1}$ . Leaving it as  $4 \pi$  rather than  $1.257 \times 10^{-6}$  is usually the faster route, because the  $\pi$  very often cancels against the  $2 \pi r$  in the straight wire formula.

8.4

EASY

Reference sheet R.4

**A solenoid is a long coil producing a uniform field inside it, and a toroid is a solenoid bent into a  $\{\{1\}\}$ .**

Answer:

A ring. Closing the coil into a doughnut confines the field entirely within the loop, so a toroid produces almost no stray field outside it, which is why it is chosen where interference would be a problem.

8.5

MEDIUM

Reference sheet R.1

**Both the Biot-Savart law and Ampere's law give the magnetic field of a current. When is Ampere's law the one to reach for?**

✓ When the arrangement has symmetry, because it is the shortcut

Correct, and symmetry is the condition that makes it usable.

✗ Always, since Biot-Savart is wrong

Wrong. Both are correct; they differ in convenience.

✗ Only for a single current element

Wrong. That is Biot-Savart's domain.

✗ When the current is changing with time

Wrong. Changing currents belong to induction, a different topic.

✗ When there is no current enclosed

Wrong. With no enclosed current the law still holds but tells you little.

Biot-Savart is general but laborious: it gives the field of one tiny current element and you must sum over the whole conductor. Ampere's law is the shortcut, but it is only SOLVABLE when you can choose a loop on which  $B$  is constant and along the path, which needs symmetry. That is exactly what Test 2 question 5 tests with its one-word answer.

8.6

MEDIUM

Reference sheet R.1

**The Biot-Savart law gives the field of a current element, and like Coulomb's law it falls as 1 over  $r$  to the power  $\{\{1\}\}$ .**

Answer:

Squared, exactly like Coulomb's law. That is the point Test 2 question 12 makes with its gap. Do not confuse it with the whole straight WIRE, whose field falls only as 1 over  $r$ , because summing the contributions of every element along an infinite line changes the power.

A circular loop of  $N$  turns and radius  $R$  carries current  $I$ . The field at its centre is:

✓  $B = \frac{\mu_0 N I}{2R}$

Correct.  $B = \frac{\mu_0 N I}{2R}$ , with no  $\pi$  in the denominator, which is what separates the flat loop from the straight wire and the toroid.

✗  $B = \frac{\mu_0 N I}{2 \pi R}$

Wrong. That  $\pi$  belongs to the toroid, not the flat loop.

✗  $B = \mu_0 n I$ , with  $n$  the turns per metre

Wrong. That is the solenoid.

✗  $B = \frac{\mu_0 I}{2 \pi R}$

Wrong. That is the long straight wire.

✗  $B = \frac{\mu_0 N I R}{2}$

Wrong.  $R$  divides here; it does not multiply.

$B = \frac{\mu_0 N I}{2R}$ , with no  $\pi$  in the denominator, which is what distinguishes it from the straight wire and the toroid. Note the  $R$

here is the radius of the LOOP itself, not a distance you choose, because this formula gives the field at one specific place: the centre.

Ampere's law states that the closed line integral of  $\mathbf{B} \cdot d\mathbf{l}$  equals  $\mu_0$  times the  $\{1\}$  current.

Answer:

The ENCLOSED current, meaning the current threading through the loop you have drawn. Currents outside the loop contribute nothing to the integral, exactly as external charges contribute nothing to the flux in Gauss's law. The two laws are the same idea in their two subjects.

The permeability of free space is written  $\mu_0$  and equals  $4 \pi$  times  $10$  to the power minus  $\{1\}$  H per m.

Answer:

Minus 7. Keeping  $\mu_0$  as  $4 \pi \times 10^{-7}$  lets the  $\pi$  cancel in  $B = \mu_0 I / (2 \pi r)$ .

**Ampère's law is most useful for conductors with high symmetry.**

Answer:

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Ampère's law says that if you walk right around any closed loop and add up the magnetic field along your path, the total equals  $\mu_0$  times the current threaded through that loop. It is ALWAYS true. But being true and being useful are different things. To actually get B out of it you need the field to have the same size everywhere on your loop and to point along it, so it can be pulled outside the sum. That only happens when the arrangement has a shape that repeats itself: a long straight wire, a solenoid, a toroid.

**Find:** The word that completes: conductors with high \_\_\_\_

#### THE WORKING, LINE BY LINE

1. The answer is SYMMETRY.
2. Why not resistance: Ampère's law contains no resistance at all, only current and field.
3. Why not permeability:  $\mu_0$  appears as a constant of the medium, but a high value does not make the law easier to solve.
4. Why not current: a big current gives a big field, but it does nothing to make the field uniform along a loop, which is the actual difficulty.

#### CHECK IT

Test it on a real case. For a long straight wire you can draw a circle around it on which B is constant and tangential, so the law collapses to  $B \times 2\pi r = \mu_0 I$  and B falls straight out. Try the same on a bent, lumpy conductor and there is no such circle, so the law still holds but gives you nothing. Symmetry is what made the difference.

Symmetry. Ampere's law is always true, but it is only USEFUL when you can choose a loop on which the field has the same magnitude everywhere and points along the loop, so that the integral collapses to B times the length. That needs symmetry: a long straight wire, a solenoid, a toroid. Without it the law still holds but cannot be solved for B, which is exactly the same reason Gauss's law is only useful for spherical, cylindrical or planar charge distributions.

According to the Biot-Savart law, the magnetic field around a current element decreases with the square of the  $\{1\}$ .

Answer:

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

The Biot-Savart law is the magnetic twin of Coulomb's law. Coulomb tells you the electric field produced by one small piece of charge; Biot-Savart tells you the magnetic field produced by one small piece of current. Both spread their influence over the surface of a sphere as they travel outwards, and a sphere's area grows as  $r^2$ , so both are diluted by the same factor.

**Find:** The word that completes: decreases with the square of the \_\_\_\_

#### FORMULA

$$dB = \frac{\mu_0 I \, dl \, \sin \theta}{(4\pi r^2)}$$

**Why this one.** The  $r^2$  sitting on the bottom is the whole answer, and it is worth writing the law out to see it rather than trying to recall a sentence about it. The same  $r^2$  appears in Coulomb's law for the same geometric reason, so the two laws can be remembered as one idea.

#### WHAT EACH SYMBOL MEANS

- $dB$  = the tiny field contributed by one small piece of the wire
- $I$  = the current in the wire, in amperes
- $dl$  = the length of that small piece, in metres
- $r$  = the distance from that piece to the point of interest, and it is SQUARED
- $\theta$  = the angle between the piece of wire and the line joining it to the point

#### THE WORKING, LINE BY LINE

1. The answer is DISTANCE.
2. Why not current: the field is directly proportional to current, not inversely, and there is no square on it.
3. Why not permeability:  $\mu_0$  is a constant and does not vary from point to point.
4. Why not angle: the angle enters through  $\sin \theta$ , which is not a square at all.

#### CHECK IT

Guard against a real trap here. One small ELEMENT of current gives a field falling as  $1/r^2$ , but a whole infinite straight WIRE gives a field falling as  $1/r$ , because it is the sum of countless elements stretching away in both directions. The question says current element, so the square is right; had it said long straight wire, it would not be.

Distance. The Biot-Savart law carries a  $1/r^2$  factor, so the contribution of a current element falls off with the square of the distance from it, exactly as Coulomb's law does for charge. Do not confuse this with the field of a whole long straight wire, which falls off as  $1/r$  rather than  $1/r^2$ , because that is the sum of every element along an infinite line.

At a distance of 5 cm from a long straight wire, the magnetic field measured is  $4.0 \times 10^{-5}$  T. Calculate the current  $I$  flowing through the wire. (Use  $\mu_0 = 4\pi \times 10^{-7}$  T/A)

✓ 10 A

Correct.  $I = 2 \pi r B / \mu_0 = 10$  A once  $\pi$  cancels.

✗ 20 A

Wrong, and exactly double: this drops the 2 in the denominator of the field formula.

✗ 3 A

Wrong. No route to 3 A from these numbers.

✗ 15 A

Wrong. No route to 15 A from these numbers.

✗ 12 A

Wrong. No route to 12 A from these numbers.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Current in a long straight wire wraps a magnetic field around itself in circles, like ripples round a stick in water. Grip the wire with your right hand, thumb along the current, and your fingers curl the way the field points. The field weakens as you move away, but only as  $1/r$ , not  $1/r^2$ , because the wire is a line rather than a point and there is always more of it stretching away to help.

**GIVEN**

- $r = 5 \text{ cm} = 0.05 \text{ m}$  (distance from the wire; convert first)
- $B = 4.0 \times 10^{-5} \text{ T}$  (the measured field)
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$

**Find:** The current  $I$  in the wire, in amperes

**FORMULA**

$$B = \frac{\mu_0 I}{(2\pi r)}$$

**Why this one.** Long straight wire is the signature phrase for this formula. Its distinguishing mark against the solenoid is the  $2\pi r$  underneath, which is the circumference of the circular field line you are standing on: the field spreads itself evenly around that circle, so the longer the circle, the weaker the field.

**WHAT EACH SYMBOL MEANS**

- $B$  = the field at distance  $r$ , in teslas
- $\mu_0$  = the permeability of free space,  $4\pi \times 10^{-7}$
- $I$  = the current in the wire, in amperes
- $r$  = the perpendicular distance from the wire, in metres, NOT squared
- $2\pi r$  = the circumference of the field circle at that distance

**THE WORKING, LINE BY LINE**

1. Convert:  $5 \text{ cm} = 0.05 \text{ m}$
2. Rearrange:  $I = (2\pi r B) \div \mu_0$
3.  $I = (2\pi \times 0.05 \times 4.0 \times 10^{-5}) \div (4\pi \times 10^{-7})$
4.  $\pi$  cancels top and bottom, which is the tidy way to do it:
5.  $I = (2 \times 0.05 \times 4.0 \times 10^{-5}) \div (4 \times 10^{-7})$
6. Top:  $2 \times 0.05 = 0.1$ , then  $0.1 \times 4.0 \times 10^{-5} = 4.0 \times 10^{-6}$
7.  $I = (4.0 \times 10^{-6}) \div (4 \times 10^{-7})$
8. Front numbers:  $4 \div 4 = 1$
9. Powers of ten:  $10^{-6} \div 10^{-7} = 10^1$
10.  $I = 1 \times 10 = 10 \text{ A}$

**CHECK IT**

Cancelling  $\pi$  before touching the calculator removes the commonest slip in this question, which is keying  $4\pi$  as 4 and then multiplying by  $\pi$  again. Sanity: 10 A is a large but ordinary current, and a field of tens of microteslas at 5 cm is the right order, being comparable to the Earth's own field.

For a long straight wire  $B = \mu_0 I / (2\pi r)$ , so  $I = 2\pi r B / \mu_0$ . Substituting with  $r = 0.05 \text{ m}$ :  $I = (2\pi \times 0.05 \times 4 \times 10^{-5}) / (4\pi \times 10^{-7})$ . The  $\pi$  cancels top and bottom, leaving  $I = (2 \times 0.05 \times 4 \times 10^{-5}) / (4 \times 10^{-7}) = 4 \times 10^{-6} / 4 \times 10^{-7} = 10 \text{ A}$ . Cancelling  $\pi$  before substituting saves the arithmetic.

8.13

MEDIUM

Reference sheet R.1

At the centre of a loop of  $N$  turns and radius  $R$  the field is  $B = \frac{\mu_0 N I}{2R}$ , a formula with no  $\pi$  in its denominator.

Answer:

No  $\pi$ . That is the quickest way to tell the flat loop apart from the straight wire,  $B = \mu_0 I$  over  $2 \pi r$ , and the toroid,  $B = \mu_0 N I$  over  $2 \pi r$ . The  $R$  here is the radius of the loop itself, since the formula gives the field at one particular place: its centre.

8.14

MEDIUM

Reference sheet R.1

The toroid formula  $B = \mu_0 N I$  over  $2 \pi r$  uses the TOTAL number of turns, not the turns per metre.

Answer: **true**

True, and the inconsistency with the solenoid is deliberate on the examiner's part. Solenoid uses small  $n$ , turns per metre. Toroid uses capital  $N$ , the total count. Reading which symbol the formula carries, before substituting, is the whole defence.

**A long straight conductor carries a current of 12 A. Calculate the magnetic field at a point 5 cm from the conductor.**

✓  $4.8 \times 10^{-5} \text{ T}$

Correct.  $2 \times 10^{-7} \times 12$  divided by 0.05.

✗  $4.8 \times 10^{-7} \text{ T}$

Wrong by a factor of 100, which is leaving r as 5 rather than 0.05.

✗  $2.4 \times 10^{-5} \text{ T}$

Wrong by a factor of two, from using  $\mu_0$  over  $4\pi$  instead of  $2\pi$ .

✗  $4.8 \times 10^{-3} \text{ T}$

Wrong by a factor of 100 the other way.

✗  $1.5 \times 10^{-4} \text{ T}$

Wrong. No route to this from these numbers.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Current in a straight wire wraps a magnetic field around itself in circles. Grip the wire with your right hand, thumb pointing along the current, and your fingers curl the way the field points. The strength falls off as 1 over r, not as the inverse square, because a wire is a line rather than a point: move away and there is always more wire stretching off in both directions to make up for the distance.

### GIVEN

- $I = 12 \text{ A}$
- $r = 5 \text{ cm} = 0.05 \text{ m}$
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$

**Find:** The magnetic field B at that point, in teslas

### FORMULA

$$B = \frac{\mu_0 I}{(2\pi r)}$$

**Why this one.** Long straight conductor is the phrase that selects this formula. The  $2\pi r$  underneath is the circumference of the field circle you are standing on: the field spreads itself around that circle, so a bigger circle means a weaker field. The solenoid formula has no  $\pi$  underneath at all, which is the quickest way to tell the two apart.

### WHAT EACH SYMBOL MEANS

- B = the field at that distance, in teslas
- $\mu_0$  = the permeability of free space,  $4\pi \times 10^{-7}$
- I = the current, in amperes
- r = the perpendicular distance from the wire, in metres, NOT squared
- $2\pi r$  = the circumference of the circular field line

### THE WORKING, LINE BY LINE

1. Convert:  $5 \text{ cm} = 0.05 \text{ m}$
2.  $B = (4\pi \times 10^{-7} \times 12) \div (2\pi \times 0.05)$
3. Cancel  $\pi$  top and bottom before anything else:
4.  $B = (4 \times 10^{-7} \times 12) \div (2 \times 0.05)$
5. Simplify  $4 \div 2 = 2$ , so  $B = (2 \times 10^{-7} \times 12) \div 0.05$
6. Top:  $2 \times 12 = 24$ , so top =  $24 \times 10^{-7} = 2.4 \times 10^{-6}$
7.  $B = (2.4 \times 10^{-6}) \div 0.05$
8.  $B = 4.8 \times 10^{-5} \text{ T}$

### CHECK IT

Learn the shortcut this cancelling reveals:  $\mu_0 \div 2\pi$  is exactly  $2 \times 10^{-7}$ , so  $B = 2 \times 10^{-7} \times I \div r$  every time for a straight wire. That one substitution removes the commonest slip, which is keying  $4\pi$  as 4 and then multiplying by  $\pi$  again. Sanity:  $4.8 \times 10^{-5} \text{ T}$  is roughly the Earth's own field, which is the right order for a domestic current a few centimetres away.

Convert first:  $5 \text{ cm} = 0.05 \text{ m}$ . Then  $B = \mu_0 I$  over  $2 \pi r$ . Keep  $\mu_0$  as  $4 \pi \times 10^{-7}$  and the  $\pi$  cancels:  $B = (4 \pi \times 10^{-7} \times 12) / (2 \pi \times 0.05) = (2 \times 10^{-7} \times 12) / 0.05 = 2.4 \times 10^{-6} / 0.05 = 4.8 \times 10^{-5} \text{ T}$ . The shortcut worth memorising is that  $\mu_0$  over  $2 \pi$  is exactly  $2 \times 10^{-7}$ , so  $B = 2 \times 10^{-7} \times I$  over r for every straight wire.

**A straight conductor carries 15 A. Calculate the magnetic field 4 cm away.**

✓  $7.5 \times 10^{-5} \text{ T}$

Correct.  $3.0 \times 10^{-6}$  divided by 0.04.

✗  $3.75 \times 10^{-5} \text{ T}$

Wrong by a factor of two, from halving  $\mu_0$  somewhere.

✗  $7.5 \times 10^{-7} \text{ T}$

Wrong by a factor of 100, from leaving the distance in cm.

✗  $6.0 \times 10^{-5} \text{ T}$

Wrong. This uses 5 cm rather than the 4 cm given.

✗  $1.9 \times 10^{-4} \text{ T}$

Wrong. No route to this from these numbers.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

The same straight wire law, with different numbers, which is the point of setting it twice: the routine should become automatic. Convert the centimetres, use  $\mu_0 \div 2\pi = 2 \times 10^{-7}$  so no  $\pi$  ever reaches the calculator, then divide by the distance to the first power.

**GIVEN**

- $I = 15 \text{ A}$
- $r = 4 \text{ cm} = 0.04 \text{ m}$
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$

**Find:** The magnetic field  $B$  at 4 cm from the wire, in teslas

**FORMULA**

$$B = \frac{\mu_0 I}{(2\pi r)}$$

**Why this one.** A straight conductor and a perpendicular distance, so the same formula as before. Nothing about a bigger current or a shorter distance changes which law applies; they only change the numbers that go into it.

**WHAT EACH SYMBOL MEANS**

- $B$  = the field, in teslas
- $I$  = the current, in amperes
- $r$  = the distance from the wire, in metres
- $\mu_0 \div 2\pi = 2 \times 10^{-7}$  exactly, worth using as a single constant

**THE WORKING, LINE BY LINE**

1. Convert:  $4 \text{ cm} = 0.04 \text{ m}$
2. Use the shortcut form:  $B = (2 \times 10^{-7} \times I) \div r$
3.  $B = (2 \times 10^{-7} \times 15) \div 0.04$
4. Top:  $2 \times 15 = 30$ , so top =  $30 \times 10^{-7} = 3.0 \times 10^{-6}$
5.  $B = (3.0 \times 10^{-6}) \div 0.04$
6.  $B = 7.5 \times 10^{-5} \text{ T}$

**CHECK IT**

Compare with the previous question as a cross-check on the physics. There, 12 A at 5 cm gave  $4.8 \times 10^{-5} \text{ T}$ . Here the current is 1.25 times larger and the distance 0.8 times as far, so the field should be  $1.25 \div 0.8 = 1.5625$  times stronger. And  $4.8 \times 1.5625 = 7.5$ , which is exactly what came out. Proportional reasoning like that catches an arithmetic slip in seconds.

Same shortcut, new numbers.  $B = 2 \times 10^{-7} \times I$  over  $r = 2 \times 10^{-7} \times 15 / 0.04 = 3.0 \times 10^{-6} / 0.04 = 7.5 \times 10^{-5} \text{ T}$ . Compare it with the 12 A at 5 cm case: more current and a shorter distance both push the field up, and it duly rises from 4.8 to  $7.5 \times 10^{-5} \text{ T}$ . A sanity check that costs nothing.

A solenoid uses  $B = \mu_0 n I$  and a toroid uses  $B = \frac{\mu_0 N I}{2 \pi r}$ . What is the difference between the  $n$  and the  $N$ ?

✓  $n$  is turns PER METRE, while  $N$  is the TOTAL number of turns

Correct, and it is the trap the examiner sets.

✗ They mean the same thing, written differently

Wrong. One is a density and the other a count, and they differ here by a factor of 2.5.

✗  $n$  is the total turns and  $N$  is turns per metre

Wrong, and exactly reversed.

✗  $n$  counts layers and  $N$  counts turns

Wrong. Neither counts layers.

✗  $n$  is for a single loop and  $N$  for many

Wrong. Both formulas describe many turns.

Small  $n$  is a DENSITY, turns per metre, found as  $N$  divided by the length. Capital  $N$  is a plain count. Test 2 question 9 punishes the confusion directly: 800 turns over 0.4 m gives  $n = 2000$  per metre, and putting 800 into the solenoid formula produces an answer that is on the option list. The solenoid also gives a UNIFORM field inside, which is why it is the standard way to make one.

A solenoid of length 0.4 m has 800 turns and produces a magnetic field of  $2.5 \times 10^{-4}$  T inside it. Calculate the current flowing through the solenoid. (Use  $\mu_0 = 4\pi \times 10^{-7}$  T/A)

✓ 0.1 A

Correct.  $n = 800/0.4 = 2000$  per metre, so  $I = B/(\mu_0 n) = 0.1$  A.

✗ 2.6 A

Wrong. This is roughly what you get using  $N = 800$  directly instead of  $n = 2000$  per metre.

✗ 5.9 A

Wrong. No route to this value from these numbers.

✗ 0.6 A

Wrong. No route to this value; it is six times the answer.

✗ 3.1 A

Wrong. No route to this value from these numbers.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A solenoid is a wire wound into a long tube. Each turn makes its own small circular field, and stacked side by side they reinforce each other inside the tube and largely cancel outside it. The result is a nearly uniform field down the middle, like a bar magnet made of wire. What matters is not how many turns you have but how tightly they are packed, which is why the formula uses turns PER METRE.

### GIVEN

- $L = 0.4 \text{ m}$  (the length of the solenoid)
- $N = 800$  turns
- $B = 2.5 \times 10^{-4} \text{ T}$  (the field inside)
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$

**Find:** The current  $I$  flowing through the solenoid, in amperes

### FORMULA

$$B = \mu_0 n I, \text{ so } I = \frac{B}{(\mu_0 n)}$$

**Why this one.** The word solenoid picks this formula out from its neighbours by a single feature: there is no  $\pi$  on the bottom. The long straight wire has  $2\pi r$  underneath and the toroid has  $2\pi r$  too, but the solenoid does not, because its field does not spread out radially. Spot the word solenoid and you have already halved the work.

### WHAT EACH SYMBOL MEANS

- $B$  = the field inside the solenoid, in teslas
- $\mu_0$  = the permeability of free space,  $4\pi \times 10^{-7}$
- $n$  = the number of turns PER METRE, found as  $N \div L$
- $I$  = the current, in amperes
- $N$  = the total number of turns, which is NOT the same as  $n$

### THE WORKING, LINE BY LINE

1. First turn  $N$  into  $n$ , turns per metre:
2.  $n = N \div L = 800 \div 0.4 = 2000$  turns per metre
3. Bottom:  $\mu_0 \times n = 4\pi \times 10^{-7} \times 2000$
4.  $4\pi = 12.566$ , so  $12.566 \times 10^{-7} \times 2000 = 2.513 \times 10^{-3}$
5.  $I = B \div (\mu_0 n) = (2.5 \times 10^{-4}) \div (2.513 \times 10^{-3})$
6. Front numbers:  $2.5 \div 2.513 = 0.995$
7. Powers of ten:  $10^{-4} \div 10^{-3} = 10^{-1}$
8.  $I = 0.995 \times 10^{-1} = 0.0995 \approx 0.1 \text{ A}$

### CHECK IT

The commonest error is feeding in 800 instead of 2000, which would make the current 2.5 times larger, about 0.25 A, and none of the options match that. Whenever a solenoid question gives a LENGTH, that length exists to convert  $N$  into  $n$ . If you never used it, you have made this mistake.

Inside a solenoid  $B = \mu_0 n I$ , where  $n$  is turns PER METRE, not the total number of turns. First  $n = N/L = 800/0.4 = 2000$  turns per metre. Then  $I = B/(\mu_0 n) = 2.5 \times 10^{-4} / (4\pi \times 10^{-7} \times 2000) = 2.5 \times 10^{-4} / 2.513 \times 10^{-3} = 0.0995$ , that is 0.1 A. Using 800 instead of 2000 is the mistake this question is built to catch.

A solenoid is 0.5 m long and has 1500 turns. What value of  $n$  goes into  $B = \mu_0 n I$ ?

✓ 3000 per metre

Correct. 1500 divided by 0.5.

✗ 1500

Wrong, and it is the trap: 1500 is  $N$ , the total count, not the density.

✗ 750 per metre

Wrong. This MULTIPLIES by 0.5 instead of dividing.

✗ 0.5

Wrong. That is the length in metres.

✗ 3000 turns in total

Wrong. The number is right but the meaning is not: there are 1500 turns in total, and 3000 is how many there would be per metre.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

In  $B = \mu_0 n I$  the letter  $n$  does not mean the number of turns. It means turns PER METRE, the packing density. This matters because the field inside a solenoid depends on how tightly the wire is wound, not on how much wire there is: stretch the same 1500 turns over twice the length and the field halves.

#### GIVEN

- length  $L = 0.5$  m
- $N = 1500$  turns in total

**Find:** The value of  $n$  to put into  $B = \mu_0 n I$ , with its unit

#### FORMULA

$$n = \frac{N}{L}$$

**Why this one.** Whenever a solenoid question gives you a LENGTH, that length is there to convert the total turns into turns per metre. If you finish a solenoid problem without having used the length, you have almost certainly fed in  $N$  by mistake.

#### WHAT EACH SYMBOL MEANS

- $n$  = turns per metre, with the unit  $\text{m}^{-1}$
- $N$  = the total number of turns, a pure count with no unit
- $L$  = the length of the solenoid, in metres

#### THE WORKING, LINE BY LINE

1.  $n = N \div L$
2.  $n = 1500 \div 0.5$
3. Dividing by 0.5 is the same as multiplying by 2
4.  $n = 3000$  per metre

#### CHECK IT

The unit is the mark here. Two of the options give the number 3000, but one of them calls it 3000 turns in total, which is wrong: there are only 1500 turns, packed at 3000 per metre because the solenoid is half a metre long. Read the unit on the option, not just the digits. Note too that  $n$  came out larger than  $N$ , which always happens when the solenoid is shorter than a metre.

$n$  is turns PER METRE, so  $n = N \text{ over } L = 1500 \text{ over } 0.5 = 3000 \text{ per metre}$ . Dividing by a number less than one makes it larger, which catches people out. Test 2 question 9 sets the same trap with 800 turns over 0.4 m, and putting the raw turn count in produces an answer that is on the option list.

## 9 Electromagnetic Induction

**Speed tip.** Nothing changing means no emf, whatever the field strength. Faraday needs a RATE, so look for a time in the question.

9.1

EASY

Reference sheet R.4

**Magnetic flux is measured in webers, and one weber is one tesla  $\text{T}$ .**

Answer:

One tesla square metre, which you can read straight off  $\Phi = BA \cos \theta$ : a flux density in tesla times an area in square metres. Keep the two units apart: the TESLA measures flux density  $B$ , the WEBER measures flux  $\Phi$ , and Test 2 question 11 depends on knowing that it is the FLUX that must change to induce an emf.

**A 2 H inductor carries 5 A of current. What is the stored energy?**

✓ 25 J

Correct.  $U = \frac{1}{2} L I^2 = 0.5 \times 2 \times 25 = 25 \text{ J}$ .

✗ 15 J

Wrong. No route to 15 J from  $L = 2 \text{ H}$  and  $I = 5 \text{ A}$ .

✗ 20 J

Wrong. This squares the inductance instead of the current:  $0.5 \times 4 \times 10$  or similar.

✗ 10 J

Wrong, and it is the common slip:  $L \times I = 2 \times 5 = 10$ , forgetting both the half and the square.

✗ 30 J

Wrong. No route to 30 J from these numbers.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A coil does not store energy in the wire, it stores it in the magnetic field filling the space around and inside the coil. Building that field takes work, because while the current is rising the coil fights back with an induced emf that opposes the change. Once the current is steady the fighting stops and the energy sits in the field, ready to be given back the instant you open the switch. That is why a switch sparks: the field collapses and dumps its energy across the gap.

#### GIVEN

- $L = 2 \text{ H}$  (the inductance, in henries)
- $I = 5 \text{ A}$  (the steady current in the coil)

**Find:** The energy  $U$  stored in the inductor, in joules (J)

#### FORMULA

$$U = \frac{1}{2} L I^2$$

**Why this one.** An inductance and a current are given and an energy is asked for, and this is the only formula joining those three. Notice its shape: a half, a property of the device, and the square of what flows through it. The capacitor version  $U = \frac{1}{2} C V^2$  has exactly the same shape, which is worth memorising as one pattern rather than two formulas.

#### WHAT EACH SYMBOL MEANS

- $U$  = the stored magnetic energy, in joules
- $L$  = the inductance, in henries (H)
- $I$  = the current, in amperes, and it is SQUARED

#### THE WORKING, LINE BY LINE

1.  $U = \frac{1}{2} \times L \times I^2$
2.  $I^2 = 5^2 = 25$
3.  $U = \frac{1}{2} \times 2 \times 25$
4.  $\frac{1}{2} \times 2 = 1$
5.  $U = 1 \times 25 = 25 \text{ J}$

#### CHECK IT

The current is squared, so doubling the current to 10 A would give 100 J, four times as much, not twice. If halving or doubling a current changes your energy by the same factor, you dropped the square.

Energy stored in an inductor is  $U = \frac{1}{2} L I^2$ , the magnetic twin of the capacitor's  $U = \frac{1}{2} C V^2$ . Substituting:  $U = \frac{1}{2} \times 2 \times 5^2 = 1 \times 25 = 25$  J. The current is SQUARED and the inductance is not, which is the only place this can go wrong.

9.3

EASY

The real test 2, question 11

**A changing magnetic field induces an emf.**

Answer:

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Faraday found that it is not the magnetic field that makes an emf, but the CHANGE in how much of it passes through a loop. Hold a magnet still inside a coil, however strong, and nothing happens; move it and a voltage appears. The quantity that has to change is the flux,  $\Phi = BA \cos \theta$ , which bundles three things at once: the strength of the field, the area of the loop, and the angle between them.

**Find:** The word that completes: a changing magnetic \_\_\_\_ induces an emf

**FORMULA**

$$\text{emf} = - \frac{N \Delta \Phi}{\Delta t}, \text{ where } \Phi = BA \cos \theta$$

**Why this one.** Writing the law out shows exactly why the missing word is flux and not field. Because flux contains B, A and  $\cos \theta$ , an emf can be produced in three different ways: change the field strength, change the area of the loop, or simply rotate the loop in a steady field. A generator uses the third, spinning a coil between fixed magnets, and no answer built on the word field can explain that.

**WHAT EACH SYMBOL MEANS**

- emf = the induced voltage, in volts
- N = the number of turns in the coil
- $\Phi$  = the magnetic flux, in webers (Wb)
- $\Delta \Phi \div \Delta t$  = the RATE of change of flux, which is what actually matters
- the minus sign = Lenz's law, saying the induced effect opposes the change that caused it

**THE WORKING, LINE BY LINE**

1. The answer is FLUX.
2. Why not force: a force is a push on a charge or a wire, and it has nothing to change through a loop.
3. Why not moment: a magnetic moment describes a current loop as a source of field, not what passes through one.
4. Why not permeability: that is a property of the material and it is not what varies with time here.

**CHECK IT**

Two words earn the marks in Faraday questions: RATE and FLUX. Say rate of change of flux, and you have stated the law. Say a strong magnetic field induces an emf and you have said something false, because a strong field held still induces nothing at all.

Flux. Faraday's law says the induced emf equals the RATE OF CHANGE of magnetic flux, not of the field itself:  $\text{emf} = - d\Phi/dt$ . The distinction earns marks, because flux is B times A times  $\cos(\theta)$ , so an emf can be induced by changing the field, by changing the area, or merely by rotating the loop, which is exactly how a generator works with a constant magnet.

9.4

EASY

Reference sheet R.4

One weber is equal to:

✓ One tesla square metre

Correct. A tesla times a square metre, exactly as  $\Phi = BA$  says.

✗ One tesla per square metre

Wrong, and inverted: the area multiplies, it does not divide.

✗ One tesla per metre

Wrong. The area is squared metres and it multiplies.

✗ One tesla per second

Wrong. That would be a rate of change of flux density.

✗ One ampere metre squared

Wrong. That is the unit of MAGNETIC MOMENT,  $m = NIA$ .

A tesla square metre, straight from  $\Phi = BA \cos \theta$ : a flux density in tesla times an area in square metres. Keep the pair apart, because the tesla measures flux DENSITY while the weber measures FLUX, and Faraday's law is about the flux changing.

9.5

MEDIUM

Reference sheet R.1

A conductor of length  $L$  moves at speed  $v$  through a magnetic field  $B$ , cutting the field lines. The emf induced is:

✓  $BLv$

Correct. All three multiply.

✗  $\frac{BL}{v}$

Wrong. A faster rod induces MORE emf, so  $v$  cannot be in the denominator.

✗  $\frac{B}{Lv}$

Wrong on both placements.

✗  $BLv$  squared

Wrong. The speed enters to the first power only.

✗  $\frac{L}{Bv}$

Wrong. This inverts almost everything.

$\text{emf} = BLv$ , the three quantities simply multiplied. It is Faraday's law in disguise: as the rod moves it sweeps out area at a rate  $Lv$ , so the flux through the circuit changes at  $BLv$ , and that rate of change IS the emf. Stop the rod and the emf stops instantly, because change is the whole word.

9.6

MEDIUM

Reference sheet R.1

The energy stored in an inductor is  $W = \text{one half } L I^2$ , in which the quantity that is SQUARED is the  $\{I\}$ .

Answer:

The current.  $W = \text{one half } L I^2$  mirrors one half  $C V^2$  for a capacitor, and the half is there for the same reason in both: the quantity climbed from zero rather than sitting at its final value throughout. Test 2 question 1 uses it, and its trap answer of 10 J is what you get by writing  $L$  times  $I$  and forgetting both the half and the square.

9.7

MEDIUM

Reference sheet R.4

**An eddy current is:**

✓ A loop of current induced inside a solid conductor by changing flux

Correct, and the name comes from the swirling pattern.

✗ The current in the primary coil of a transformer

Wrong. That is simply the primary current.

✗ A current that flows only in superconductors

Wrong. They appear in ordinary conductors.

✗ The leakage current through an insulator

Wrong. That is leakage, a different thing.

✗ A current that alternates at very high frequency

Wrong. Frequency is not what defines them.

Changing flux induces an emf in ANY conducting loop, and a solid block of metal is full of possible loops, so swirling currents appear inside it. They are Faraday's law applied where nobody wired a circuit. They waste energy as heat, which is why transformer cores are built from thin laminated sheets rather than solid metal, breaking up the paths the eddies would take.

9.8

MEDIUM

Reference sheet R.1

**A loop sits in a strong but perfectly STEADY magnetic field. What emf is induced in it?**

✓ None, because it is the rate of CHANGE of flux that induces an emf

Correct. A steady flux induces nothing.

✗ A large one, because the field is strong

Wrong. Strength is irrelevant; only change matters.

✗ A small but non-zero one

Wrong. It is exactly zero.

✗ One proportional to the area of the loop

Wrong. Area enters the flux, but a constant flux still has zero rate of change.

✗ One proportional to the field strength

Wrong, for the same reason.

Faraday's law is  $\text{emf} = \text{minus } N \, d\Phi/dt$ . If the flux is not changing, its rate of change is zero and so is the emf, no matter how strong the field is. CHANGE is the whole word: the field can be enormous and still induce nothing.

9.9

MEDIUM

Reference sheet R.2

**In Faraday's law the key word is  $\{\{1\}\}$ : a steady flux induces nothing.**

Answer:

Change. The law is about  $d\Phi/dt$ , the rate of change of flux, not about the flux itself.

9.10

MEDIUM

Reference sheet R.2

**The – sign in Faraday's law expresses Lenz's law, and it is required by conservation of energy.**

Answer: **true**

True. Without the opposition, an induced current would strengthen the very change that created it, and the system would generate energy from nothing.

**Faraday's law is  $\text{emf} = \text{minus } N d\Phi/dt$ , so what induces an emf is not the flux itself but its rate of {{1}}.**

Answer:

Change. A coil parked in the strongest steady magnet you can find generates nothing at all. Move it, turn it, or switch the field, and it does. That single word is why a generator must spin and why a transformer cannot run on DC.

A rod 0.25 m long moves at 4 m/s through a 0.2 T field, cutting the lines squarely. What emf is induced?

✓ 0.2 V

Correct.  $0.2 \times 0.25 \times 4$ .

✗ 5 V

Wrong. This divides somewhere rather than multiplying throughout.

✗ 0.05 V

Wrong. This omits the speed.

✗ 1.25 V

Wrong. No route to this from these numbers.

✗ 0.8 V

Wrong. This drops the field strength.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Push a rod across a magnetic field and the free electrons inside it are dragged sideways by the magnetic force, piling up at one end. That separation of charge is a voltage, and it lasts as long as the motion does. This is motional emf, and it is Faraday's law in disguise: as the rod moves, the area of the circuit it sweeps out grows, so the flux through that circuit changes.

**GIVEN**

- $L = 0.25 \text{ m}$ , the length of the rod
- $v = 4 \text{ m s}^{-1}$
- $B = 0.2 \text{ T}$ , and the rod cuts the lines SQUARELY

**Find:** The induced emf, in volts

**FORMULA**

$$\text{emf} = BLv$$

**Why this one.** Cutting the lines squarely means all three are mutually perpendicular, so there is no sine or cosine to apply and the three simply multiply. This is the shortcut form of Faraday's law for a straight conductor; the general  $\Delta\Phi \div \Delta t$  version would need the swept area and the time, which the question does not give directly.

**WHAT EACH SYMBOL MEANS**

- emf = the induced voltage, in volts
- $B$  = the flux density, in teslas
- $L$  = the length of rod inside the field, in metres
- $v$  = the speed at which it moves, in  $\text{m s}^{-1}$

**THE WORKING, LINE BY LINE**

1.  $\text{emf} = B \times L \times v$
2.  $\text{emf} = 0.2 \times 0.25 \times 4$
3.  $0.2 \times 0.25 = 0.05$
4.  $\text{emf} = 0.05 \times 4$
5.  $\text{emf} = 0.2 \text{ V}$

**CHECK IT**

Check by units, which is the reliable guard here: tesla  $\times$  metre  $\times$  (metre per second) works out to volts, so the three quantities were meant to be multiplied. Sense check the size too: a fifth of a volt from a quarter-metre rod at walking pace is a reminder of why real generators use many turns and spin fast, since one rod moving slowly produces very little.

$\text{emf} = BLv = 0.2 \times 0.25 \times 4 = 0.2 \text{ V}$ . All three quantities simply multiply. It is Faraday's law wearing a different hat: the rod sweeps area at a rate  $Lv$ , so the flux changes at  $BLv$ , and that rate IS the emf.

**9.13**

MEDIUM

Reference sheet R.1

**The – sign in  $\text{emf} = \text{minus } L \text{ d}\Phi/\text{d}t$  belongs to Lenz's law, which says the induced effect opposes the change that caused it.**

Answer:

Lenz's law, and it is conservation of energy in disguise. If the induced effect reinforced the change instead of opposing it, the smallest nudge would grow without limit and make energy from nothing. Nature always fights back.

**A 4 H inductor carries 3 A. What energy is stored in it?**

✓ 18 J

Correct. Half of  $4 \times 9$ .

✗ 12 J

Wrong. That is  $L$  times  $I$ , forgetting both the half and the square.

✗ 36 J

Wrong. That is  $L I^2$  with the half forgotten.

✗ 6 J

Wrong. This halves  $L$  times  $I$  rather than squaring the current.

✗ 1.5 J

Wrong. This squares nothing and halves twice.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

An inductor stores energy in the magnetic field it creates, and the field cannot appear instantly: while the current is climbing, the coil induces a back-emf opposing the climb, and the work done against that back-emf is what ends up stored. Open the switch and the field collapses, returning the energy, which is why switching off an inductive load makes a spark.

**GIVEN**

- $L = 4 \text{ H}$
- $I = 3 \text{ A}$

**Find:** The energy  $U$  stored in the inductor, in joules

**FORMULA**

$$U = \frac{1}{2}LI^2$$

**Why this one.** An inductance and a current given, an energy wanted, and this is the only relation between the three. Notice the family resemblance to the capacitor's  $U = \frac{1}{2}CV^2$ : a half, the device property, and the square of what flows through or sits across it. Remembering one pattern gives you both.

**WHAT EACH SYMBOL MEANS**

- $U$  = the stored energy, in joules
- $L$  = the inductance, in henries
- $I$  = the current, in amperes, and it is SQUARED

**THE WORKING, LINE BY LINE**

1.  $U = \frac{1}{2} \times L \times I^2$
2.  $I^2 = 3^2 = 9$
3.  $U = \frac{1}{2} \times 4 \times 9$
4.  $\frac{1}{2} \times 4 = 2$
5.  $U = 2 \times 9 = 18 \text{ J}$

**CHECK IT**

The option 12 J is what you get by forgetting the square and computing  $\frac{1}{2} \times 4 \times 3 \times 2$ , and 36 J is what you get by forgetting the half. Both wrong routes are in the list, so make sure both the half and the square survived. Guard by proportion: triple the current and the energy would rise ninefold, not threefold.

$W = \text{one half } L I^2 = 0.5 \times 4 \times 9 = 18 \text{ J}$ . The CURRENT is squared and the inductance is not. It mirrors one half  $C V^2$  for a capacitor, and the half is there for the same reason: the current climbed from zero rather than sitting at 3 A throughout.

9.15

MEDIUM

Reference sheet R.4

**Eddy currents are loops of current induced inside a solid conductor by changing flux, and laminating a transformer core reduces them.**

Answer: **true**

True on both. A solid block of metal is full of possible loops, so changing flux sets swirling currents going inside it, wasting energy as heat. Building the core from thin insulated sheets breaks up the paths those eddies would take, which is why transformer cores are laminated rather than solid.

9.16

MEDIUM

Reference sheet R.1

**For a conductor moving through a magnetic field the induced emf is  $BLv$ , where  $B$  is the flux density,  $L$  is the length of the conductor and  $v$  is its  $\{\{1\}\}$ .**

Answer:

Speed. Three quantities, all multiplied, no squares and no constants: park the rod and  $v$  is zero, so the emf is zero, which is Faraday's law again. Standing still through the strongest field on earth induces nothing.

A coil of 200 turns experiences a change in magnetic flux from 0.08 Wb to 0.02 Wb in 0.10 s. Calculate the induced emf.

✓ 120 V

Correct. 200 turns times a rate of 0.6 Wb per second.

✗ 12 V

Wrong by a factor of ten in the division by 0.10 s.

✗ 0.6 V

Wrong. 0.6 V is the rate of change of flux, before multiplying by the 200 turns.

✗ 200 V

Wrong. 200 is the number of turns, not a voltage.

✗ 20 V

Wrong. This uses a flux change of 0.01 Wb rather than 0.06 Wb.

**BACKGROUND, SO THIS IS NOT JUST A FORMULA**

Faraday's law is about the RATE at which flux changes, not about how much flux there is. A coil sitting in the strongest field in the world produces nothing if nothing is changing. What is more, every turn of the coil experiences the same change, and their emfs add in series, so a coil of 200 turns produces 200 times the emf of a single loop. That is the whole reason coils are wound rather than left as single rings.

**GIVEN**

- $N = 200$  turns
- Flux falls from  $\Phi_1 = 0.08$  Wb to  $\Phi_2 = 0.02$  Wb
- $\Delta t = 0.10$  s

**Find:** The magnitude of the induced emf, in volts

**FORMULA**

$$\text{emf} = \frac{N \Delta\Phi}{\Delta t}$$

**Why this one.** A flux change, a time and a number of turns are exactly the three quantities Faraday's law needs. The minus sign belongs to Lenz's law and states the direction, meaning the induced effect opposes the change that caused it; since the question asks only for the size, work with the magnitude of the change.

**WHAT EACH SYMBOL MEANS**

- emf = the induced voltage, in volts
- $N$  = the number of turns, which MULTIPLIES the result
- $\Delta\Phi$  = the change in flux, in webers, taken as final minus initial
- $\Delta t$  = the time over which the change happens, in seconds

**THE WORKING, LINE BY LINE**

1. Change in flux:  $\Delta\Phi = 0.08 - 0.02 = 0.06$  Wb
2. Rate of change:  $\Delta\Phi \div \Delta t = 0.06 \div 0.10 = 0.6$  Wb  $\text{s}^{-1}$
3. Multiply by the turns:  $\text{emf} = 200 \times 0.6$
4.  $\text{emf} = 120$  V

**CHECK IT**

The option 0.6 V is the rate of change before multiplying by the turns, which is the commonest omission, and it is sitting there waiting. Ask two questions of every Faraday problem: did I divide by the TIME, and did I multiply by the TURNS. Note also that a modest flux change of 0.06 Wb yields 120 V purely because of the 200 turns, which is exactly how a real generator reaches useful voltages.

Faraday's law, size first. The CHANGE in flux is  $0.08 - 0.02 = 0.06$  Wb, and it happens in 0.10 s, so the rate of change is  $0.06 / 0.10 = 0.6$  Wb per second. Multiply by the 200 turns:  $\text{emf} = N \times \text{rate} = 200 \times 0.6 = 120$  V. Two traps live here. The flux VALUES do not go into the formula, only their difference. And the number of turns multiplies, because each turn is its own loop cut by the same changing flux.

**The self-induced emf in an inductor is  $\text{emf} = \text{minus } L \, dI/dt$ . What does the  $-$  sign express?**

✓ Lenz's law: the induced effect opposes the change that produced it

Correct, and the same  $-$  sign appears in Faraday's law for the same reason.

✗ That the current always flows backwards through an inductor

Wrong. The current flows whichever way the circuit drives it.

✗ That the inductance  $L$  is a negative quantity

Wrong. Inductance is always positive.

✗ That energy is lost as heat

Wrong. Heat loss is resistance, not inductance.

✗ Nothing physical; it is a sign convention only

Wrong. It carries the physics, and dropping it would permit perpetual motion.

Lenz's law, and it is conservation of energy wearing a  $-$  sign. If the induced effect REINFORCED the change instead of opposing it, the smallest disturbance would grow without limit and make energy from nothing. Nature always fights back: try to increase the current and the inductor pushes against you, which is why a current cannot change instantly in an inductor.

**What does Lenz's law say, and which conservation principle is behind it?**

✓ The induced effect opposes the change that produced it, and it follows from conservation of energy

Correct on both.

✗ The induced effect reinforces the change that produced it, and it follows from conservation of charge

Wrong. Reinforcement would create energy from nothing.

✗ The induced current is always clockwise, by convention

Wrong. The direction depends on the change, not on a convention.

✗ The induced emf equals the flux rather than its rate of change

Wrong. It is the rate of change that matters.

✗ It applies only to transformers

Wrong. It applies to every induced emf.

Lenz opposes. The  $-$  sign in  $\text{emf} = \text{minus } N \, d\Phi/dt$  is Lenz's law, and the reason is conservation of energy: if the induced effect reinforced the change instead, the smallest disturbance would grow without limit and produce energy from nothing. Nature always fights back.

---

## 10 Transformers

**Speed tip.** Step the voltage up and the current steps down. Power in equals power out if it is ideal.

**A transformer has 100 turns on the primary and 1000 on the secondary, with 240 V applied to the primary. Is it a step up or a step down, and what is the secondary voltage?**

✓ Step up, 2400 V

Correct. Ten times the turns gives ten times the voltage.

✗ Step down, 24 V

Wrong direction. More secondary turns means MORE voltage, not less.

✗ Step up, 240 V

Wrong. That is the primary voltage unchanged.

✗ Step down, 2.4 V

Wrong on both direction and size.

✗ Step up, 24000 V

Wrong by a factor of ten, from a turns ratio of 100.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A transformer trades voltage against current using only the ratio of turns on its two coils. The alternating current in the primary makes a changing flux in the iron core, the core carries that flux to the secondary, and each secondary turn picks up an equal share. More turns therefore means more volts. Nothing is created: what the voltage gains, the current gives up.

### GIVEN

- $N_p = 100$  turns on the primary
- $N_s = 1000$  turns on the secondary
- $E_p = 240$  V applied to the primary

**Find:** Whether it steps up or down, and the secondary voltage  $E_s$

### FORMULA

$$E_s / E_p = N_s / N_p$$

**Why this one.** The turns ratio and one voltage are given and the other voltage is wanted. Read the ratio before calculating:  $N_s$  is ten times  $N_p$ , so this is a step UP and the answer must exceed 240 V. That reading alone eliminates most of the options.

### WHAT EACH SYMBOL MEANS

- $E_s$  = the secondary or output voltage, in volts
- $E_p$  = the primary or input voltage, in volts
- $N_s$  = turns on the secondary coil
- $N_p$  = turns on the primary coil

### THE WORKING, LINE BY LINE

1. Turns ratio:  $N_s \div N_p = 1000 \div 100 = 10$
2. More turns on the secondary, so it STEPS UP
3.  $E_s = E_p \times 10$
4.  $E_s = 240 \times 10$
5.  $E_s = 2400$  V

### CHECK IT

Decide the direction first and the arithmetic becomes a formality: step up rules out both step-down options immediately, and among what remains the ratio is 10, not 1 and not 100. Remember that stepping the voltage up steps the current DOWN by the same factor, so this transformer would deliver a tenth of the primary current, since power cannot be manufactured by winding wire.

The voltage ratio equals the turns ratio:  $V_s \text{ over } V_p = N_s \text{ over } N_p = 1000 \text{ over } 100 = 10$ . So  $V_s = 240 \times 10 = 2400$  V. Decide step up or step down FIRST, from whether  $N_s$  exceeds  $N_p$ , and three of the five options disappear before any arithmetic. Test 2 question 15 is exactly this.

In one type of transformer,  $N_s$  is greater than  $N_p$ . If  $N_s = 1000$ ,  $N_p = 100$  and  $E_p = 240$  V, what is the value of  $E_s$ ?

✓ 2400 V

Correct.  $E_s = E_p \times N_s/N_p = 240 \times 10 = 2400$  V.

✗ 24 V

Wrong, and it steps DOWN by ten, which contradicts  $N_s$  being greater than  $N_p$ .

✗ 240 V

Wrong. This is the primary voltage unchanged, which would need  $N_s = N_p$ .

✗ 2.4 V

Wrong. This steps down by a hundred, the opposite of what the stem describes.

✗ 24000 V

Wrong. This steps up by a hundred, using the turns as 1000 and 10.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A transformer trades voltage against current using nothing but the number of turns on each coil. The primary's alternating current makes a changing flux in the iron core, the core carries that flux to the secondary, and every turn on the secondary picks up the same share of it. More turns therefore means more volts. Nothing is gained: whatever the voltage goes up by, the current comes down by, because power in equals power out in the ideal case.

#### GIVEN

- $N_s = 1000$  turns (secondary)
- $N_p = 100$  turns (primary)
- $E_p = 240$  V (primary voltage)

**Find:** The secondary voltage  $E_s$ , in volts

#### FORMULA

$$E_s / E_p = N_s / N_p$$

**Why this one.** The turns ratio and one voltage are given and the other voltage is wanted, which is exactly what this relation provides. The question also tells you which way the answer must go before any arithmetic:  $N_s$  is greater than  $N_p$ , so this is a STEP-UP transformer and the secondary voltage must be larger than 240 V.

#### WHAT EACH SYMBOL MEANS

- $E_s$  = the secondary or output voltage, in volts
- $E_p$  = the primary or input voltage, in volts
- $N_s$  = the number of turns on the secondary coil
- $N_p$  = the number of turns on the primary coil

#### THE WORKING, LINE BY LINE

1. Rearrange:  $E_s = E_p \times (N_s \div N_p)$
2. Turns ratio:  $N_s \div N_p = 1000 \div 100 = 10$
3.  $E_s = 240 \times 10$
4.  $E_s = 2400$  V

#### CHECK IT

Decide step-up or step-down BEFORE calculating and half the options die immediately.  $N_s$  is bigger than  $N_p$ , so the answer must be bigger than 240 V, which eliminates 24 V, 240 V and 2.4 V in one stroke. That leaves only 2400 V and 24000 V, and the ratio is 10, not 100.

The turns ratio equals the voltage ratio:  $E_s/E_p = N_s/N_p$ . So  $E_s = E_p \times (N_s/N_p) = 240 \times (1000/100) = 240 \times 10 = 2400 \text{ V}$ . The stem tells you  $N_s$  is greater than  $N_p$ , which means this is a STEP-UP transformer, so the answer must be larger than 240 V. That single check eliminates three of the five options before any arithmetic.

10.3

MEDIUM

Reference sheet R.1

**In a transformer the voltage ratio equals the turns ratio,  $V_s$  over  $V_p = N_s$  over  $N_p$ , and it works on AC only.**

Answer:

AC only, because a transformer needs a CHANGING flux to induce anything in the secondary. Deciding step up or step down before any arithmetic, simply from whether  $N_s$  exceeds  $N_p$ , eliminates most of the options on a multiple-choice paper.

10.4

MEDIUM

Reference sheet R.4

**The two coils of a transformer are connected to each other electrically.**

Answer: **false**

They are not. The coils are coupled by MUTUAL INDUCTION through the shared core: the changing flux from the primary induces an emf in the secondary without any conducting path between them. That electrical isolation is a feature rather than an accident, and it is why transformers are used to separate circuits as well as to change voltage.

10.5

HARD

Reference sheet R.4

**A transformer is two coils coupled by mutual induction. Why does it work on AC but not on DC?**

✓ Because it needs a CHANGING flux to induce anything, and a steady direct current gives a steady flux

Correct, and it is the same sentence that governs every induction question on the paper.

✗ Because DC would melt the windings

Wrong. Heating is not the reason, and a transformer on DC simply does nothing.

✗ Because the coils are only connected electrically on AC

Wrong. The coils are never electrically connected; that is the point of a transformer.

✗ Because DC has no voltage

Wrong. DC certainly has a voltage.

✗ Because iron cores only magnetise under AC

Wrong. Iron magnetises perfectly well under DC.

Faraday's law again:  $\text{emf} = \text{minus } N \, d\Phi/dt$ , so it is the RATE OF CHANGE of flux that induces, not the flux itself. An alternating primary current makes a constantly changing flux in the core, which the secondary responds to. A steady direct current makes a steady flux, its rate of change is zero, and the secondary sees nothing at all. Note the two coils are never connected electrically in either case; the core carries the flux between them.

**A 240 V step-down transformer is used to light up eight 12 V, 24 W lamps and draws a current of 1 A in the primary. Calculate its efficiency.**

✓ 80 %

Correct. Output  $8 \times 24 = 192$  W over input  $240 \times 1 = 240$  W, giving 80 per cent.

✗ 60 %

Wrong. This would need an output of 144 W, that is six lamps rather than eight.

✗ 40 %

Wrong. This would need an output of 96 W, that is four lamps.

✗ 20 %

Wrong. This uses one lamp only:  $24/240$  wrongly scaled.

✗ 0 %

Wrong. A transformer lighting eight lamps is plainly transferring power, so zero cannot be right.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

A transformer is judged by how much of the power fed in actually comes out the other side. Nothing is perfect: some energy heats the core through eddy currents, some is lost reversing the magnetisation, some warms the windings. Efficiency is simply the fraction that survives the trip, written as a percentage. Note that a transformer changes voltage and current but can never create power.

### GIVEN

- Primary voltage  $V_p = 240 \text{ V}$
- Primary current  $I_p = 1 \text{ A}$
- Eight lamps, each rated  $12 \text{ V}$  and  $24 \text{ W}$

**Find:** The efficiency, as a percentage

### FORMULA

$$\text{efficiency} = \frac{P_{\text{out}}}{P_{\text{in}}}$$

**Why this one.** Both powers can be built from what is given, so no transformer turns ratio is needed at all. The lamp voltage of  $12 \text{ V}$  is a distractor here: the lamps are rated  $24 \text{ W}$  each, and a wattage is already a power, so it can be used directly without touching the voltage.

### WHAT EACH SYMBOL MEANS

- $P_{\text{out}}$  = the total power the lamps actually consume, in watts
- $P_{\text{in}} = V_p I_p$ , what the mains delivers to the primary, in watts
- $V_p, I_p$  = the primary voltage and current, in volts and amperes
- efficiency = the ratio of the two, multiplied by 100 to make a percentage

### THE WORKING, LINE BY LINE

1. Output power = 8 lamps  $\times$  24 W each
2. Output power = 192 W
3. Input power =  $V_p \times I_p = 240 \times 1$
4. Input power = 240 W
5. efficiency =  $192 \div 240 = 0.8$
6. As a percentage:  $0.8 \times 100 = 80 \%$

### CHECK IT

Efficiency can never exceed 100 %, because a transformer has no source of energy of its own. If your fraction comes out above 1, you divided input by output instead of output by input. Here 192 is comfortably less than 240, so the answer is safely under 100 %.

Efficiency is output power over input power. Output: eight lamps at  $24 \text{ W}$  each is  $8 \times 24 = 192 \text{ W}$ . Input: the primary side,  $P = VI = 240 \times 1 = 240 \text{ W}$ . So efficiency =  $192/240 = 0.8$ , that is 80 per cent. The  $12 \text{ V}$  rating of each lamp is not needed, because the wattage already states the power each one consumes.

---

## 11 Maxwell and Electromagnetic Waves

**Speed tip.** In a vacuum everything is  $c = 3 \times 10^8$ .  $E/B = c$  for a plane wave, and  $c = f\lambda$  ties frequency to wavelength.

A radio transmitter produces an electromagnetic wave with an electric field amplitude of 0.080 V/m. What is the corresponding magnetic field amplitude?

✓  $2.67 \times 10^{-10} \text{ T}$

Correct. 0.080 divided by  $3.0 \times 10^8$ .

✗  $2.4 \times 10^7 \text{ T}$

Wrong. This MULTIPLIES by  $c$  instead of dividing, which is the commonest slip here.

✗  $2.67 \times 10^{-8} \text{ T}$

Wrong by a factor of 100 in the exponent.

✗ 0.080 T

Wrong. That is the electric amplitude repeated, in the wrong unit.

✗  $3.0 \times 10^8 \text{ T}$

Wrong. That is the speed of light itself, labelled as a field.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

An electromagnetic wave carries an electric field and a magnetic field locked together, at right angles to each other and to the direction of travel. They rise and fall in step, and their sizes are not independent: the ratio of the two is always the speed of light. That is why the magnetic part of light is so small a number, and why light is usually described by its electric field alone.

#### GIVEN

- $E_0 = 0.080 \text{ V m}^{-1}$ , the electric field amplitude
- $c = 3.0 \times 10^8 \text{ m s}^{-1}$

**Find:** The magnetic field amplitude  $B_0$ , in teslas

#### FORMULA

$$E_0 / B_0 = c, \text{ so } B_0 = E_0 / c$$

**Why this one.** For any plane wave in free space the two amplitudes are tied by the speed of light, so one gives the other immediately with no wavelength, frequency or power needed. Dividing by such a large number is what makes  $B_0$  tiny, which is the physical point rather than an arithmetic accident.

#### WHAT EACH SYMBOL MEANS

- $E_0$  = the electric field amplitude, in  $\text{V m}^{-1}$
- $B_0$  = the magnetic field amplitude, in teslas
- $c$  = the speed of light in vacuum,  $3.0 \times 10^8 \text{ m s}^{-1}$

#### THE WORKING, LINE BY LINE

1.  $B_0 = E_0 \div c$
2.  $B_0 = 0.080 \div (3.0 \times 10^8)$
3. Write the top in standard form:  $0.080 = 8.0 \times 10^{-2}$
4. Front numbers:  $8.0 \div 3.0 = 2.67$
5. Powers of ten:  $10^{-2} \div 10^8 = 10^{-10}$
6.  $B_0 = 2.67 \times 10^{-10} \text{ T}$

#### CHECK IT

Multiplying instead of dividing gives  $2.4 \times 10^7 \text{ T}$ , which is offered as an option and is physically absurd: the strongest laboratory magnets manage a few tens of teslas. Whenever you divide by  $c$  the answer must come out very SMALL, and whenever you multiply by  $c$  it must come out very large. That single habit sorts this whole family of questions.

In an electromagnetic wave the two fields are locked together by  $E = cB$ , so  $B = E \text{ over } c = 0.080 / 3.0 \times 10^8 = 2.67 \times 10^{-10} \text{ T}$ . The magnetic amplitude always comes out tiny beside the electric one, by a factor of the speed of light, which is why radio work quotes  $E$  rather than  $B$ .

**A laser pointer emits 5.0 mW of power in a beam with cross-sectional area  $3.0 \times 10^{-6} \text{ m}^2$ . Calculate the intensity.**

✓ 1667 W/m<sup>2</sup>

Correct.  $5.0 \times 10^{-3}$  divided by  $3.0 \times 10^{-6}$ .

✗  $1.5 \times 10^{-8} \text{ W/m}^2$

Wrong. This MULTIPLIES the power by the area instead of dividing.

✗ 1.667 W/m<sup>2</sup>

Wrong by a factor of 1000, from leaving the power in milliwatts.

✗  $6.0 \times 10^{-4} \text{ W/m}^2$

Wrong. No route to this from these numbers.

✗ 1000 W/m<sup>2</sup>

Wrong. 1000 W/m<sup>2</sup> is the intensity of SUNLIGHT, the figure being compared against.

#### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Intensity is power spread over area, which is why a laser feels dangerous and a light bulb of the same wattage does not. The bulb throws its power over a whole room; the laser concentrates it into a spot a few millimetres across. Same watts, wildly different intensity, and it is the intensity that does the damage.

#### GIVEN

- $P = 5.0 \text{ mW} = 5.0 \times 10^{-3} \text{ W}$
- $A = 3.0 \times 10^{-6} \text{ m}^2$

**Find:** The intensity  $I$ , in  $\text{W m}^{-2}$

#### FORMULA

$$I = \frac{P}{A}$$

**Why this one.** Intensity is defined as power per unit area, so this is the definition rather than a derived result. The only real work is the conversion of milliwatts to watts, which must happen before the division.

#### WHAT EACH SYMBOL MEANS

- $I$  = the intensity, in watts per square metre
- $P$  = the power, in watts
- $A$  = the cross-sectional area of the beam, in  $\text{m}^2$

#### THE WORKING, LINE BY LINE

1. Convert the power:  $5.0 \text{ mW} = 5.0 \times 10^{-3} \text{ W}$
2.  $I = P \div A$
3.  $I = (5.0 \times 10^{-3}) \div (3.0 \times 10^{-6})$
4. Front numbers:  $5.0 \div 3.0 = 1.667$
5. Powers of ten:  $10^{-3} \div 10^{-6} = 10^3$
6.  $I = 1.667 \times 10^3 = 1667 \text{ W m}^{-2}$

#### CHECK IT

The option 1.667 is what you get by leaving the power in milliwatts, and it is the one most people pick. Feel the size of the answer:  $1667 \text{ W m}^{-2}$  is above the roughly  $1000 \text{ W m}^{-2}$  of full sunlight, from a 5 mW pointer, which is precisely why you must never aim one at an eye. If your answer had come out around  $1 \text{ W m}^{-2}$ , the conversion did not happen.

Intensity is power per unit area:  $I = P \text{ over } A = 5.0 \times 10^{-3} / 3.0 \times 10^{-6} = 1667 \text{ W/m}^2$ . The interpretation the slide draws is worth keeping: despite a tiny power of only 5 mW, this beats the roughly 1000 W per square metre of full sunlight, because the energy is concentrated into a very small area. That is why a laser pointer is dangerous to the eye and a 5 mW lamp is not.

11.3

MEDIUM

Lecture slides, module 5, slide 32

Select the properties of electromagnetic waves the slides list.

✓ Transverse: E and B oscillate perpendicular to the direction of propagation

Correct, transverse.

✓ In vacuum they all travel at  $c = 3 \times 10^8 \text{ m/s}$

Correct, and it is the same c for every part of the spectrum.

✓ They need no medium, unlike sound

Correct, and it is the property that distinguishes them from sound.

✓ E and B are linked at every point by  $E = cB$

Correct, which is what makes B tiny compared with E.

✓ They carry energy, with the flux given by the Poynting vector

Correct, and the Poynting vector  $S = \text{one over } \mu_0, \text{ times } E \text{ cross } B$ .

✗ They travel faster through a dense medium than through vacuum

Wrong. Nothing exceeds c, and a medium slows light rather than speeding it.

Five properties, and the sixth option is false in a way worth knowing: nothing outruns c, and light in fact SLOWS in a medium such as glass or water. Note also that all electromagnetic waves share the same speed in vacuum, from radio at the long end of the spectrum to gamma rays at the short one, and only their frequency and wavelength differ, tied together by  $c = f \lambda$ .

11.4

HARD

Lecture slides, module 5, slide 40

In an electromagnetic wave the electric and magnetic contributions to the energy density are equal.

Answer: true

True, and it follows from  $E = cB$ . The energy density is  $u = \text{one half } \epsilon_0 E^2 \text{ plus } B^2 \text{ over } 2 \mu_0$ . Substituting  $B = E/c$  and  $c = 1 \text{ over the square root of } \mu_0 \epsilon_0$  makes the second term identical to the first, so the energy is shared exactly evenly between the two fields and  $u = \epsilon_0 E^2 = B^2 \text{ over } \mu_0$ . The magnetic amplitude LOOKS tiny beside the electric one, but it carries just as much energy.

11.5

HARD

Lecture slides, module 5, slide 33

In  $E_y = E \sin(kx - \omega t)$ , the wave number is  $k = 2 \pi \text{ over } \lambda$  and the angular frequency is  $\omega = 2 \pi f$ .

Answer:

$\lambda$ , the wavelength. Both symbols carry the same  $2 \pi$ : k packs  $2 \pi$  into one wavelength of SPACE,  $\omega$  packs  $2 \pi$  into one period of TIME.  $E_y$  and  $B_z$  share the identical  $\sin(kx - \omega t)$ , which is why the two fields peak together and vanish together.

11.6

HARD

Lecture slides, module 5, slide 33

In the plane wave  $E_y = E_0 \sin(kx - \omega t)$  and  $B_z = B_0 \sin(kx - \omega t)$ , the electric field points along y, the magnetic field along z, and the wave travels along x.

Answer: **true**

True, and the three directions are mutually perpendicular by construction. The subscript on each field names its own direction, and the x inside the bracket names the direction of travel. Hold your thumb, first finger and second finger at right angles: E, B and the direction of travel, never two of them along the same line.

11.7

HARD

Lecture slides, module 5, slide 42

The intensity of an electromagnetic wave,  $I = \frac{1}{2} \epsilon_0 c E_0^2$ , is measured in watts per square {{1}}.

Answer:

Watts per square metre: power spread over area. The half is there because the field oscillates, so the AVERAGE of the squared sine over a cycle is one half, not one. The same one half turns up in  $\frac{1}{2} C V^2$  and  $\frac{1}{2} L I^2$ , always for the same reason.

11.8

HARD

Lecture slides, module 5, slide 42

Doubling the peak electric field of a wave doubles its intensity.

Answer: **false**

It QUADRUPLES it.  $I = \frac{1}{2} \epsilon_0 c E_0^2$ , and the field is squared, so twice the field gives four times the intensity. That is why a lamp twice as bright does not feel twice as fierce, and why halving the field cuts the power to a quarter.

11.9

HARD

Lecture slides, module 5, slide 41

The Poynting vector  $S = \frac{1}{\mu_0} E \times B$ , gives the rate at which {{1}} is transported per unit area by the wave.

Answer:

Energy. The name is a happy accident: the vector POINTS the way the energy travels, which is the direction of  $E \times B$ , at right angles to both fields. Its size is the power crossing each square metre, and averaging it over a cycle is exactly what gives the intensity  $I$ .

In free space Maxwell's equations give a wave equation whose speed is  $v = 1$  over the square root of  $\mu_0 \epsilon_0$ . Evaluating it with  $\mu_0 = 4 \pi \times 10^{-7}$  and  $\epsilon_0 = 8.85 \times 10^{-12}$  gives:

✓  $2.998 \times 10^8$  m/s, which is exactly the speed of light

Correct, and the agreement with the measured speed of light is the whole point.

✗  $3.34 \times 10^{-9}$  m/s

Wrong.  $3.34 \times 10^{-9}$  is the square root of the product, the number you still have to invert.

✗  $1.11 \times 10^{-17}$  m/s

Wrong.  $1.11 \times 10^{-17}$  is the product itself, before the square root.

✗  $8.99 \times 10^9$  m/s

Wrong.  $8.99 \times 10^9$  is Coulomb's constant  $k$ , a different combination of the same constants.

✗  $2.998 \times 10^6$  m/s

Wrong by a hundred in the exponent.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

Maxwell's greatest result was not a new experiment but a piece of arithmetic. He combined his four equations, found they predicted a wave, and computed its speed from two constants measured in entirely electrical and magnetic experiments, with no light involved. The number that came out was the measured speed of light, which is how light was discovered to be an electromagnetic wave.

### GIVEN

- $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$

**Find:** The wave speed  $v$ , in  $\text{m s}^{-1}$

### FORMULA

$$v = 1 / \sqrt{\mu_0 \epsilon_0}$$

**Why this one.** The two constants come from measurements with magnets and with capacitors, and neither has anything to do with light, which is what makes the result so striking. Do the product first, then the square root, then the reciprocal; changing that order is the usual source of error.

### WHAT EACH SYMBOL MEANS

- $v$  = the speed of the electromagnetic wave in free space, in  $\text{m s}^{-1}$
- $\mu_0$  = the permeability of free space, from magnetic measurements
- $\epsilon_0$  = the permittivity of free space, from electric measurements
- $c$  = the accepted symbol once you recognise the answer as the speed of light

### THE WORKING, LINE BY LINE

1. Multiply first:  $\mu_0 \epsilon_0 = 4\pi \times 10^{-7} \times 8.85 \times 10^{-12}$
2.  $4\pi = 12.566$ , and  $12.566 \times 8.85 = 111.2$
3. Powers of ten:  $10^{-7} \times 10^{-12} = 10^{-19}$
4.  $\mu_0 \epsilon_0 = 111.2 \times 10^{-19} = 1.112 \times 10^{-17}$
5. Square root:  $\sqrt{1.112 \times 10^{-17}} = 3.336 \times 10^{-9}$
6. Reciprocal:  $v = 1 \div (3.336 \times 10^{-9})$
7.  $v = 2.998 \times 10^8 \text{ m s}^{-1}$ , which is the speed of light  $c$

### CHECK IT

Taking a square root of a power of ten needs the exponent to be EVEN, so rewrite  $1.112 \times 10^{-17}$  as  $11.12 \times 10^{-18}$  before rooting, and the root is  $\sqrt{11.12 \times 10^{-18}} = 3.336 \times 10^{-9}$ . The option  $3.34 \times 10^{-9}$  is what you get by stopping before the reciprocal, and  $8.99 \times 10^9$  is the Coulomb constant wandering in from another topic. You should recognise  $3 \times 10^8$  on sight, which makes this question answerable in five seconds.

$\mu_0$  times  $\epsilon_0 = 4\pi \times 10^{-7} \times 8.85 \times 10^{-12} = 1.113 \times 10^{-17}$ . Its square root is  $3.336 \times 10^{-9}$ , and one divided by that is  $2.998 \times 10^8$  m/s. This is the result the slides call the moment that shook nineteenth-century physics: two constants measured in laboratory experiments about magnets and static charge, with no light involved at all, combine to give the measured speed of light exactly. Maxwell's own conclusion was that we can scarcely avoid the inference that light consists in transverse undulations of the same medium which is the cause of electric and magnetic phenomena.

Continuing the laser problem, the intensity is  $1667 \text{ W/m}^2$ . Using  $I = \text{one half } \epsilon_0 c E\text{-nought squared}$ , find the electric field amplitude.

✓ about 1120 V/m

Correct. The square root of 3334 divided by  $2.655 \times 10^{-3}$ .

✗ about 1.12 V/m

Wrong by a factor of 1000, a slip in the exponent before the square root.

✗ about 3334 V/m

Wrong. 3334 is the numerator  $2I$ , not the answer.

✗ about  $2.655 \times 10^{-3} \text{ V/m}$

Wrong.  $2.655 \times 10^{-3}$  is the denominator  $\epsilon_0 c$ .

✗ about 628 V/m

Wrong. This forgets the factor of 2 when rearranging the one half.

### BACKGROUND, SO THIS IS NOT JUST A FORMULA

The energy in an electromagnetic wave is carried by its fields, and the intensity is what the fields deliver per second per square metre. The relation involves  $E_0$  SQUARED, so doubling the field quadruples the intensity, which is the same square law that appears in every wave in physics. Going backwards from intensity to field therefore needs a square root.

### GIVEN

- $I = 1667 \text{ W m}^{-2}$ , carried over from the previous part
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
- $c = 3.0 \times 10^8 \text{ m s}^{-1}$

**Find:** The electric field amplitude  $E_0$ , in  $\text{V m}^{-1}$

### FORMULA

$$E_0 = \sqrt{2I / (\epsilon_0 c)}$$

**Why this one.** It is  $I = \frac{1}{2}\epsilon_0 c E_0^2$  rearranged for  $E_0$ . The half moves across as a doubling on top,  $\epsilon_0 c$  moves to the bottom, and the square comes off as a square root. Do it as one rearranged formula rather than in pieces, because each move is a chance to lose the factor of two.

### WHAT EACH SYMBOL MEANS

- $E_0$  = the electric field amplitude, in  $\text{V m}^{-1}$
- $I$  = the intensity, in  $\text{W m}^{-2}$
- $\epsilon_0$  = the permittivity of free space
- $c$  = the speed of light, in  $\text{m s}^{-1}$
- the 2 = what is left of the half after rearranging

### THE WORKING, LINE BY LINE

1. Bottom first:  $\epsilon_0 \times c = 8.85 \times 10^{-12} \times 3.0 \times 10^8$
2. Front numbers:  $8.85 \times 3.0 = 26.55$
3. Powers of ten:  $10^{-12} \times 10^8 = 10^{-4}$
4. Bottom =  $26.55 \times 10^{-4} = 2.655 \times 10^{-3}$
5. Top:  $2 \times I = 2 \times 1667 = 3334$
6. Divide:  $3334 \div (2.655 \times 10^{-3}) = 1.2558 \times 10^6$
7. Square root:  $E_0 = \sqrt{1.2558 \times 10^6}$
8.  $E_0 = 1120 \text{ V m}^{-1}$

### CHECK IT

The option  $2.655 \times 10^{-3}$  is the value of  $\epsilon_0 c$ , offered to catch anyone who stopped mid-calculation, and  $628 \text{ V m}^{-1}$  is roughly what dropping the factor of 2 would give. Cross-check forwards in one line:  $\frac{1}{2} \times 8.85 \times 10^{-12} \times 3 \times 10^8 \times 1120^2 = \frac{1}{2} \times 2.655 \times 10^{-3} \times 1.2544 \times 10^6 = 1665 \text{ W m}^{-2}$ , which is the intensity we started from. Note too that  $\sqrt{1.2558 \times 10^6}$  is manageable by hand, since  $10^6$  roots to  $10^3$  and  $\sqrt{1.2558}$  is about 1.12.

Rearrange in symbols first: E-nought = square root of (2I over  $\epsilon_0 c$ ). Top:  $2 \times 1667 = 3334$ . Bottom:  $8.85 \times 10^{-12} \times 3.00 \times 10^8 = 2.655 \times 10^{-3}$ . Dividing gives  $1.256 \times 10^6$ , and its square root is about 1120 V/m. The square root at the end is the step to watch, since E-nought appears SQUARED in the intensity formula.